

REVIEW

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A systematic review of real-time detection and classification of power quality disturbances

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Abstract

This paper offers a systematic literature review of real-time detection and classification of Power Quality Disturbances (PQDs). A particular focus is given to voltage sags and notches, as voltage sags cause huge economic losses while research on voltage notches is still very incipient. A systematic method based on scientometrics, text similarity and the analytic hierarchy process is proposed to structure the review and select the most relevant literature. A bibliometric analysis is then performed on the bibliographic data of the literature to identify relevant statistics such as the evolution of publications over time, top publishing countries, and the distribution by relevant topics. A set of articles is subsequently selected to be critically analyzed. The critical review is structured in steps for real-time detection and classification of PQDs, namely, input data preparation, preprocessing, transformation, feature extraction, feature selection, detection, classification, and characterization. Aspects associated with the type of disturbance(s) addressed in the literature are also explored throughout the review, including the perspectives of those studies aimed at multiple PQDs, or specifically focused on voltage sags or voltage notches. The real-time performance of the reviewed tools is also examined. Finally, unsolved issues are discussed, and prospects are highlighted.

Keywords Bibliometric analysis, Classification, Detection, Power quality (PQ), Real-time, Systematic review, Voltage sag, Voltage notch

1 Introduction

Power Quality (PQ) is defined as the set of characteristics of electricity at a given point in an electrical system [1], and these characteristics are evaluated against a set of agreed reference parameters. Thus, PQ Disturbances (PQDs) are deviations in these characteristics from the reference parameters, deviations which can be perceptible to the users of the electrical power grid (producers and consumers) [2]. Therefore, PQDs are classified and studied as the combination of voltage quality and current quality [2]. The impact of PQDs will depend on the

severity of these deviations interfering with the expected operation of the electrical systems. Noticeable PQDs will directly affect the interaction between consumers and producers of electricity, leading to energy inefficiency, limited generation/consumption of electricity, malfunction and damage of sensitive equipment, maloperation of control-based industrial processes, etc. [3].

The detection and classification of PQDs are among the main components of PQ monitoring systems in the smart grid paradigm [3, 4]. The detection process helps to indicate the time and location of deviations in voltage and current, whereas the classification process contributes to the identification of disturbances and sources of disturbance, and the selection of adequate mitigation techniques to overcome current and/or future equipment malfunction. The real-time approach of PQ monitoring systems (synchronized, continuous, single, or multi-point measurements) can help in the understanding of PQD

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propagation in the power grid and allow accurate and faster decisions regarding the mitigation of PQ issues.

Some PQDs are more likely to appear and cause equipment malfunctioning than others. Voltage sag, also referred to as voltage dip, is the decrease of the RMS voltage between 0.1 and 0.9 pu lasting from 0.5 cycle to 1 min [5, 6]. Voltage sags are mainly caused by faults in the power system, transformer energizing, motor starting, and switching of heavy loads [5, 6]. The frequency of occurrence of voltage sags is between a few tens and one thousand events per year, with typical durations of less than 1 s, and voltage drops above 40% [6]. It is widely known as one of the PQDs that produce the highest economic losses in industrial, commercial, and microgrids/isolated networks [4, 7–11]. Other widely studied PQDs are swells, interruptions, imbalance, flicker, and harmonics [4, 9, 11], disturbances for which specialized algorithms have been proposed in the literature. In contrast, there are PQDs that have not yet been widely explored, especially in the scenario of high penetration of renewable energy resources and their interfacing power electronics circuits.

Voltage notches are steady-state sub-cycle waveform distortions produced by power electronic converters when current is commutated from one phase to another leading to short-duration overcurrent [5, 9, 12, 13]. A voltage notch is characterized by its depth, i.e., the average distance of the line voltage to the ideal sine wave during notch; width, i.e., the notch duration with values of less than half cycle; and area, i.e., the product of the notch depth times the notch width [12]. There is still a lack of comprehensive assessment and accurate techniques for real-time detection, classification, characterization, and aggregation of cycle-based voltage notches when considering the operation of power grids. Thus, notches have been mainly assessed with non-specific, multi-disturbance detection and classification techniques (“multiple PQDs” techniques).

The detection and classification of PQDs in both offline and real-time applications have been addressed from different perspectives. Several methods for the classification of three-phase unbalanced voltage sags due to faults are described in [14]. Signal processing and Artificial Intelligence (AI) techniques for classification of several PQ events are reviewed in [13], where Fourier Transform (FT), Short-Time FT (STFT), and Wavelet Transform (WT) are considered as the main signal processing techniques, and expert systems, fuzzy systems, Artificial Neural Networks (ANNs), and Genetic Algorithms (GA) as the main AI techniques. In [13], the authors also propose an initial structure for the detection and classification of PQDs, including feature extraction and classification (decision making). Some of the techniques

used for signal analysis are described in [15]: FT, STFT, WT, Gabor Transform (GT), Stockwell Transform (ST), Kalman Filter (KF), etc., as well as some automatic classification techniques such as ANNs, fuzzy logic, Support Vector Machines (SVM), and Bayesian classifiers. The detection and analysis of voltage events are reviewed in [9], which focuses on offline and real-time techniques using a combination of WT and ANNs, SVM, and Fuzzy Expert Systems (FES). Reference [16] introduces the category of optimization techniques for the classification of PQ events and also presents a comparative chart taking into account input data (synthetic/practical) and input noise. In [17], WT is compared with other techniques for the detection of transient disturbances in voltage supply systems. Methods for the identification of voltage sag sources are explored in [18], which classifies them into single and multi-monitor (multi-point) based measurements. This first set of works (2003–2013) are focused on PQD detection and classification techniques, and most do not include pre-or post-processing techniques.

Comprehensive reviews on signal processing, AI, and optimization techniques applied in the detection and classification of PQDs are developed in [19, 20]. These reviews present two complementary structured methodologies for PQD detection and classification, i.e., input data space, feature extraction, feature selection, classification, and decision space [19], and pre-processing and post-processing stages [20]. A holistic taxonomy for signal processing, AI, and optimization techniques are also presented, highlighting the need for more methods in the detection and classification of real-time, noisy, three-phase, single and multiple PQDs.

The RMS method, WT, ST, ANNs, SVM, and some indices for voltage sag disturbance recognition are reviewed in [21]. In [4], WT and SVM are reviewed and applied for the detection and classification of sags, swells, and harmonics, including a table that relates PQDs and corresponding standards. In [22], a review of PQDs measurement and analysis on shipboard power systems is presented, which mainly refers to voltage and frequency fluctuations, fault detection and classification, voltage sags and swells, transients and voltage notching, harmonic distortion, and voltage imbalance. In addition to signal processing techniques, reference [10] classifies fuzzy logic, ANNs, SVM, particle swarm optimization, and GA as soft-computing techniques for feature extraction and classification of PQDs.

A comprehensive review and comparison of PQD detection and classification are presented in [23], which describes the advantages and disadvantages of different techniques. Signal processing techniques are classified into eight categories: FT, WT, ST, GT, KF, Hilbert-Huang Transform (HHT), Mathematical Morphology (MM),

and others including Variational Mode Decomposition (VMD). For the classification of PQDs, seven categories are presented: ANNs, SVM, FES, neuro-fuzzy system, Extreme Learning Machine (ELM), deep learning, and miscellaneous pattern recognition techniques. In [24], a taxonomy is proposed for digital signal processing techniques, one which presents the categories of non-parametric and parametric techniques including in the latter KF, rotational invariance techniques, multiple signal classification, and autoregressive-moving-average, etc. These works contribute to widening the state-of-the-art techniques used for the detection and classification of PQDs, while outlining the advantages and disadvantages for the techniques. However, search rules used in the literature reviews are not presented.

Reference [25] presents a detailed assessment of the theory and application of ST for the detection and classification of PQDs. A mitigation stage is proposed in [3], where the input data is from the stages of detection and classification of PQDs in the context of renewable energy resources.

Real-time techniques are explicitly addressed for the detection and classification of PQDs in [11], which presents the search rules, the evolution of PQDs publications, the internal structure of a typical embedded system for PQD classification, and other comparative analyses. Moreover, the need for testing the detection and classification algorithms with no synthetic (measured) PQDs is highlighted in [26], and it concludes that such an aspect will be crucial for applications in smart grids. In this same approach, references [27] and [28] present a review of the potential applications of deep learning in smart grids, where the structured scheme of consecutive steps for detection and classification of PQDs is blurred into a compact, general, black-box approach. Finally, in [29], the main advantages and disadvantages of several techniques for the detection and classification of PQDs are summarized, including an accuracy assessment of the methods proposed in different publications.

Based on the above synthesis, there is extensive literature on detection and classification of PQDs but some relevant topics still require further research and analysis from different perspectives, e.g., algorithms for real-time applications, the need of applications in the field, etc. Other topics have been marginally explored such as characterization and classification of voltage notches. In this context, the proper selection of relevant literature to analyze the current state of the art is challenging. Therefore, this paper proposes a systematic and reproducible approach to identify relevant literature in both extensively and marginally explored topics. Moreover, the extensive literature is worth studying from the quantitative perspective with a bibliometric analysis to identify

research trends. The bibliometric-based methodology proposed in this paper is then applied to carry out a systematic literature review of the real-time detection and classification of PQDs. The main contributions of this paper are:

- The proposal and application of a novel methodology for reproducible and systematic literature reviews.
- The application of a bibliometric analysis to the metadata retrieved from bibliographic databases.
- The description of techniques for real-time detection and classification of PQDs.
- The categorization of techniques used in the stages of PQD detection and classification, and their corresponding advantages and disadvantages in the context of electrical engineering.
- The identification and emphasis on marginally explored PQDs (voltage notches).

Figure 1 illustrates the structure and logical framework of the paper. In line with Fig. 1, the remainder of the paper is structured as follows. Section 2 shows the concepts and theoretical background for the understanding of PQDs and their classification, while Sect. 3 presents the stages of the systematic methodology for the literature review. Sections 4 and 5 develop quantitative and qualitative analyses of the literature in the bibliometric analysis and the literature review, respectively. Section 5 describes and analyzes the stages for PQD detection and classification. The discussion of the main findings related to the literature review is summarized in Sect. 6, and

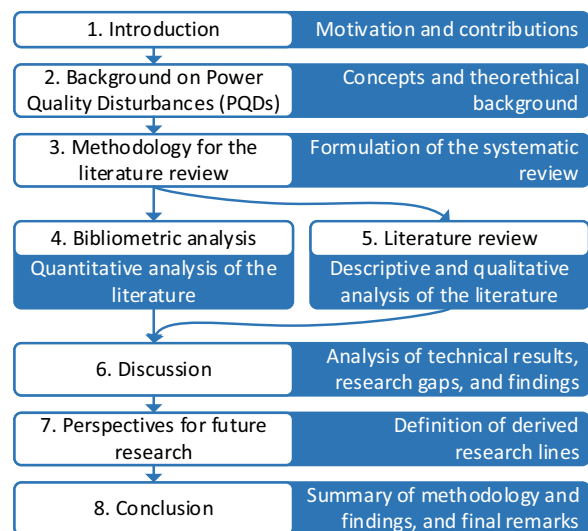


Fig. 1 Structure and logical framework of the paper

Sect. 7 proposes some perspectives for future research. Finally, Sect. 8 presents the conclusions.

2 Background on power quality disturbances (PQDs)

The characteristics of electricity at a given point on an electrical system, compared to a set of reference technical parameters are widely known as PQ [1]. The reference (ideal) signals in three-phase systems are three pure sinusoid waveforms with constant phase angles and amplitudes, and a 120° phase shift between them. A PQD therefore refers to any deviations from the reference voltages and currents. In this context, PQ is assessed as the combination of voltage quality and current quality [2].

Table 1 summarizes the features of typical PQDs in the power grid. The types of PQDs are grouped into variations (continuous deviations) and events (finite deviations).

Variations are usually produced in a typical operation of a power system, while events are generally unpredictable phenomena. Given the dynamic nature of the power system (power frequency changes, slow voltage changes, imbalance), the main sources of common variations (waveform distortion, notches, fluctuations) are the power electronics interfaces, e.g., for renewable energy sources, energy-saving equipment, electric vehicle chargers, and mains communication systems. Switching maneuvers, faults, and lightning strikes are

typical sources of events. By using these two categories, the taxonomy of typical PQDs (continuous and finite deviations) can be structured for detection, processing, classification, and further processing.

According to Table 1, voltage notches are classified as variations (continuous deviations), while other disturbances in this category are power frequency variations which refer to deviations of the power system fundamental frequency from the nominal value (50 Hz or 60 Hz). Also, among variations, slow voltage changes refer to the increase (overvoltage) or decrease (under-voltage) of about 20% of the rated RMS voltage lasting more than 1 min. Waveform distortions including harmonics, interharmonics, and suprahharmonics, are periodic deviations from the ideal power frequency sine wave, characterized by the spectral content of the deviation. Fluctuations are systematic changes of the voltage envelope and can be perceived as flicker. In addition, imbalance refers to differences between voltage and current magnitudes of phases in a three-phase system, and/or a deviation from the ideal 120° phase shift between phases [5].

Voltage sags are events (finite deviations) according to Table 1. Other events include swells, which refer to the increase of the RMS voltage above 1.1 pu lasting from half cycle to 1 min. Interruptions are characterized by the complete loss of voltage, i.e., less than 0.1 pu. Finally, transients are sudden changes in voltage or current that occur over a short period of time [5].

Table 1 Classification of typical PQDs

Type of disturbance	Deviation		Evaluation	Main cause
Variations (continuous deviations)	Frequency		–	Imbalance in generation and demand
	Slow voltage changes		Voltage magnitude	Variation in demand and generation
	Waveform distortion	Frequency domain	DC Offset Harmonics Interharmonics Suprahharmonics	Electrical machines (transformers, generators, motors), power electronic converters, narrowband PLC
	Notches	Time domain	Momentary amplitude deviation	
	Fluctuation		Flicker	Welding machines, multi-cycle control
	Imbalance		Symmetrical components	Unbalanced (single-/two-phase) loads or generators
Events (finite deviations)	Rapid voltage changes		Voltage magnitude	Connection of heavy loads, fault clearance, climatic hazards
	Sags			
	Swells			
	Interruptions			
	Transients			Switching operations, lightning strikes

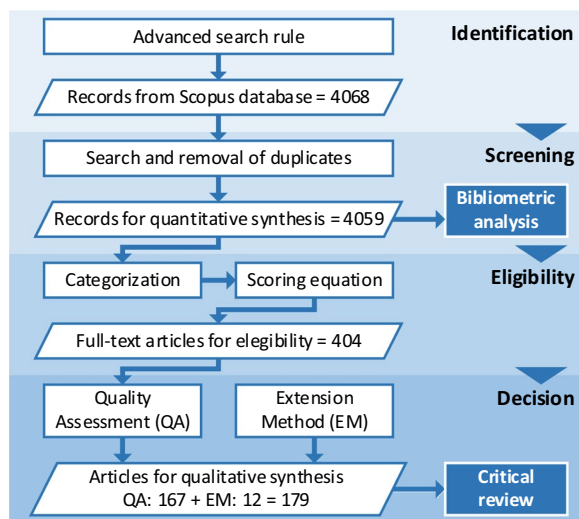


Fig. 2 Stages of the systematic literature review

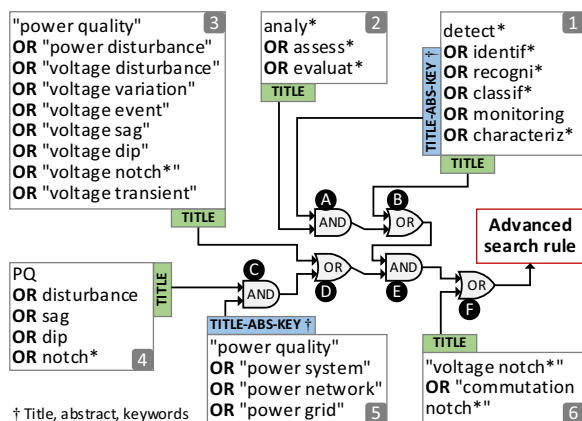


Fig. 3 Advanced search rule applied in the Scopus database

3 Methodology for the literature review

The general schema to conduct the literature review is based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [30]. According to Fig. 2, the process is performed through four stages, namely, identification, screening, eligibility, and decision. Details on the stages are given in the following subsections.

3.1 Identification and screening

In the identification stage, an advanced search rule is formulated in the Scopus database to obtain bibliographic metadata of publications related to the detection and classification of PQDs. The search rule is shown in Fig. 3 and is formed by logical operators

and sets of terms to be searched only in titles or titles, abstracts, and keywords. Set 1 has general terms such as detection and classification, while terms with a broader meaning are listed in Set 2. To avoid misleading results with these broader terms, the constraints including terms of Set 1 in title, abstract, or keywords are merged in Gate A. Set 3 contains terms related to PQ with a focus on sags and notches, while Set 4 also refers to PQ with more vague terms. Thus, to avoid misleading results, the constraints including terms of Set 5 are merged in Gate C. The search merged in Gate E must include terms of general topics (B) and PQ (D). Finally, the terms of Set 6 are merged in Gate F to include more publications related to voltage notches because the results in this topic are very limited.

The search rule was applied on March 2nd, 2022, and limited to publications up to 2021. The search yielded 4068 records. Then, in the screening stage, duplicates were removed. The resulting 4059 records, including 2140 conference papers (53%), 1841 journal articles (45%), 61 reviews (2%), and 17 other types of publications, e.g., books, book chapters, etc. (<1%), are studied quantitatively in the bibliometric analysis of Sect. 4.

3.2 Eligibility and decision

The eligibility and decision stages present the selection procedure to identify a reduced set of relevant and diverse research papers to be analyzed qualitatively in the critical review. Only journal and conference papers are considered because they summarize very well the state of research. The selection procedure is carried out in four steps as follows.

1. *Categorization* Two dimensions are defined to classify papers, namely, period according to date of publication (i.e., ≤ 2005 , 2006–2010, 2011–2015, and 2016–2021) and type of PQD (i.e., multiple PQDs, voltage sag, and voltage notch). All papers are classified according to both dimensions to obtain a final selection of papers distributed in time and type of disturbance. The classification in period is straightforward, and the classification according to the type of disturbance is made by searching the terms sag (or dip) and notch in titles. Papers not classified as sag or notch are included in the category of multiple PQDs. The resulting distributions of papers in both dimensions are shown in the light-blue rectangles in Fig. 4.
2. *Scoring equation* An equation is formulated to measure the relevance of papers from different perspectives, with five proposed indices. The title similarity index is calculated as the ratio of word coincidences to the total of words in the paper title, while the coincidences are given by predefined terms related

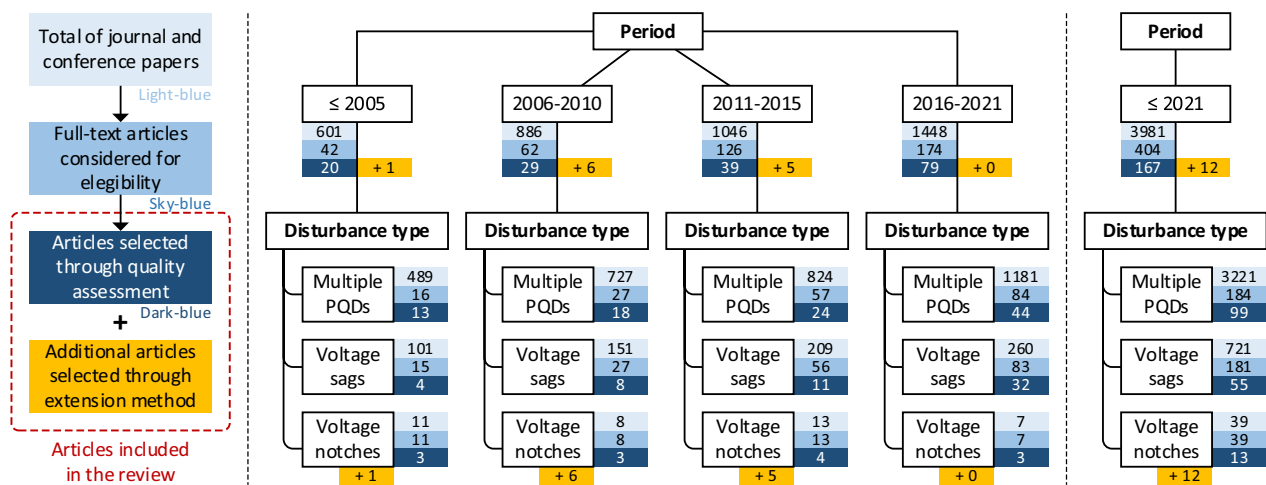


Fig. 4 Number of papers through the process of selection for the critical review, organized by period and type of PQD

Table 2 Weights for indices of the scoring equation

Index	Weight
Title similarity	0.3083
Abstract similarity	0.2114
Cross-citation	0.2883
Review cross-citation	0.1010
Journal	0.0910

to the topic, e.g., real-time, sag, detection. Likewise, an abstract similarity index is calculated in abstracts, whereas the cross-citation index is defined as the ratio of cross-citations (i.e., the number of citations by papers in the search list) to the number of citations of the most cited paper of the same year or later. Similarly, the review cross-citation index is calculated considering only citations by review papers in the list. Finally, the journal index is defined as 1 for journals and 0 for conference papers.

Weights are estimated for each of the five indices using the Analytic Hierarchy Process (AHP) [31]. AHP is a multicriteria decision-making approach in which factors are arranged in a hierarchic structure. Thus, weights are obtained by estimating the relative magnitude of pairwise comparisons and further computations as described in [31]. The resulting weights for the proposed five indices are reported in Table 2. The total score is calculated for each paper by summing the indices multiplied by the corresponding weights. A total of 404 top papers (10% of the papers in the search list) are selected according to scores considering a larger proportion of papers for periods between 2011 and 2021 (i.e., 2011–2015, and

Table 3 Questions for the quality assessment and their weights

Question	Weight
Q1: Does the paper address the purpose of our research?	0.3680
Q2: Does the paper present new techniques in the field?	0.2778
Q3: Does the paper have comparative analyses?	0.1321
Q4: Are the results of the study reproducible?	0.1321
Q5: Are the limitations and validity discussed?	0.0900

2016–2021) than for periods before 2011 (i.e., ≤ 2005, 2006–2010). This prioritization is done to consider more papers from recent periods. Also, the number of papers selected for types of disturbances is distributed as evenly as possible. The results are shown in the sky-blue rectangles in Fig. 4.

- Quality assessment** An assessment is conducted on the top 404 full-text papers to reduce the selection for the critical review, while the five questions in Table 3 are formulated. A set of three qualifications is defined for the questions: {0 (not at all), 0.5 (moderately), 1 (absolutely)}. AHP is then applied to estimate the weight for each question, and the results are reported in Table 3. The questions are evaluated for each paper, and the total score is calculated as the sum of the results of each question multiplied by the corresponding weights. Finally, 167 papers with a score higher than 0.8 are selected for the critical review, as reported by the dark-blue rectangles in Fig. 4.
- Extension method** Twelve additional papers related to voltage notches are identified and included because very few (only 13) are initially detected for this dis-

turbance. The extension method applied to identify additional papers is based on the concepts of bibliographic coupling and co-citations as explained in [32]. The resulting number of papers is reported in the yellow rectangles in Fig. 4.

The selection procedure conducted using the above four steps results in the papers reported in Table 4, where they are organized according to the proposed categories, i.e., period, and type of PQD.

4 Bibliometric analysis

The number of publications related to the detection and classification of PQDs over time is shown in Fig. 5, which also shows the number of those publications that include the term “real-time” (or “online” or “on-line”) in the title, abstract, or keywords.

The increasing interest in PQ detection and classification is observed especially from 2000 to 2010 (see Fig. 5). An apparent decrease occurred in 2014, but then the trend became positive again up to 2019. While the trend of the total number of publications decreased in 2020 and 2021, those related to real-time show an apparent increase. Moreover, Fig. 5 shows that a small number of publications include the term “real-time” in title, abstract, or keywords, reaching significant appearances only in the last decade. This situation highlights the need for further studies in the real-time detection and classification of PQDs.

To observe the state of research by country, Table 5 reports the number of publications from different countries, according to the affiliation of the first author. The top seven contributors, i.e., those countries with more

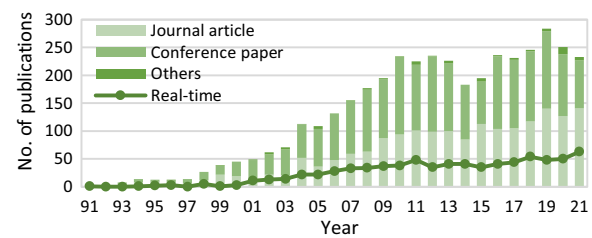


Fig. 5 Evolution over time of the number of publications per type related to the detection and classification of PQDs

than 100 publications, are shown. China has the largest contribution with a significant lead over the rest, and is followed by India, USA, Brazil, and Spain. Table 5 also shows the number of those publications with the term “real-time”. The corresponding ranking is similar to the ranking for the total number of publications, with the exception of Brazil which appears in sixth place. Table 5 also reports the percentages of contribution of the top seven countries to the total number of publications, with the top three, i.e., China, USA, and India, contributing more than 50% of the total number of publications and more than 60% of the publications with the term “real-time”.

Regarding the original language of publications, English is dominant with 85%. Moreover, a significant 13% were written in Chinese, while the remaining 2% correspond to other languages such as Spanish, Polish, and Portuguese.

A set of categories from the search are also identified from titles and the results for the number of publications in each category are presented in Fig. 6. The number of publications with or without the term “real-time” in the title, abstract, or keywords, are also identified for each category.

Figure 6a reports the results for general topics and shows that the largest number of publications is related to detection. The term detection usually refers

Table 4 References identified for disturbance types and periods

Type of PQD	Period	References	No. of references	
			Per period	Total
Multiple PQDs	≤ 2005	[33–45]	13	99
	2006–2010	[46–63]	18	
	2011–2015	[64–87]	24	
	2016–2021	[88–131]	44	
Voltage sags	≤ 2005	[132–135]	4	55
	2006–2010	[136–143]	8	
	2011–2015	[144–154]	11	
	2016–2021	[155–186]	32	
Voltage notches	≤ 2005	[187–190]	4	25
	2006–2010	[191–199]	9	
	2011–2015	[200–208]	9	
	2016–2021	[209–211]	3	

Table 5 Top seven countries according to the total number of publications on the topic

Country	Total no. of publications	Ranking for total	No. of publications related to real-time	Ranking for real-time
China	1281 (32%)	1	275 (36%)	1
India	549 (14%)	2	127 (17%)	2
USA	300 (7%)	3	62 (8%)	3
Brazil	174 (4%)	4	28 (4%)	6
Spain	153 (4%)	5	37 (5%)	4
Malaysia	139 (3%)	6	30 (4%)	5
Italy	139 (3%)	7	18 (2%)	7

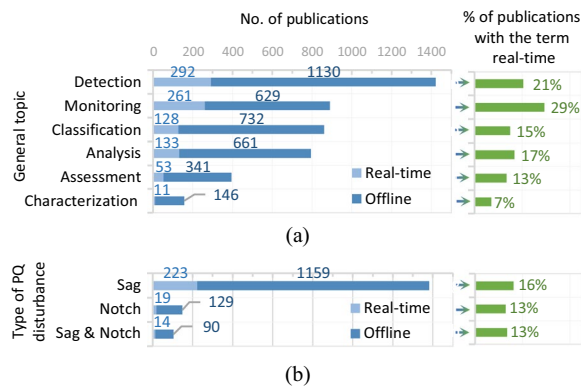


Fig. 6 Number of publications by **a** topic and **b** type of PQD, with or without the term “real-time”, and respective percentages

to algorithms for identifying the occurrence of a PQD. The second category according to the number of publications is monitoring, which refers to the tracking of power signals, including the PQDs. In this case, hardware implementation emerges as an important aspect. The third category is classification which refers to the recognition of the type of PQD, e.g., sag, notch, harmonics, transient, flicker. The category analysis is broader in meaning and comprises several aspects described in the rest of the categories. The category assessment includes methods for impact analysis of PQDs. Characterization refers to the quantification of parameters that define a PQD, e.g., depth and duration of voltage sags.

The corresponding percentages of publications with the term “real-time” in general topics are also shown in Fig. 6a. Monitoring has a substantial percentage because real-time implementation, i.e., algorithms and hardware, is usually discussed in this category. Detection is also important in real-time applications, e.g., for the operation of Dynamic Voltage Restorers (DVR) and protection systems. As the term “analysis” is a broader concept, it also includes some aspects of real-time applications. Classification, assessment, and characterization are more time-consuming tasks and are generally used in offline applications.

Figure 6b presents the results for types of PQDs. The large number of publications related to voltage sags demonstrates that research on the topic is already mature. On the other hand, only a few studies are related to notches, highlighting that research on the topic is incipient and requires further investigation given its relevance to the industrial sector. In the case of studies considering sags and notches at the same time, the terms are searched not only in titles but also in abstracts and keywords. However, few publications

are identified. In this context, tools to analyze simultaneously sags and notches can improve the state of research and be useful for the industry.

Regarding the corresponding percentage of publications with the term “real-time”, Fig. 6b shows that more real-time applications have been implemented for voltage sags, e.g., for the operation of fault protection systems. However, the percentage is still low in this category. In the case of the simultaneous analysis of sags and notches, very few applications in real-time have been developed.

5 Literature review

A comprehensive critical literature review of real-time detection and classification of PQDs is developed in this section, based on 179 selected papers (see Sect. 3.2). The categorization of the type of PQDs according to Sect. 3.2 is used along with the development of the literature review to analyze the articles from the perspective of multiple PQDs, voltage sags, and voltage notches.

The category of multiple PQDs includes studies oriented towards the classification of the disturbances. Thus, most articles in this category present methods to distinguish between a set of PQDs identified from voltage waveforms. At the beginning of the research on the detection and classification of PQDs, articles were focused on tools to extract the features to distinguish the types of PQDs. For instance, reference [33] proposes WT as the means to transform and extract the features, while [40] utilizes ST to extract distinctive features. Thereby, in this first period, references [33] and [40], and similar studies in [34–37], laid the foundations for the classification of multiple PQDs. Afterward, studies were focused on the automatic classification of PQDs, mostly using AI techniques such as ANN [38, 39], and SVM [50]. In this context, other studies included the classification of combined PQDs, e.g., sag with harmonics, flicker with sag, [61]. Recently, real-time performance in the detection and classification of PQDs has also been a concern [11], because of the applications in fault protection systems and root-cause detection and mitigation. Another real-time application is the analysis of the propagation of PQDs using measurements in various nodes of the system.

The second category according to the type of PQDs refers to voltage sags. This set of articles is generally focused on the classification of voltage sags into different categories. For instance, some studies use three-phase classifications of sags [140, 158], which are strongly related to the type of faults that cause the voltage sags. Another aspect analyzed in this category is the classification of sags according to the root causes [133, 147, 149, 150, 180], e.g., faults, the starting of induction motors and heavy loads, and transformer energizing. The

characterization of voltage sags, i.e., the quantification of the parameters such as duration, magnitude, starting and ending phase angle, etc., is also performed in some studies [132, 173, 174]. Finally, this set of articles also includes research on the detection of sags for applications such as DVR and protection systems operation [134, 136, 153]. In these applications, real-time detection is of utmost importance.

The third category according to the type of PQD is associated with voltage notches. These studies mainly deal with the detection of notches in voltage signals [189, 190, 194, 205, 206, 209]. The characterization of voltage notches is also performed in some cases. However, this characterization is usually incipient, ambiguous, and misses important features of voltage notches. Moreover, the classification according to types of voltage notches has not been addressed in the literature.

The categorization of articles in the type of disturbances aims to address different topics of interest: the classification of PQD types; the classification, characterization, and detection of voltage sags; and the classification, characterization, and detection of voltage notches. However, studies in the category of multiple PQDs include voltage sags [34–36, 38–58, 60–75, 77–80, 82–119, 121–131], and/or notches [33, 37, 40, 43, 49, 52, 54–56, 61, 62, 65–68, 74, 75, 77, 79, 82, 84, 85, 87, 91, 94–100, 102, 103, 105–110, 112–116, 118, 121, 123, 124, 130], though in such cases, there is no particular focus on those disturbances.

From a different perspective, the process for real-time detection and classification of PQDs can be analyzed from the stages and steps, as shown in the proposed schema in Fig. 7. In this literature review, four major stages are identified, namely: (i) input space, (ii) preprocessing, (iii) feature engineering, (iv) decision space. Moreover, each of the major stages includes the steps

to achieve their goals. Eight steps are identified in the comprehensive process. However, not all the steps are always necessary, and hence the proposed schema should be considered flexible. In the context of this review, the steps are defined as follows:

1. *Input data preparation (i)* The step for obtaining data from different sources for designing and training algorithms. Sources of data include laboratory experiments, field measurements, simulations, and synthetic data generated with equations representing the diverse PQDs. This step constitutes the input space stage.
2. *Data preprocessing (ii)* The step for preparing input data to improve efficiency in the subsequent stages. Preprocessing has tasks such as segmentation, normalization, and denoising. Data validation to ensure reliability and correctness of data, i.e., data quality, is also part of the data preprocessing step. However, no extensive details have been described in the reviewed literature regarding data validation. This step accounts for the preprocessing stage.
3. *Transformation (iii)* The process of converting raw data from one domain, e.g., time, frequency, time–frequency, to another. Transformations used in PQD detection and classification include FT, WT, and ST, among others. The transformation step is the first of the feature engineering stage.
4. *Feature extraction (iii)* The computation of numerical indices (from the transformation outputs/coefficients) that are usable for tools in the decision space. These indices usually comprise statistical variables. The feature extraction step is part of the feature engineering stage.
5. *Feature selection (iii)* The process of selecting features and reducing dimensionality to enhance the efficiency in the decision space. Selection can be manual or automatic using optimization methods and dimensionality reduction techniques. The feature selection step provides the output of the feature engineering stage.
6. *Detection (iv)* The identification of states different from the normal operation through thresholds and triggers. Detection is a step in the decision space.
7. *Classification (iv)* The distinction of the type of PQD. Classification is mostly conducted automatically using AI techniques. Classification is a step in the decision space.
8. *Characterization (of single PQDs) (iv)* The calculation of the parameters that characterize a single PQD (e.g., for voltage sags: Point-On-Wave (POW) of initiation/ending, duration, magnitude, phase angle

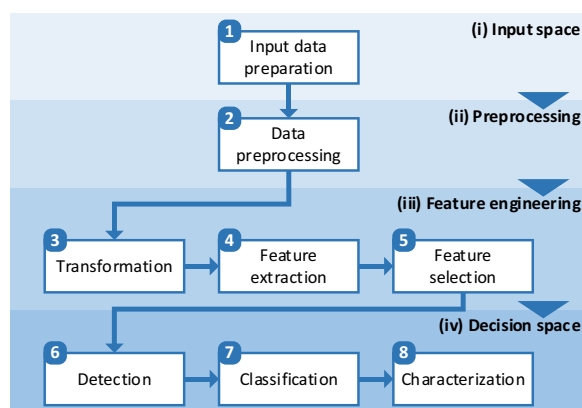


Fig. 7 General scheme for real-time detection and classification of PQDs and transversal topics to be dealt with in the review

jump). Characterization is a step in the decision space.

The literature review is structured and developed according to the eight steps to conduct the real-time detection and classification of PQDs. The categories according to the type of disturbance, i.e., multiple PQDs, voltage sags, and voltage notches are analyzed throughout the comprehensive review. Aspects of the real-time operation are also considered.

5.1 Input data preparation

The process of detection and classification of PQDs starts with the acquisition of signals. This process can be classified into two general groups according to the literature: measurement-based (laboratory, field) and model-based (equations, simulations) techniques. In most cases, the model-based techniques are used for algorithm training and measurement-based techniques for the deployment of the final detector/classifier.

Table 6 summarizes the advantages and drawbacks of each approach in the input data preparation step. The table also gives the references for where the techniques are applied. These references are classified according to the “type of disturbance” categories, i.e., multiple PQDs, sags, and notches. To give an idea of the importance and usage of the techniques, the percentage of the total number of references per category is also presented (see the total number of references per category in Table 4).

5.1.1 Field measurements

The acquisition of real signals under real operating conditions is the main goal of this approach. The same specialized equipment used in laboratory measurements is also required in field measurements, but the signals are directly provided by the power grid. The availability of events is therefore limited because of their unpredictability. Disturbances having noise, simultaneity between deviations and variations, and the evolution and propagation of PQDs affected by power demand variations characterize field measurements. These realistic factors require the development of robust algorithms for PQD detection and classification. Usually, measurements aim at catching the highest amount of information but sometimes they are focused on specific PQDs such as voltage sags and notches.

5.1.2 Equations

Instead of specialized equipment for the generation and measurement of PQDs, some can be abstracted into mathematical models that emulate real waveform distortions. In this waveform-level approach, the signals are initially built as ideal sinusoidal stored in computational

structures (i.e., arrays and matrices) and then the different kinds of variations and events are superimposed. Although it is highly flexible and easy to change any parameters of these equations, it is rather challenging to exactly replicate a real disturbance and its evolution in time without knowledge of the system features. These equations can be extended to represent different kinds of disturbances, or specific disturbances such as voltage sags and notches.

5.1.3 Simulations

Simulations in specialized software also rely on mathematical models. However, this system-level approach allows not only the reproduction of different PQDs from different operating conditions of power systems, but also the assessment of the interactions and evolution of disturbances in both time and space. Therefore, simulations can be used to replicate operational conditions that produce a certain type of disturbance, modify those conditions to stress the system response, combine different disturbances in a controlled environment, including power system variations to forecast the future behavior of PQDs, etc. The simulation results are strongly dependent on how closely the models represent the real system, so a validation stage between real measurements and simulated results is necessary before forecasting results for other operating scenarios.

5.2 Data preprocessing

After the acquisition of PQDs, some techniques are used to handle the data before further processing. These techniques are optional and should not modify the useful information contained in the acquired signals. The additional hardware and/or software required at this stage may improve the performance of detection and classification algorithms in subsequent stages. Table 7 summarizes the advantages and drawbacks of these steps, indicates references where applications are observed, and reports usage percentages according to the “type of disturbance” categories.

5.2.1 Segmentation

This technique is used for the isolation in the time domain of the most relevant sections of the acquired signals, especially when events (see Table 1) are to be analyzed. The computational burden of subsequent processing stages may be reduced when segmentation is correctly applied since non-relevant data is discarded. However, the decision on relevance can vary when different disturbances coexist at the same measured signals. Therefore, a set of thresholds are generally used when dealing with the detection and classification of multiple PQDs. Moreover, segmentation is less

Table 6 Summary of input data sources

Data source	Advantages	Disadvantages	Application			
			Multiple PQDs		Sags	
			Refs	%	Refs	%
Laboratory measurements	Emulation of realistic conditions Controlled conditions Adaptable	Specialized supply and measurement setups required (high cost) Measurement uncertainty to be considered	[66, 73, 75, 77, 82, 84, 85, 91–93, 96–101, 106, 107, 111, 113–115, 120, 121, 124, 125, 127, 128, 130, 131]	30	[136, 139, 142, 143, 145, 153, 159, 160, 164, 170, 182]	20
Field measurements	Real operating conditions Real behavior of equipment Real signals: distortion, noise	Specialized measurement setup required (high cost) Measurement uncertainty to be considered Uncontrolled conditions	[33, 35, 39, 48, 67, 69, 72, 76, 77, 99, 103–105, 118, 120, 126]	16	[132, 135, 139, 141, 144, 148, 151, 152, 157, 158, 160, 162, 163, 165, 166, 172, 174, 175, 179–181]	38
Equations	Low-computation burden (low cost) Easily modifiable	Idealized assumptions Validation of equations against real phenomenon Only for waveform-oriented assessments (regardless of root causes)	[34, 42, 46, 47, 49, 50, 53, 55–58, 60–65, 71–73, 75, 80, 84, 85, 88, 89, 91–100, 103–106, 108–119, 122–126, 128, 129, 131]	61	[139, 140, 143, 144, 150, 151, 157, 161, 162, 177]	18
Simulations	Emulation of realistic conditions Controlled conditions System-oriented assessments (considering disturbances' root causes)	Specialized software required (medium cost) Validation of interactions between grid and equipment models	[36, 40–45, 47, 48, 51, 52, 54, 59, 64–68, 70, 72, 74, 77, 78, 80–83, 87, 90, 95, 102, 103, 108, 110, 124]	35	[132, 133, 137, 138, 145, 147–150, 153–156, 160, 167–169, 171–174, 176–186]	58
					[189, 191, 193, 200, 201, 205, 207, 211]	32

Table 7 Summary of data preprocessing techniques

Set of techniques	Advantages	Disadvantages	Application					
			Multiple PQDs		Sags		Notches	
			Refs	%	Refs	%	Refs	%
Segmentation	Isolation of relevant data The reduced computational burden for subsequent stages	Thresholds required Subjective isolation for simultaneous PQDs Additional equipment/processing required	[38, 42, 44, 47, 57, 59, 61, 63, 69, 72, 75, 76, 81, 89, 97, 99, 101, 108, 124, 128, 130]	21	[159, 162, 168, 174, 175, 184]	11	[191, 194, 195, 202, 205, 207, 211]	28
Denoising/filtering			[62, 91, 102, 126, 128, 130]	6	[141, 152, 175]	5	[188, 199, 204, 208]	16
Normalization	Amplitude-independent techniques Thresholds in the percentage of the original signal	Interpretation of results is relative Loss of real-signal amplitude features	[42, 48, 50, 54, 55, 58, 59, 64, 68–72, 74, 76, 80–82, 88, 91–93, 97, 98, 101, 103, 108–111, 115, 119, 120, 122]	34	[140, 141, 143, 144, 146–148, 150, 152–159, 161, 162, 164–166, 168, 181, 182]	44	[209, 211]	8

frequently used for specific disturbances, as can be seen in the literature for voltage sags and notches in Table 7.

5.2.2 Denoising/filtering

Similar to the segmentation technique, the filtering or denoising technique aims at discarding the non-relevant information in the acquired signals. In this case, the isolation is performed over the spectral components of the signals. For this reason, additional hardware (analog filters) and/or processing steps (digital filters) may be required. This technique is most useful when the bandwidth of the PQD of interest is previously known or when only emissions at certain frequencies are of interest (e.g., voltage and currents at the fundamental frequency only). Compared to segmentation and normalization techniques, denoising has been less used in the context of PQD detection and classification.

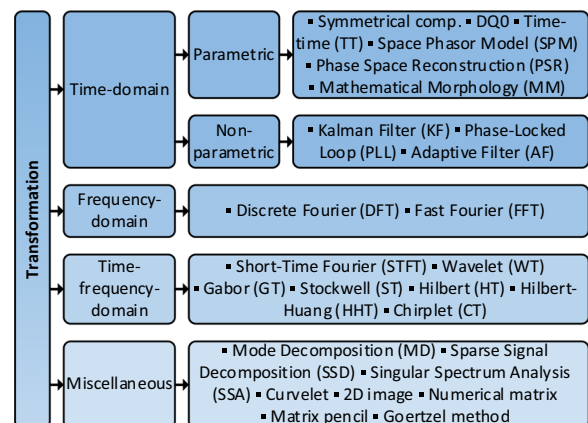
5.2.3 Normalization

Normalization is the most frequently used data preprocessing technique for multiple PQDs and voltage sags but is used less frequently in specific detection and classification of notches (see Table 7). Similar to segmentation and denoising, the normalization step reduces computational burden by avoiding the calculation of large numbers. This is achieved by the division of all quantities against a numerical base, i.e., the rated or peak value of signals. However, the scale is lost and the distinction between relevant and non-relevant amplitudes will depend on the relative choice of the numerical base.

5.3 Transformation

Following the acquisition of data and the eventual application of some preprocessing techniques, transformation is traditionally the most used stage for signal processing purposes. This stage aims at using a different representation of original data, usually by changing the analysis domain, to unveil hidden features and patterns.

As shown in Fig. 8, most of the transformations used in PQD detection and classification can be grouped into time, frequency, and time–frequency domains. Other miscellaneous transformation techniques are widely used in other domains (speech recognition, arrhythmia classification, denoising, image compression, etc.) and have been adopted by researchers to decompose and/or cluster voltage and current data from power systems for detection and classification of PQDs. The groups of transformation techniques applied to the detection and classification of PQDs

**Fig. 8** Taxonomy of transformation techniques

are depicted in Fig. 8, where time-domain transformations are further divided into parametric and non-parametric techniques. Further descriptions of each of the time-, frequency-, and time–frequency-domain techniques are provided in the following sections.

The most relevant advantages and disadvantages of the transformation techniques are presented in Table 8. The table also reports references of applications and the respective percentages per category according to the type of PQD.

5.3.1 Time domain

Time-domain transformations are widely used to track the evolution of monitored signal features through time. They are commonly used to analyze PQ events like interruptions, transients, rapid voltage changes, swells, and sags because they are in principle unpredictable and non-periodic phenomena (see Table 1). Nevertheless, PQ deviations like notches are also processed in the time domain since their spectral signature is spread over a wide frequency range. Time-domain transformations shown in Fig. 8 can be subdivided into non-parametric, e.g., tracking of symmetrical components, DQ0, Time-time Transform (TT), Clarke transform in complex domain Space Phasor Model (SPM), Phase Space Reconstruction (PSR), MM, etc., and parametric, e.g., KF, Phase-Locked Loop (PLL), Adaptive Filters (AF), etc. Non-parametric transformations decompose the original measured signals into components that clearly show how the PQDs are progressing in time, whilst parametric transformations use assumptions about the statistical distribution of the population from which the samples (signals) were taken. Parametric transformations are mostly used to track and statistically estimate magnitudes, phase angles, frequency, etc. Although transformations relying on phasor theory, like symmetrical components, belong in principle to frequency domain abstraction, the evolutions of the computed amplitudes and phase angles are usually analyzed in the time domain.

5.3.2 Frequency domain

The frequency-domain transformations are essentially applied to steady-state signals using FT. This transformation decomposes distorted signals into a summation of pure sinusoids having different frequencies. The Fast FT (FFT) is a widely known technique for Discrete FT (DFT) computation. The computation of DFT is carried out as [66]:

$$V^n[k] = \sum_{i=0}^{N-1} v[i + (n-1) \cdot N] \cdot \exp[-j(2\pi ki)/N] \quad (1)$$

where N is the number of samples in one cycle, n is the order number of the signal cycles, and $n=1, 2, \dots, 10$,

which gives the absolute value of the argument. $V^n[k]$ is the DFT for the samples contained in the n th cycle, $v[i]$ represents the sampled input signal, $i=0, 1, 2, \dots, L-1$, with L being the length of the signal.

For the detection of multiple PQDs, FFT has been used mostly for the computation of fundamental or specific harmonic amplitude, phase angle shift between harmonic components, RMS values, and total harmonic distortion [66, 89, 90, 97, 102]. Although FFT and DFT may yield inaccurate results for non-stationary signals, some studies have adopted this transformation for voltage sag detection, while some have used FFT for notch analysis [188] because the spectral components of this disturbance are rather spread over the whole frequency range. However, FFT has also been used in combination with WT [61, 80, 189], 192, 196] and Mode Decomposition (MD) [99] techniques to cope with non-stationary PQDs.

5.3.3 Time–frequency domain

Transformations classified in the time–frequency domain keep most of the advantages of time and frequency techniques. These techniques can provide time and frequency information at the same time, which helps improve the accuracy of PQD detection and classification algorithms.

STFT is the computation of FT over a section of the signal assumed to be stationary. STFT has been used for detection of multiple PQDs [35, 102] and voltage sags [137, 160].

WT, also known as a multiresolution analysis, decomposes the original signal into various scales of a short-term waveform called the “mother wavelet”. Discrete WT (DWT) is the discrete realization of WT and has been widely used for detection and classification of multiple PQDs as the main transformation technique [33, 34, 37–39, 42, 44–46, 48, 51, 53, 55, 59, 60, 62–64, 69, 76, 83, 95, 101, 110, 123] or combined with other techniques [35, 36, 41, 61, 73, 80, 102, 117, 126]. It has also been used as the main transformation technique for the detection and classification of voltage sags [133, 142, 146, 154, 168, 169] and notches [187, 190, 191, 193, 194, 197–199, 204, 208], or combined with other techniques for voltage sags [160, 181, 186] and notches [189, 192, 196, 202]. DWT is given by [33]:

$$DWT_{\psi} x(m, n) = \int_{-\infty}^{\infty} x(t) \psi_{m,n}^*(t) dt \quad (2)$$

where $\psi_{m,n}(t)$ is the mother wavelet.

ST is a hybrid technique that includes a phase correction to the WT and a variable Gaussian window to STFT to get a combination of these two techniques. ST can be calculated by multiplying the continuous WT with a phase factor as [40]:

Table 8 Summary of transformation techniques

Technique	Advantages	Disadvantages	Application		Sags		Notches	
			Multiple PQDs					
			Refs	%	Refs	%	Refs	%
Time domain								
Parametric								
120	Decomposition of unbalanced systems	One cycle or more should be analyzed	[36, 47, 50, 65, 67, 74, 82, 91, 102, 106, 108, 109, 111, 114, 115]	15	[136, 138–140, 151–153, 155–158, 161, 163, 167, 172, 174, 186]	31	[195, 200, 201, 206, 209, 211]	24
DQ0	Decomposition of time-varying signals	The main frequency varies over time						
TT	Provide transient characteristic of signal	High computational burden						
SPM	Instantaneous data from 3ph systems	Single-phase information not available						
PSR	Instantaneous data from 3ph systems	Repeated information after transformation						
MM	Low computational burden	Not suitable for high-frequency distortions						
Non-parametric								
KF	Accurate for parameter estimation	Used for single PQD detection until now (very specific applications)						
AF	Suitable for parameters self-tuning							
PLL	Accurate for parameter tracking							
Frequency domain								
FFT	A computationally efficient technique for steady-state signals	Inaccurate for non-periodic signals	[61, 66, 80, 89, 90, 97, 99, 102, 128, 130]	10	[160, 184]	4	[188, 189, 192, 196, 202, 211]	24
DFT	Realization of FT in discrete time	Inaccurate for non-periodic signals High computational burden						

Table 8 (continued)

Technique	Advantages	Disadvantages	Application							
			Multiple PQDs		Sags		Notches			
			Refs	%	Refs	%	Refs	%		
Time–frequency domain										
STFT	Successful time–frequency decomposition of stationary signals	Trade-off between time and frequency resolution	[33–46, 48, 49, 51–65, 68–73, 75–81, 83–85, 88, 92, 94–96, 98, 101, 102, 104, 105, 107, 109, 110, 113, 114, 117, 118, 120, 122, 123, 125, 126, 129, 130]	72	[133, 137, 142, 143, 146, 147, 149, 150, 154, 160, 168, 169, 173, 176, 178, 181, 184, 186]	33	[187, 189–194, 196–199, 202, 204, 208]	56		
WT	Information on non-stationary signals	Highly affected by noise								
GT	Good time–frequency resolution	High computational burden Spectral cross-interference can appear								
ST	Combine advantages of WT and STFT	High computational burden Performance depends on the windowing function								
HT	Information on non-stationary signals	Focused only on narrowband conditions								
CT	Combine advantages of WT and STFT Detection of linear and non-linear frequency variations over time	High computational burden								
Miscellaneous										
MD	Low computational burden (iterative)	Exhaustive search technique	[86–88, 93, 99, 100, 103, 112, 124, 129, 130]	12	[134, 153, 166, 176, 179, 181]	11	-	0		
SSD	Several PQDs processed simultaneously	Computation time (optimization problem)								
SSA	Noise suppression and forecasting	High computational burden								

$$S(\tau, f) = \exp(i2\pi f\tau) \cdot W(\tau a) \quad (3)$$

ST has been used for the detection and classification of multiple PQDs as the main transformation technique [40, 43, 49, 52, 54, 57, 68, 70–72, 75, 77, 78, 81, 85, 92, 94, 96, 104, 105, 118, 120, 125] or in combination with a spline wavelet [41], TT [65, 109, 114], VMD [88], WT [117], and others [102]. Similarly, it has been used for the assessment of sags as the main technique [143, 149, 178] or combined with VMD [176] and FT [184].

Finally, GT, Hilbert Transform (HT), and Chirplet Transform (CT) are other time–frequency techniques used for the detection and classification of PQDs. GT is mostly known as an accurate tool for phasor estimation and has been used as a measurement tool for the analysis of events (short-term deviations). It has been mainly applied in combination with the Wigner distribution function, resulting in the so-called Gabor-Wigner Transform (GWT) [58], and with time–frequency representation. HT of a real-valued time-domain signal produces an orthogonal, real-valued time-domain signal that is 90° ($\pi/2$ radians) shifted from the original. This technique has been widely applied alongside Empirical Mode Decomposition (EMD), resulting in the so-called HHT. Researchers have applied HHT for the detection and classification of multiple PQDs [56, 73, 79, 84, 102, 107, 113], and specifically for voltage sags [147, 150, 173], but not yet to the specific assessments of voltage notches. CT can be seen as the generalization of FT, STFT, and WT. Analogous to the mother wavelets in WT, chirplets are usually generated from a single mother chirplet which is a windowing function. It has been applied for the assessment of multiple PQDs [98] and voltage notches [202]. Voltage sags have not yet been specifically addressed nor characterized with CT.

5.3.4 Miscellaneous

Transformation techniques widely used in other domains have been adopted by researchers for the detection and classification of PQDs. The MD technique is mainly composed of EMD [86, 93, 129] and VMD [88, 99, 100, 124, 130] for the assessment of multiple PQDs, and also for the specific assessment of voltage sags [176]. EMD takes the linear or non-linear input signal and iteratively decomposes it into a series of smaller components known as Intrinsic Mode Functions (IMF). VMD is based on a constrained variational optimization problem and is a non-recursively adaptive technique that decomposes a linear or non-linear input signal into a finite number of sub-signals or modes having specific sparsity properties (compactly band-limited IMF). There are also other techniques adapted for PQD detection and classification such as Sparse Signal

Decomposition (SSD) on an overcomplete hybrid dictionary matrix [87], Singular Spectrum Analysis (SSA) along with Curvelet technique [103], 2D image techniques (gray-scale images) [112, 179, 181], numerical pencil [134], matrix pencil [153], and Goertzel method [166].

5.3.5 Quantitative analysis of transformation techniques

Figure 9 depicts how transformation techniques are distributed according to the analysis domain, how they are used (main technique or in combination with other techniques), type of disturbance, and real-time applications. Specifically, Figs. 9a and b rank respectively the most popular transformation techniques for detection and classification of PQDs, according to the absolute and relative number of articles found in this literature review. Techniques classified into the time–frequency (52%) and time domains (19.7%) represent 71.7% of the total number of transformation techniques, whereas WT and ST, which are time–frequency domain techniques, represent 44.1% of the total transformation techniques. WT is used either as the main technique or in combination with other techniques, such as FFT, ST, and other time-domain techniques.

The evolution of transformation techniques is described in Fig. 9c. WT has been intensively used as the main time–frequency technique for the detection and classification of PQDs since 1996 [33], and from 2011 onwards it has been used in combination with other techniques. ST and other time–frequency techniques have increased their participation in PQDs detection and classification, as well as time-domain techniques such as SPM, PSR, and (extended) KF/AF. Nevertheless, the combination of several techniques has seen increased interest from researchers, especially in the last six years.

In terms of the PQDs to be detected and classified, Fig. 9d shows that WT, time-domain techniques, and the combination of several techniques have been used for the assessment of multiple PQDs as well as voltage sags and notches. Although WT has been the preferred technique for the assessment of voltage notches, time-domain techniques are becoming relevant for assessing this type of disturbance. The trend regarding transformation techniques is to develop one technique, or a combination of several, that can be used for the accurate detection and classification of the highest number of PQDs (variations and events, see Table 1).

Figure 9e shows that WT, ST, and time-domain techniques (SPM, PSR, and KF/AF) are used for real-time detection. Real-time detection and classification are also performed by the combination of transformation techniques such as WT and FFT, WT and ST, etc.

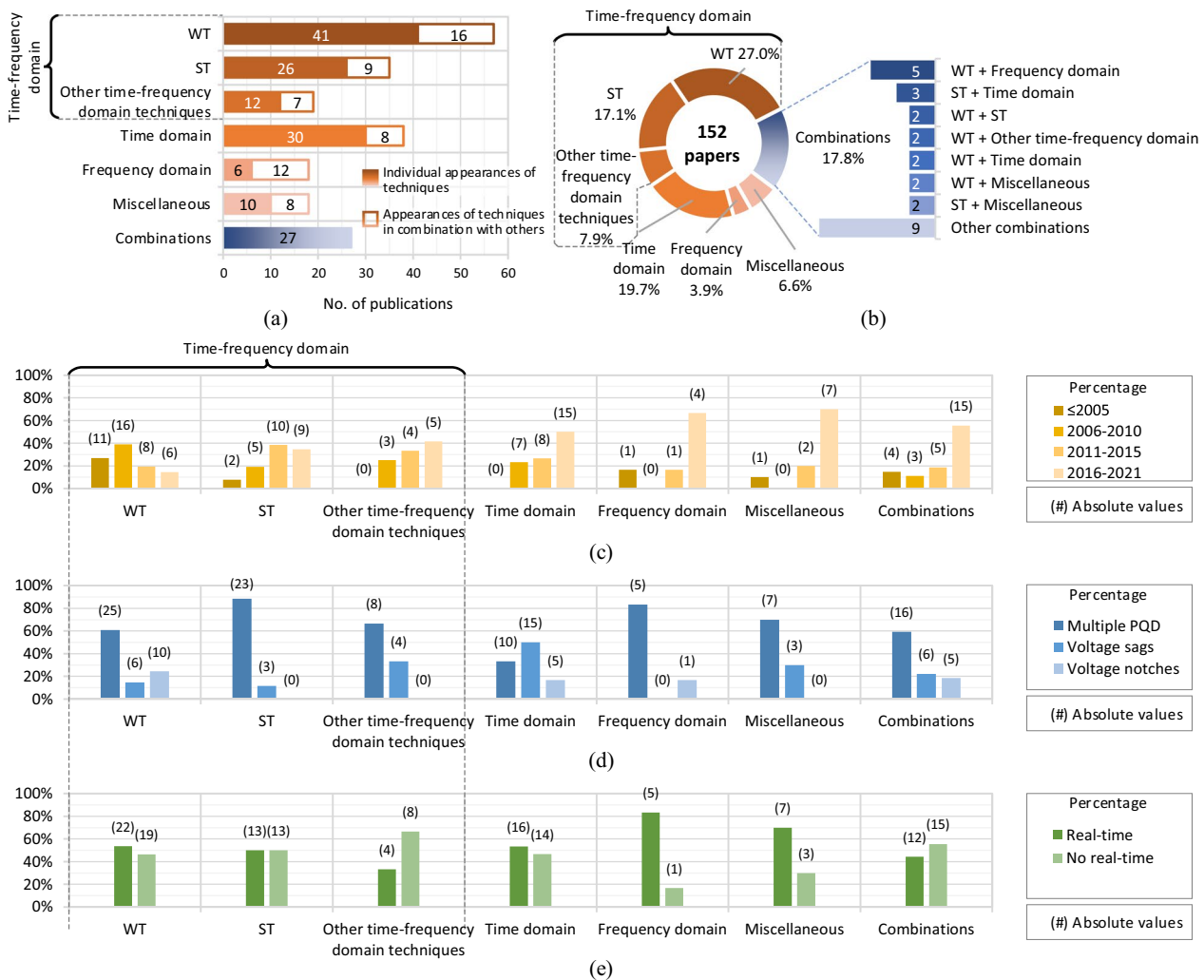


Fig. 9 Distribution of papers in the identified transformation techniques. **a** The number of papers, **b** percentages and number of papers that use a combination of techniques, **c** percentages and number of papers from the time perspective, **d** the type of disturbance perspective, and **e** the real-time perspective

5.4 Feature extraction

Feature extraction aims to reduce the amount of data from the transformation stage that will be processed for the detection and classification of PQDs. A set of statistical, time series, spectral and image features can be used for this purpose. A set of features describing one PQD, e.g., variations, may not be suitable for describing another type of PQD, e.g., events. Therefore, establishing a set of comprehensive, robust, and accurate features that allow the detection of different PQDs is one of the most challenging tasks in the process.

Figure 10 depicts the categories into which the feature extraction stage can be further divided. The main advantages and disadvantages are listed in Table 9, which also includes references with applications and the respective percentage of usage according to the type of disturbance.

5.4.1 Statistical features

Mean and median are the set of most used statistical features for central tendency. Arithmetic mean (also known as arithmetic average) is a central tendency measure for a finite number of values from an observation process (sampling). This is calculated as the sum of all values divided into the amount of data and is a relatively simple way of computation (low computational cost) but sensitive to outliers (data with atypical values). This metric is widely used for feature extraction in multiple PQDs, and specifically for voltage sags and notches (see Table 9). Less sensitive to outliers but with a compulsory “ordering” process, the median is the other central tendency widely used for feature extraction. When dealing with a large enough dataset, the underlying population distribution may be assumed as normal (Gaussian). Therefore,

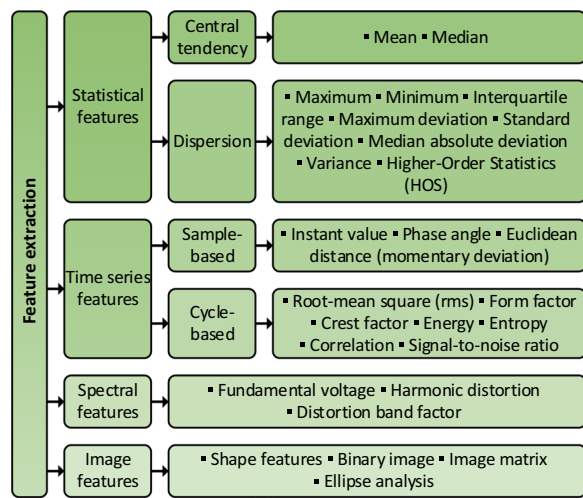


Fig. 10 Taxonomy of features for detection and classification

similar values for both mean and median are obtained. However, many processes do not follow a normal distribution, and thus median may be a more accurate measure of central tendency than the mean.

In contrast to central tendency, a measure of dispersion is achieved by many different indices. Maximum and minimum values are easy-to-compute metrics that give general information about the analyzed dataset. The interquartile range is a descriptive metric defined as the difference between the 75th and 25th percentiles and needs an ordering process that may be challenging for online applications in embedded hardware. Deviation metrics in the form of maximum deviation, standard deviation, mean absolute deviation and median absolute deviation compute the distance between the observed value of a variable and a central tendency metric. Variance and Higher-Order Statistics (HOS) such as skewness and kurtosis aim at describing the shape of the underlying probability distribution function. These metrics are widely used for detection and classification of multiple PQDs, voltage sags and voltage notches (see Table 9).

5.4.2 Time series features

Time series are sequences of data points ordered in the time domain. The sampling process is usually performed at a fixed frequency, and therefore the time between successive samples is theoretically the same. In the context of electrical engineering, time-varying electrical variables such as voltages and currents are sampled through analog–digital converters, which convert the real-life analog signals into discrete signals. The cyclic nature of AC systems is defined by a cycle, which is the time that a signal repeats its values in the time domain. Therefore, it

is possible to characterize a signal by extracting sample-based or cycle-based features.

Sample-based features take advantage of the evolution of discretized signals in the time domain. The features are the extracted samples and therefore very detailed information on signal evolution can be retrieved. However, a large amount of data can result from this stage if a high sampling frequency is used. Sample-based feature extraction techniques such as instantaneous values, phase angle and momentary deviation (Euclidean distance) are used for multiple PQDs, voltage sags and notches according to the references indicated in Table 9.

Cycle-based features are focused on the evolution of periodic signals over time. In this sense, the features are extracted in multiples of one cycle of the main signal and therefore the amount of processed data is much less than that using sampled-based features. However, information on sub-cycle PQDs like notches, spikes and transients is no longer available in this approach. RMS value, crest and form factors, energy, entropy, correlation, and signal-to-noise ratio are usual metrics computed from a cycle-based approach. These metrics are used for feature extraction of multiple PQDs, voltage sags and notches (see Table 9).

5.4.3 Spectral features

Spectral features are a set of indices that naturally result after the use of frequency or time–frequency domain transformation techniques. In the context of power systems, the fundamental frequency is the nominal frequency at which most of the electric power is generated and transmitted (theoretically 50 Hz or 60 Hz). In contrast, spectral distortion is the result of the nonlinear, nonconstant behavior of electrical equipment that indicates a deviation from the ideal (reference) pure sinusoidal signals. In the context of power systems, spectral distortion can be generally classified into the harmonic range (below 2 kHz), the so-called supraharmmonic range (between 2 and 150 kHz) or the high-frequency range (above 150 kHz). Fundamental frequency and harmonic distortion are used for the detection and classification of multiple PQDs, voltage sags and notches (see Table 9). There is a special case of harmonic distortion, called Distortion Bands, where the distortion is computed in other ranges different from harmonic, supraharmmonic or high-frequency ranges [197, 202].

5.4.4 Image features

Image features are mostly related to 2D functions and/or representations of PQDs. Taking advantage of the steady-state cyclic variation of voltage and current in power systems, some transformation techniques (SPM, PSR, instantaneous symmetrical components, etc.) describe

Table 9 Summary of feature extraction techniques

Set of features	Advantages	Disadvantages	Application		Sags		Notches		%
			Multiple PQDs						
			Refs	%	Refs	%	Refs	%	
<i>Statistical features</i>									
Central tendency									
Summary of typical values in a specific time frame		Based only on quantitative, historical data	[60, 62, 63, 65, 67, 68, 71–77, 80, 81, 83, 85, 87, 89, 91, 95–97, 100, 102, 105, 108–110, 112, 114, 117, 120, 123, 126, 128]	36	[132, 143, 149, 168]	7	[187, 196, 201]		12
Low computational burden		Not able to generalize the findings to a broader population of PQDs							
Dispersion									
Summary of variability in a specific time frame		Based only on quantitative, historical data High values might not be useful to describe the PQDs behavior	[34, 40, 41, 43, 48, 49, 52, 54–57, 60–63, 65, 67, 68, 70–72, 74–76, 79, 81, 83–85, 87–89, 91–98, 100, 102, 105–110, 112, 114, 115, 117, 118, 120, 123, 126, 128, 130]	58	[132, 143, 144, 148–150, 168, 169, 177, 178, 183, 184]	22	[187, 189, 191, 193, 196, 211]		24
<i>Time series features</i>									
Sample-based									
Independent of sampling frequency Detailed information		Further processing might be required	[61, 64, 65, 67, 75, 77, 87, 91, 92, 96, 99, 115, 128]	13	[133, 145, 186]	5	[205–209, 211]		24
Cycle-based									
Suitable for steady-state disturbances		Limitations with sub-cycle disturbances (e.g., notches)	[33, 36–39, 42, 44–46, 52, 53, 56, 57, 59, 64–66, 69, 72, 79, 82–84, 88, 93, 96, 99, 101, 102, 106, 109, 110, 114, 118, 122, 123, 126, 128, 130]	61	[132, 135, 141, 146–148, 150, 155, 159, 167, 174, 175]	27	[189, 192, 194, 196, 199, 204, 208, 211]		32
<i>Spectral features</i>									
Non-evident periodicities might be revealed		Frequency-domain transformation is compulsory Inaccurate for PQ events	[61, 62, 66, 70–72, 77, 82, 89, 97, 102, 110, 115, 118, 128]	15	[132, 149, 156, 161]	7	[188, 189, 197, 202, 211]		20
<i>Image features</i>									
Processing of 2-D signals Signal processing methods can be applied		Problems from "non-perfect" modulation (swing of main frequency)	[50, 111, 112, 130]	4	[140, 151–153, 157, 158, 163, 167, 169, 172, 181, 186]	22	-		0

these signals as phase vectors in the complex plane (phasors). From this, the most popular 2D feature extraction technique relies on Ellipse features since the resultant circumferences can give useful information about the features of the variations and events listed in Table 1. Ellipse features have been used for specific detection and classification of voltage sags [151–153, 157, 158, 163, 172]. Other features taken from the 2D representation of PQDs are shape features and factors (center of mass, eccentricity, convexity, centroid distance, chord length, etc.) [140, 169], binary image [50, 112], and image matrix [111], among others.

5.5 Feature selection

The step of feature selection involves identifying as few characteristics as possible to obtain enough information that can yield suitable results in the decision space stages (detection, classification, and/or characterization of PQDs). Hence, the ways of selection and the selected features depend on what suitable results mean in the context of each study. The feature selection also reduces the computational burden in the decision space and usually leads to more accurate results.

Different categories have been proposed for feature selection techniques in the literature. For instance, reference [76] proposes three main categories, namely, filtering, wrapper, and embedded, which are related to the level of dependence of the feature selection on the decision (learning) algorithms. Thus, filtering approaches are very independent, and embedded approaches intertwine the selection and decision algorithms. In this literature

review, three groups of methods are identified, including handcrafted, optimization methods, and dimensionality reduction algorithms.

Table 10 summarizes the advantages and drawbacks of the most relevant feature selection techniques identified in the literature review according to the proposed classification. The table also reports the references where the methods are applied according to the type of disturbance categories and presents the usage percentages.

It is worth mentioning that deep learning techniques perform an automatic process of feature selection. It is conducted in the first layers of the classifiers where the best features for the decision space are automatically selected. Therefore, no category is included in this section for this type of tool because no external intervention is required.

5.5.1 Handcrafted/empirical

In handcrafted feature selection, a detailed manual analysis is performed on the extracted features to observe the differences according to the type of PQD. For instance, this approach is observed in [33] where coefficients resulting from the WT and multiresolution analysis are analyzed for sample signals with different PQDs. The study in [33] and many others, e.g., [34–41], show how the analysis can be mostly supported by visual inspection of signals (time-domain waveforms) and extracted features (time-domain indices). Most of the handcrafted feature selections observed in the literature apply a contextual approach where a physical meaning is given to the extracted features. Some examples of contextual

Table 10 Summary of feature selection techniques

Set of techniques	Advantages	Disadvantages	Application					
			Multiple PQDs		Sags		Notches	
			Refs	%	Refs	%	Refs	%
Handcrafted	Physical interpretation of contextual features Ease understanding	Time-consuming Detailed knowledge of features and phenomena is usually required	[33–41, 43–54, 56–62, 64, 66–74, 77–94, 96–99, 101, 102, 104, 106–108, 110, 112, 113, 115, 117, 118, 120, 122, 126, 128]	77	[132–140, 142–157, 160, 161, 163, 166, 169, 172–174, 176, 178, 181, 182, 185, 186]	71	[187–210]	96
Optimization methods	Yield optimal sets of features Sound theoretical background Diverse methods	Not necessarily a physical interpretation of features High computational burden They rely on the proper a priori selection of relevant features	[55, 95, 102, 109, 114, 123, 127]	7	[168, 177, 185]	5		-0
Dimensionality reduction	Physical interpretation of indices Diverse methods and indices	May lead to some data loss Handcrafted rules are required in most cases	[42, 62, 75, 76, 98, 100, 102, 110, 112]	9	[141, 162, 170, 177]	7		-0

feature selection are [33–41]. The literature also shows that other empirical approaches such as sequential forward selection, sequential backward selection, and random mutation have been mainly used for the detection and classification of multiple PQDs. In these empirical approaches, a set of features are obtained beforehand based on expert knowledge or using similar features to previous studies. Then, in forward selection methods, e.g., [60, 88, 89, 91, 96, 102, 112], features from the established set are included one at a time, and the performance of the classifier is verified. The process is repeated until the performance has no apparent improvement. Conversely, backward selection, e.g., [88], uses the complete set of features and removes one at a time until the desired trade-off between accuracy and computational performance is reached. On the other hand, random mutation tests random subsets of features and selects the one with the best performance [57, 62, 64, 71, 117].

Those studies from the literature focused on voltage sags mainly use contextual feature selection. For instance, reference [132] extracts directly the features from the voltage waveforms according to standard definitions, i.e., initial phase angle shift, recovery period, voltage change. In other cases, coefficients from transforms, e.g., WT [133] and ST [143], are analyzed contextually according to their correspondence to the standard parameters that characterize voltage sags, i.e., magnitude, duration, etc. A combination of forward and backward selections with GA to select the best features to identify voltage sag source location (upstream or downstream) is proposed in [185].

In the studies focused on voltage notches, contextual feature selection is dominant for handcrafted methods [187–210]. In this case, selected contextual features are analyzed visually from the coefficients of WT.

5.5.2 Optimization methods

Optimization methods have also been used for feature selection purposes in PQD detection and classification. According to [212], a mathematical optimization method which consists of finding the best possible solution by changing variables that can be controlled, is often subject to constraints. Optimization methods can be classified as deterministic (exact) or stochastic (approximate), and stochastic methods can be further divided into heuristic and metaheuristic. Most methods used for PQD detection and classification are metaheuristic. Metaheuristic methods include exploratory search methods such as GA, and swarm optimization algorithms such as particle swarm optimization. For instance, GA is used in [55, 102, 109] to find the optimal set of features to classify multiple PQDs with high accuracy. This is based on the mechanics of selection and survival of the fittest, and consists of

three operations including reproduction, crossover, and mutation [55]. The studies in [55, 102, 109, 127] show that GA maintains or improves the classification accuracy while it decreases the computational burden (reducing the number of features). A variation of GA is presented in [109], where a fast and elitist nondominated sorting is used to generate Pareto-optimal solutions. This modification offers better speed, solution spread, and convergence. The artificial bee colony optimization algorithm is used to improve the performance of a multiple PQDs classifier in [95], while the artificial bee colony is used in [123] in combination with particle swarm optimization to improve the accuracy of a PNN in PQD classification. The artificial bee colony algorithm is a swarm intelligence optimization technique where different types of bees apply strategies for finding the best sources of food (solutions). This algorithm uses three groups of bees, namely, employed bees, onlooker bees, and scout bees. Employed bees search for food in specific sources and share the information to onlooker bees, whereas scout bees search for new sources of food. This optimization method provides an optimal subset of features with fast convergence [95]. Other swarm colony algorithms have also been used for feature selection. For instance, ant colony optimization [114] is inspired by the foraging behavior of ants, and offers high accuracy in the classification of multiple PQDs and a faster solution to an optimal feature subset than other studies.

Feature selection based on optimization techniques is also used in studies focused on voltage sags. The study in [168] uses ant lion optimization to improve performance in classifying the underlying causes of voltage sags. This optimization technique mimics how ant lions hunt and consume their prey and provides advantages such as good population diversity, storing good solutions, exploitation, exploration, and flexibility [168]. Similarly, a teaching–learning-based optimization technique is used in [177]. The teaching–learning-based method is a population algorithm that mimics the influence of a teacher on the output of learners in a class. Results in [177] demonstrate good accuracy in classifying causes of voltage sags in noisy signals with a reduced subset of optimal features. GA is used in [185] for voltage sag source location classification.

5.5.3 Dimensionality reduction

In machine learning, dimensionality reduction refers to decreasing the number of input variables in a model, i.e., selecting a subset of the original variables (features) and converting the data to a lower-dimensional space [213]. Dimensionality reduction may be useful for efficiently storing and processing data. Extensive techniques are used in machine learning for dimensionality reduction.

Principal Component Analysis (PCA) is one of the most used techniques. PCA is a multivariate technique that analyzes a data table with several variables and extracts the important information to represent the table as a reduced set of new orthogonal variables called principal components [214]. PCA is used for feature selection in multiple PQD classification in [62], while multiway PCA is used for the classification of voltage sags in [141]. Likewise, Independent Component Analysis (ICA) has been also used in applications of voltage sag classification [162, 177]. Other dimensionality reduction techniques observed in the literature for applications of multiple PQD classification include information gain measurement [42], Fischer linear discriminant analysis [100], k-means based apriori [76], Gini index-based threshold for selection of features [75, 98, 110, 112], and maximum relevance minimum redundancy [102]. In the case of voltage sag classification, k-means-singular value decomposition is also used [170], whereas decision trees and the Gini index are used in [211] for binary classification of voltage notches (non-notch/notch).

5.6 Detection

In this review, detection refers to the identification of states different from the ideal conditions of voltage and current waveforms (signals with no disturbance) through thresholds, triggers, and other techniques.

According to the above definition, detection may overlap in some cases with the step of classification because the classification process usually includes a category for ideal conditions of voltage and current waveforms. This situation mainly occurs in the classification of multiple PQDs, e.g., [118, 121]. In such cases, the detection should be considered as part of the classification algorithm and is mostly used in offline applications, where techniques are applied to stored signals. For real-time applications, given that no threshold or trigger occurs, the complete algorithms are constantly executed with a certain periodicity and the process requires high computational performance and storage capacity. There are exceptions in the classification of multiple PQDs, where detection is implemented as a previous and independent step of the process using techniques such as AF [36, 115], sine wave inference [64], and Euclidean square distance [106].

In the case of voltage sag classification, detection is mostly achieved as a previous independent step using techniques such as empirical/handcrafted definition of thresholds [132, 133, 137, 139–141, 148, 151–153, 163, 172], statistical-based sequential method [135], and ICA [177].

Detection has been described as an important step in combination with classification in the decision space. However, detection as the only aim of the decision space

is also of interest for some applications such as the operation of DVR [134, 184], voltage sag compensators [136, 145], and protection systems [146]. In these applications, the process includes some characterization of single PQDs. For voltage sags, techniques used for detection include adaptive notch filter [136], KF [138, 156, 161], integrator model [155], harmonic footprint [160], ICA [162], and Goertzel algorithm [166]. When the interest of the study is the characterization of voltage notches, detection has been conducted using the Teager energy operator and threshold algorithm [201], and the Euclidean norm [206, 209].

5.7 Classification

The main purpose of classification is to categorize the PQDs observed in voltage and/or current signals according to the types of deviations from the ideal waveforms. For instance, Table 1 presents some of the deviation types that identify the categories of PQDs. Among the categories of PQDs there are voltage sags and swells, harmonic distortion, transients and voltage notches, flicker, imbalance, etc. Classification is mostly conducted according to those categories. However, some approaches also categorize the phenomena according to the root causes of the PQD. For instance, reference [81] presents a categorization according to different causes, namely, fault, self-extinguishing fault, line energizing, non-fault interruption, and transformer energizing. Similarly, voltage sags are usually classified according to the main underlying causes, i.e., faults, motor starting, and transformer energizing [175–180].

Several techniques for the classification of PQDs have been identified from the analysis of the selected literature. A taxonomy of these techniques is proposed in Fig. 11, which shows three major categories, namely, handcrafted, probabilistic, and AI-based methods. The latter category is divided into various subcategories as it is the most widely used in the literature for the detection and classification of PQDs. Especially, machine learning tools have been widely used for the task of classification. Machine learning is a field of AI that focuses on the development of algorithms to make computer systems able to learn from data. Machine learning can be further divided into supervised learning, i.e., when algorithms need labeled data for training, and unsupervised learning, when no labels are required but are automatically identified by algorithms. In this literature review, only supervised learning algorithms are analyzed because they are the most used ones.

Table 11 reports the advantages and drawbacks of classification techniques, references of applications and percentage of usage according to the type of disturbance.

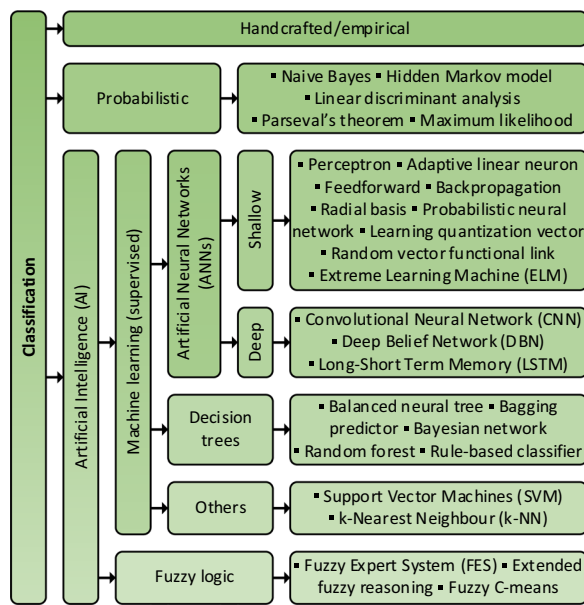


Fig. 11 Taxonomy of methods for classification of PQDs

5.7.1 Handcrafted/empirical

In handcrafted classification, thresholds to identify the categories of PQDs are defined by the observation of the extracted features during diverse experiments or by using expert knowledge on the physical interpretation of the phenomena. For instance, in [33] the coefficients of the WT are used to detect a variety of PQDs obtained from field measurements. Other approaches use handcrafted classification of voltage sags, e.g., reference [137] presents a method to classify sags according to their root causes including faults, motor starting, and transformer energizing based on thresholds for the STFT. Reference [148] classifies voltage sags according to the type of fault, based on defined ranges for indices calculated with symmetrical components. Other methods [153, 163, 172], analyze visually the ellipses generated through the SPM in the complex plane to define manually the ranges of the ellipse parameters for the types of sags.

5.7.2 Probabilistic

The probability of a signal containing a certain PQD is determined from the probability density functions of the extracted features associated with the disturbance. Examples of probabilistic methods include Parseval's theorem [34], the maximum likelihood [36], and the definition of ranges for statistical variables [40, 41]. Probabilistic methods are also used to identify the categories for classifying sags according to root causes and location (upstream or downstream). These methods include the singularity detection theory [133], the statistical-based

sequential method [135], and the energy-based method [139].

5.7.3 Shallow artificial neural networks (ANNs)

ANNs are computational models of reasoning inspired by the human brain [215], and comprise a set of processors (neurons) interconnected through weights passing signals from one neuron to another. An ANN can model complex nonlinear functions using extensive simple operations. Shallow ANNs are formed by an input layer, one or two hidden layers, and an output layer. ANNs are typically used in classification problems where each neuron of the output layer represents a category and is activated according to the respective inputs. In the problem of multiple PQDs and voltage sag classification, several types of shallow ANNs have been used taking advantage of their flexibility and adaptability to problems where labels are well identified, for instance, the learning vector quantization [37–39], probabilistic neural network [44, 52, 56, 65, 71, 84, 95, 102, 123, 147, 168], self-organizing learning array [46], radial basis function [47, 83], multilayer perceptron [48, 53, 57, 65, 89, 93, 102, 168, 177, 178], adaptive linear network [82], feedforward [82, 85, 102], backpropagation [90], random vector functional link [113], and modular ANN [143]. Most recently, a learning algorithm known as ELM has been gaining popularity because of its remarkable efficiency. ELM randomly chooses hidden nodes and analytically determines the output weights of a single-layer feedforward network [216]. Examples of ELM applications are [81, 94, 100, 107, 108, 169, 176, 178].

5.7.4 Deep artificial neural networks (ANNs)

Deep learning has emerged as a new machine learning paradigm where deep ANNs are composed of multiple processing layers to learn representations of data with multiple levels of abstraction [217]. This paradigm can dramatically improve the automatic classification abilities in diverse areas such as speech recognition, image processing, and detection and classification of PQDs. A remarkable improvement provided by deep learning, and the main motivation to apply these types of algorithms to PQD detection and classification, is the ability of models to automatically extract the best set of features from raw data to conduct classification. Convolutional Neural Networks (CNN) have been widely used in multiple PQDs and voltage sag classification [103, 111, 116, 119, 121, 125, 130, 131, 158, 164, 167, 171, 179, 183]. Other deep learning models have been particularly used in the classification of voltage sags according to the root causes, e.g., deep feedforward ANNs [124], Long Short-Term Memory (LSTM) [129, 159, 180], Deep Belief Networks (DBN)

Table 11 Summary of classification techniques

Type of technique	Advantages	Disadvantages	Application					
			Multiple PQDs		Sags		Notches	
			Refs	%	Refs	%	Refs	%
Handcrafted	Simple application Physical interpretation Fast operation	Inaccuracy Expert knowledge needed Time-consuming design	[33]	1	[137, 148, 153, 163, 172]	9	–	0
Probabilistic	Simple application Sound theoretical foundation Physical interpretation	Difficult modeling and implementation	[34, 36, 40, 41, 44, 70, 102]	7	[133, 135, 139, 141]	7	–	0
Shallow ANN	Flexibility Detailed knowledge of the phenomena is not required Able to solve nonlinear functions	Time-consuming training Extensive data for training Handcrafted feature extraction	[37–39, 44, 46–48, 52, 53, 56, 57, 62, 64, 65, 71, 79, 81–85, 89, 90, 93–95, 100, 102, 106–108, 113, 123]	33	[143, 147, 168, 169, 176–178]	13	–	0
Deep ANN	Flexibility Detailed knowledge of phenomena is not required Able to solve nonlinear functions Automatic feature extraction	The very high computational burden for training Extensive computations during operation hinder real-time applications	[103, 111, 116, 119, 121, 124, 125, 129–131]	10	[158, 159, 164, 165, 167, 171, 175, 179, 180, 182, 183]	20	–	0
Decision tree	Very simple application Efficiency for real-time application Robustness to outliers and noisy data	Complex decision trees can be difficult to understand Complexity increases exponentially with the size of the tree	[42, 49, 66, 70, 72, 75, 77, 79, 80, 85, 87, 89, 91, 96–99, 101, 102, 104–106, 109, 112, 114, 117, 122, 126, 128]	30	[140, 151, 152, 185]	7	[200]	4
SVM	Sound theoretical foundation Only a dozen examples for training are required	Computational inefficiency Low scalability	[47, 50, 51, 59, 60, 63, 69, 74, 76, 86, 88, 92, 98, 101, 102, 110, 114, 115, 127]	19	[149, 150, 168, 170, 177, 185]	11	[211]	4
k-NN	Very simple application Ease understanding	Reduced accuracy Sensitive to the choice of k	[55, 65, 102, 114]	4	[171, 185]	4	–	0
Fuzzy logic	Better representation of expert knowledge Physical interpretation of events Robustness to noisy data	Reduced accuracy No systematic It depends on human knowledge A lot of testing is necessary	[43, 45, 48, 54, 61, 64, 67, 68, 96, 120]	10	[132]	2	[189]	4

[165, 175], and independently recurrent neural networks [182].

5.7.5 Decision trees

Decision trees are knowledge-based systems obtained by inductive inference from examples [218]. Then, these systems are driven by the explicit representation of knowledge. Simple in application, decision trees allow for high efficiency which is essential for real-time applications. Moreover, these models provide good physical interpretation of the phenomena. These advantages have motivated widespread use of decision

trees, especially in the classification of multiple PQDs, though with lower usage in the classification of voltage sags and notches. Simple decision trees are used in the classification of multiple PQDs [42, 49, 66, 70, 75, 77, 85–87, 89, 96–99, 101, 102, 109, 114, 117, 118, 128]. Some variants such as random forest [102, 105, 112, 126] and bagging predictor [97, 98] and rule-based classifiers [72, 80, 91, 122], have been also used in the literature for multiple PQD classification. Rule-based classifiers are also used in [140, 151, 152] for voltage sag classification and in [200] for voltage notches, whereas decision trees are used in ensemble models as weak

classifiers in combination with other methods [185] for voltage sag classification.

5.7.6 Support vector machines (SVM)

SVM are robust supervised learning models applied in classification and regression problems. The basic idea behind SVM is to maximize the gap between different classes [219]. Based on this feature, SVM can be trained using a reduced number of examples. This makes this tool promising in cases where extensive training data is not available as in some applications of PQD detection and classification. SVM have been widely used for detection and classification of PQDs and voltage sags, while having lower usage for voltage notches (see Table 11). Variants of SVM such as multiclass SVM [51, 110, 115], least square SVM [76, 101, 170], rank SVM [86], and directed acyclic graph SVM [92, 127] have been also used.

SVM are used for notch identification in [211], where a classifier, i.e., SVM, is used to obtain a binary categorization of voltage signals, namely, non-notch or notch.

5.7.7 *k*-nearest neighbor (*k*-NN)

The *k*-Nearest Neighbor (*k*-NN) algorithm finds a group of *k* objects in the training set that are closest to the test object, and bases the assignment of a label on the predominance of a particular class in this neighborhood [220]. Given the ease of implementation of the model, it has been applied in the classification of multiple PQDs and voltage sags [55, 65, 102, 114, 171]. Simplicity of the method has been the main motivation for application of *k*-NN in PQD detection and classification. However, it has been mainly used in combination with other methods because of its reported drawbacks such as the reduced accuracy and high sensitivity to the constant *k*. For instance, in [185], *k*-NN is used as a weak classifier in an ensemble model in combination with decision trees.

5.7.8 Fuzzy logic

Fuzzy logic approaches are based on the sound theoretical foundations of fuzzy sets [221]. The basic idea of fuzzy logic lies in the definition of true fuzzy logic values of variables between 0 and 1. Fuzzy logic allows a closer representation of human reasoning where the information usually has a level of uncertainty. In some applications of PQD detection and classification, data for training have high uncertainties and thus, in those cases, fuzzy logic represents an alternative approach. For automatic classification of PQDs, fuzzy logic has been used from the beginning of research on the topic. Some examples include [43, 48, 61, 132]. FES have also been implemented [67, 189], as well as extended fuzzy reasoning [45], and fuzzy C-means [54, 68, 96, 120].

5.7.9 Quantitative analysis of classification techniques

Figure 12 presents a quantitative analysis of the selected literature according to the classification techniques described above. In Fig. 12a, the absolute number of individual appearances of each technique is shown, i.e., the number of papers where only one technique (or set of techniques in the same category) is used for classification. It also presents the number of appearances of the techniques in combination with others and the total number of papers where classification is achieved with a combination of techniques from different categories. As seen, shallow ANNs are the most popular models in the reviewed literature, followed by decision trees, SVM, deep ANN, fuzzy logic, probabilistic and handcrafted method, and *k*-NN. Also, a significant number of publications (19) implement combined methods for classification. The number of papers that do not perform classification is 46, and hence, a total of 133 papers are categorized in techniques for the classification of PQDs.

Figure 12b shows the corresponding percentages of classification techniques according to the individual appearances. The distribution of combined techniques is also reported in absolute values. In this case, shallow ANNs are the most popular in combinations. For instance, shallow ANNs are used in combination with decision trees, SVM, fuzzy logic, probabilistic methods, and *k*-NN. Moreover, some methods include the combination of three and four types of classification approaches.

A report on the distribution of classification techniques from different perspectives is included in Fig. 12. Figure 12c shows the distribution of techniques in time divided into the periods defined for the categorization of papers, i.e., ≤ 2005 , 2006–2010, 2011–2015, and 2016–2021. This indicates the absolute value of individual appearances of techniques in each period and the corresponding percentage. For instance, shallow ANNs present an increasing trend, where most of the papers were published in the period from 2016 to 2021. Similar behavior is also observed in decision trees with a more pronounced trend. An apparent observation is that all deep ANN approaches had been published in the last period, in agreement with the proliferation of deep learning. The combination of techniques exhibits a constantly increasing trend. Conversely, the use of probabilistic and fuzzy logic methods has been decreasing.

Figure 12d shows the distribution of papers according to the type of PQD, i.e., multiple PQDs, voltage sags, and voltage notches. Most papers performing classification are in the category of multiple PQDs as expected because the types of PQDs are distinguished in these cases. Classification is also used for voltage sags as they can be categorized according to the root causes. No categories are

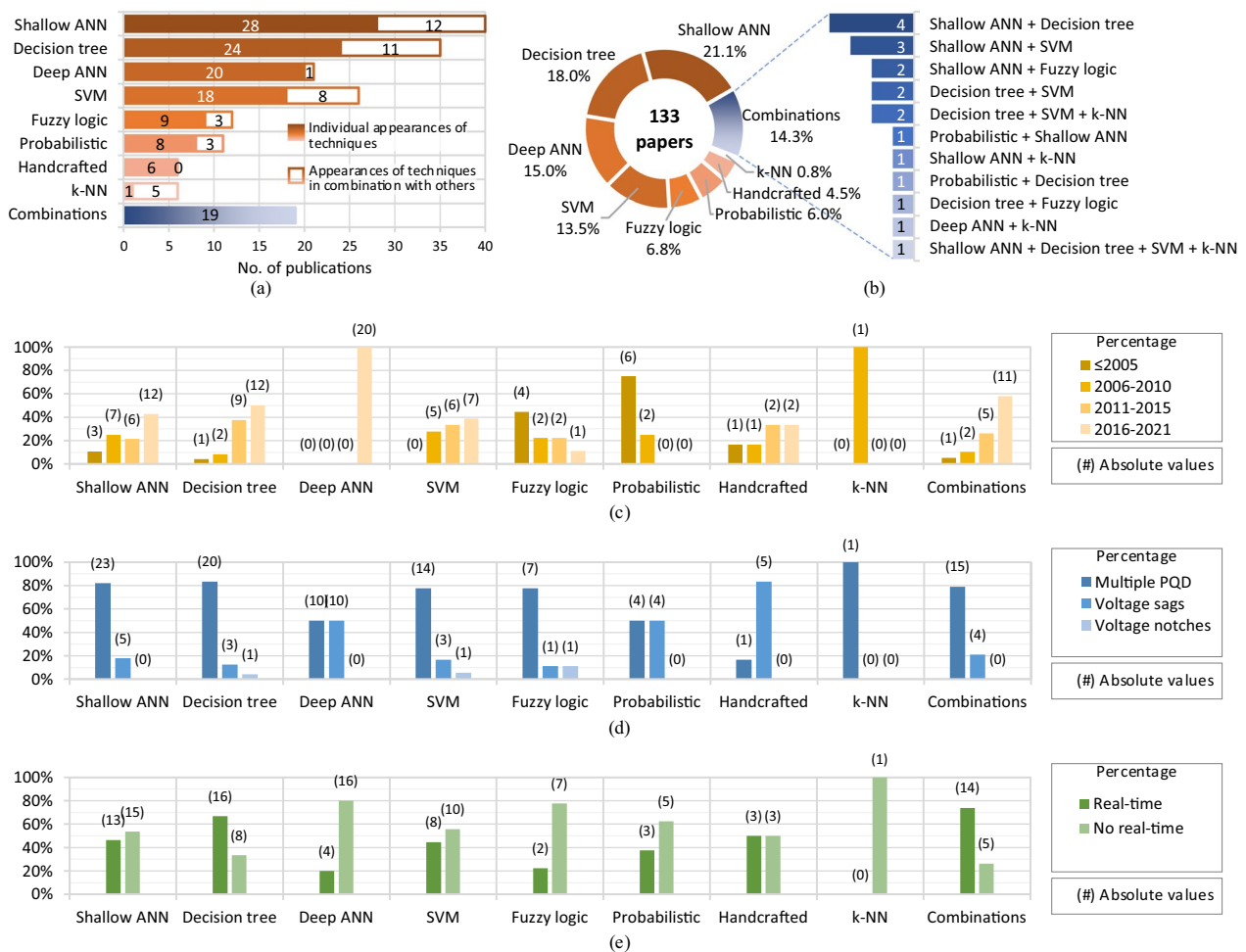


Fig. 12 Distribution of papers in the identified classification techniques. **a** The number of papers, **b** percentages and number of papers that use a combination of techniques, **c** percentages and number of papers from the time perspective, **d** the type of disturbance perspective, and **e** the real-time perspective

identified for voltage notches; therefore, classification is incipient in this case. It is also observed that deep ANN models are evenly used for multiple PQDs and voltage sag classification.

In Fig. 12e, the distribution of techniques according to the real-time application is presented. For instance, decision trees are more used in real-time approaches, which is attributable to the simple operation of algorithms. Similarly, the combination of techniques is mostly used for real-time applications, where the methods take advantage of each technique in terms of efficiency. Conversely, deep ANNs, fuzzy logic, and probabilistic methods are less used for real-time applications because of the cost of computation during the operation of algorithms.

5.8 Characterization

The characterization of single PQDs refers to the quantification of features that distinguish the disturbance. The

single-PQD characterization referred to in this section usually differs from the characterization conducted in the stages described previously. The feature extraction stage presented in Sect. 5.4 explains methods to obtain statistical, time-series, spectral, and image-based characteristics that are useful as inputs for the classification techniques to identify the category of PQD. However, these characteristics are not necessarily representative of the physical features that describe the phenomena according to deviations and standard limits. For instance, in [168], the extracted features include the mean, variance, kurtosis, skewness, entropy, etc. of the coefficients from the WT to classify voltage sags according to their causes. Thereby, these features are not explicitly related to magnitude, duration, POW characteristics, and phase angle jump, which are the physical features that identify voltage sags. The abovementioned difference occurs because the purpose of feature extraction is to obtain characteristics that

allow the most accurate and efficient results in the classification stage. Conversely, in the single-PQD characterization referred to in this section, which is mainly useful for assessing the severity and impact of the disturbance, the physical representation and interpretation of the phenomena is the most important aspect.

This review gives a particular focus to voltage sags and notches. Hence, only the aspects observed in the selected literature related to their characterization are briefly described in the following sections.

Table 12 summarizes the parameters used to characterize voltage sags and notches. A brief definition of each parameter is provided, and its relevance is highlighted. References of the selected literature that analyze each of the parameters are also indicated, as well as the usage percentage.

5.8.1 Voltage sag characterization

A voltage sag is a decrease in the RMS voltage to between 0.1 and 0.9 pu lasting from 0.5 cycle to 1 min [5]. Voltage sag characterization is useful in assessing equipment sensitivity. For instance, sag magnitude is the main factor that determines if a piece of equipment will malfunction or stop working. Moreover, sag duration is important to establish the impact on industrial processes. Thus, for voltage sags lasting longer, the probability that industrial processes stop is higher. Other voltage sag characteristics including POW of initiation/ending and phase angle jump may affect the performance of power electronic equipment that uses phase angle information [6]. In the literature, single-event characteristics including magnitude and duration of sags have been widely addressed. However, POW characteristics and phase angle jump still require further research to be more accurately computed and to better determine the impact on equipment.

Approaches to characterizing voltage sags usually include the analysis of time domain waveforms and profiles (RMS value), or the analysis of ellipse parameters obtained by using the SPM, e.g., [140, 151–153, 157, 163, 172]. Interest in multistage characterization is emerging [157, 174], as the usual single-event characteristics may not be enough to describe voltage sag events in real conditions.

5.8.2 Voltage notch characterization

Voltage notches are sub-cycle waveform distortions characterized by a periodic voltage reduction lasting less than half a cycle. Notches may be caused by the normal operation of power electronic converters when current is commutated from one phase to another leading to short-duration overcurrent [5]. A voltage notch is defined by its depth, width, and area [12]. Moreover, the number of voltage notches occurring per cycle or half-cycle is an

important feature in suitably assessing the aggregated impact of the disturbance. The number of notches also allows for identification of the type of converter causing the disturbance.

According to the analyzed literature, width is usually obtained in the process of voltage notch characterization. However, depth and area have been less studied. The number of notches per cycle can be extracted from various methodologies in the literature, but it is not explicitly performed. In general, the literature related to voltage notches is still incipient and there is a lack of research on the characterization of the phenomena to analyze, for instance, severity and impact on end-user equipment.

6 Discussion

The discussion is organized into remarks on the methodology for the review and the bibliometric analysis, and the development of the literature review throughout the stages for detection and classification of PQDs described in Fig. 7 i.e., input space, preprocessing, feature engineering, and decision space. Detailed technical aspects related to these stages are also presented, analyzed, and discussed.

6.1 Methodology for the review and bibliometric analysis

6.1.1 Methodology for the literature review

The methodology proposed in Sect. 3 to find the relevant literature is reproducible and scalable. However, a significant effort is required to retrieve and structure the bibliographic metadata to conduct the process. In this context, a gap in the process is observed and a possible improvement of the methodology lies in a higher level of automatization in data retrieval and cleaning. In addition, the indices formulated in the scoring equation (see Sect. 3.2—Scoring equation) to facilitate the selection of the most relevant literature are subject to improvement. Especially, the title and abstract similarity indices may be enhanced using more advanced language processing techniques.

6.1.2 Bibliometric analysis

The bibliometric analysis has revealed the increasing trend in publications related to the detection and classification of PQDs. Likewise, publications including real-time aspects are also increasing, but with a more conservative trend. Thereby, the need for more research on real-time aspects is highlighted. Data of origin of publications indicated that most of the research on the topic (more than 50% of publications) is performed in China, India, and the USA. This situation suggests that higher efforts in researching the topic should be performed worldwide because the context of power systems varies from region to region. Distribution on general topics

Table 12 Summary of parameters to characterize voltage sags and notches

PQD	Parameter	Definition	Relevance	References	% to the total
Voltage sag	Magnitude	The minimum remaining RMS voltage during the sag event, with values between 0.1 and 0.9 pu [5]	Essential to define the impact on end-user equipment and industrial processes [7]	[49, 77, 132, 134–138, 140, 142, 143, 146, 148, 156, 157, 161, 163, 173, 174, 184, 186, 201, 204]	13
	Duration	The time interval where voltage is less than 0.9 pu. Durations are between 0.5 cycles and 1 min [5]		[34, 44, 87, 90, 132, 134–138, 142, 146, 154, 156, 157, 160–163, 166, 173, 174, 184, 186, 200–202, 204]	15
	POW of initiation	The phase angle of the voltage waveform at the start of the sag event [7]	Useful in determining equipment sensitivity, and the exact voltage sag duration [163]	[87, 132, 134, 137, 138, 146, 154–157, 160–163, 166, 173, 174, 184, 186, 200, 201]	12
	POW of recovery	The phase angle of the voltage waveform at the end of the sag event [7]		[87, 132, 134, 137, 138, 146, 154–157, 160–163, 166, 173, 174, 184, 186, 200, 201]	12
	Phase angle jump	Phase shift in the voltage waveform at the start of the sag event [7]	It may affect power electronic converters using phase-angle information [6]	[132, 134–136, 152, 154, 157, 163, 173, 174, 184, 186]	7
	Ellipse characterization	It refers to the analysis of sag parameters through the features of the ellipses obtained with the SPM [140]	It allows for reducing the number of variables at each stage of the analysis [140]	[140, 151–153, 157, 163, 172, 200]	6
	Multistage characterization	Calculating the parameters that identify voltage sags with segments of different behavior during the sag event	Single-event features (magnitude, duration) are not enough to characterize a sag [157]	[157, 174]	1
	Depth	The average depth of the line voltage notch from the sine wave of voltage [12]	High levels of voltage deviations downgrade electronic equipment and damage inductive components [190]	[190, 191, 194, 201, 209, 210]	3
	Width	Duration of the notch switching disturbance [12]	It also determines the severity of the notch related to the duration of the momentary short circuit	[87, 187, 190–192, 194–196, 199–201, 203–205, 207–210]	10
	Area	The product of the notch depth times the width of the notch [12]	It allows quantifying the severity of notches	[190, 191, 194, 201, 209, 210]	3
Voltage notch	Number of notches per cycle	Occurrences of notch switching disturbances in one cycle	It allows quantifying the aggregate severity of notches and identifying the producing converter	[37, 87, 187, 190, 193, 195, 200, 201, 203–207, 209]	8

indicates that real-time aspects are more related to monitoring and detection, and offline applications are more common in the characterization, assessment, and classification of PQDs. Looking at the type of disturbance, general-purpose approaches (multiple PQDs) and tools focused on voltage sags have extensive literature and are very mature. By contrast, research on voltage notches is still incipient. Additional quantitative analyses have been presented for the main steps in the process of real-time detection and classification of PQDs, i.e., transformation and classification. The results of these analyses indicate the most widely used techniques.

6.2 Input space and preprocessing

6.2.1 Input data preparation

The most popular techniques for the input space are the equation- and simulation-based signals. In the case of laboratory and field measurements, the signal acquisition is mainly carried out using DSP, FPGA, microcontrollers, and computers. Real-time digital simulators are a new trend in PQDs simulation. More research should be performed on the use of laboratory and field measurements for the assessment of tools for real-time detection and classification of PQDs, to deal with the uncertainties associated with real conditions.

6.2.2 Data preprocessing

Segmentation and normalization are usually applied for preprocessing of multiple PQDs and voltage sags. In addition, denoising/filtering is mostly applied to specific disturbances such as voltage notches. No extensive details regarding data validation to ensure reliability of data from field measurements have been found in the literature on PQD detection and classification. Therefore, further analysis of data validation in the context of PQD detection and classification may be useful.

6.3 Feature engineering

6.3.1 Transformation

Time–frequency and time-domain techniques are the two most popular for the transformation of PQDs in general, and voltage sags and notches. The techniques widely used in time–frequency domain are WT, ST, and HHT, or their combination with other techniques. The most used non-parametric transformation techniques in the time domain are SPM, PSR, and TT, whilst the most used parametric transformation techniques are KF (especially for voltage sags) and AF (e.g., adaptive linear network). There are still opportunities to apply real-time oriented transformation techniques to perform detection, classification, and characterization of specific disturbances such as voltage sags and notches using non-parametric techniques such as SPM, PSR, and MM, as well as parametric

techniques such as Extended KF (non-linear filtering) and AF.

6.3.2 Feature extraction

The feature extraction stage is mostly performed using statistical and time series techniques. The mean value is the clear dominant statistical metric for central tendency, whilst statistical dispersion is usually measured through standard deviation, maximum/minimum value, and statistical values such as variance and HOS (skewness, kurtosis, etc.). On time series techniques, cycle-based indices as coefficients of energy (especially for voltage notches), RMS values, and Shannon entropy are the most common, whereas the most popular sample-based indices are absolute/maximum and event duration for multiple PQDs and voltage sags, and derivative as well as Euclidean distance specifically for voltage notches. The latter might be potentially useful for the assessment of sub-cycle disturbances (e.g., notches, transients).

6.3.3 Feature selection

The step of feature selection is only involved in designing the tools for detection and classification of PQDs but not in the operation. Thereby, the computation times for selecting the optimal set of features for specific applications is a process before the real-time operation. An adequate selection of features allows an enhanced operational efficiency and accuracy. Most approaches in the literature for feature selection are handcrafted based on expert and widely accepted knowledge, with the advantage of obtaining a better physical interpretation of the phenomena. However, handcrafted feature selection can be a time-consuming process. Optimization methods have also been used to provide enhanced results but usually become complicated when selecting a suitable set of features. Finally, dimensionality reduction approaches have shown promising results in the feature selection process providing a good compromise between simplicity of implementation and accuracy and efficiency of results. More effort should be expended on the automatization of dimensionality reduction techniques and the proper physical interpretation of the selected features.

6.4 Decision space

6.4.1 Detection

As a prior step in the process of classifying PQDs, detection may be essential in real-time operation because its independent implementation allows for improving efficiency and general performance. To that end, the algorithms for the detection of states different from the ideal conditions of voltage waveforms are performed constantly in a simple process in real-time. If a PQD is subsequently detected, a more complex process of classification

is activated, thus reducing the computational burden of embedded systems. The abovementioned approach has not been extensively implemented and analyzed in the literature while such approach is likely to provide promising results for the implementation of PQD monitoring systems. Furthermore, in the case of voltage sags, detection is useful for applications such as the suitable operation of DVR and protection systems. In those cases, some characterization of the disturbance is necessary.

6.4.2 Classification

Techniques for automatic classification of PQDs have been widely studied in the literature, mainly focused on the categorization of multiple PQDs (sags, swells, notches, transients, harmonics, flicker, etc.). Other advances are in the classification of complex (combined) PQDs, the classification according to root causes, and real-time applications. According to the review, the latter two topics still require further research. The most popular technique for classification is shallow ANNs because of their flexibility in learning any pattern from any set of features. However, ANNs require extensive data and computational effort for training. Decision trees have also been extensively used for classification because of their simplicity, being proper for real-time applications. Recently, deep learning techniques are gaining interest because of the high level of automatization (feature extraction is performed automatically as a process within the technique). However, in these approaches, the physical interpretation of features is lost. Moreover, great potential is observed in the use of unsupervised learning techniques because they have not been yet extensively studied in the context of PQD classification.

6.4.3 Characterization

Single PQD characterization results in the physical interpretation of the phenomena and provides relevant information for the assessment of severity and impact on end-user equipment. Therefore, characterization is useful for analyzing disturbances from the electromagnetic compatibility perspective. Characterization of voltage sags has been widely addressed concerning magnitude and duration. However, POW characteristics and phase angle jump that impact on power electronic equipment still require further research. Multistage sag characterization is of emerging interest because of its occurrence in real conditions and needs further work. Voltage notch characterization is still incipient, especially regarding severity assessment. For instance, the definition of notch depth is ambiguous in the literature. Moreover, the analysis of notching ringing as described in [5] has not yet been addressed in the literature and the characterization

of voltage notches may be much more challenging in this context.

6.5 Discussion of technical aspects

Technical issues associated with the steps for PQD detection and classification are discussed. The main technical issue related to input space and data preprocessing is associated to modeling of uncertainties occurring in field measurements. This requires applying data analysis techniques including data plausibility, data cleansing, statistical inference, etc. In the feature engineering stage, the real-time application of processing techniques is still challenging because of computation times in the most widely applied time–frequency-domain transforms such as WT and ST. Although promising results are obtained in the real-time application of time-domain transforms such as SPM, the challenge is the accurate representation of the phenomena and the analysis of single-phase voltages and currents. Feature selection using optimization methods is still challenging while the improvement in accuracy is limited. Therefore, a proper trade-off between complexity and accuracy should be considered in such cases. In the decision space, a challenging technical issue is the formulation of a comprehensive method considering all steps in the decision space, i.e., detection, classification, and characterization that may be useful for real-time applications, e.g., the analysis of PQD propagation to identify and localize root causes, the operation of protection systems, and the automatic implementation of mitigation measures.

7 Perspectives for future research

On transformation techniques, the research trend shows that time-domain and combinations of techniques from different domains are becoming relevant in PQD detection and classification in general. The trend also shows that research is towards the application of one transformation technique, or a combination of several, that can accurately detect the highest number of PQ variations and events rather than specific methodologies for specific disturbances.

WT, ST, and time-domain techniques (non-parametric techniques such as SPM, SPR and MM, as well as parametric techniques such as extended KF and AF) seem to have potential for real-time detection and classification of either PQDs in general (multiple) or the specific (e.g., voltage sags and notches). These techniques, among other characteristics, are flexible in detecting different PQDs and can be used in devices with restricted computational resources. Nevertheless, it is also acknowledged that the combination of different state-of-the-art techniques can also be of benefit for the detection and classification of PQDs. Opportunities

for future research also exist in real-time detection, classification, characterization, and possibly in feature aggregation of sub-cycle disturbances, such as voltage notches and transients.

The use of powerful image classification techniques after the 2D transformation of signals is also a promising field of research. This approach allows the use of tools from the ever-increasing potential of the image processing and classification field, e.g., the attention mechanism to improve classification accuracy and transfer learning to reuse pre-trained models [167]. Also, a 2D transformation of signals allows the use of deep learning tools such as CNN. Alternatively, the use of simpler and robust techniques to analyze 2D figures in the complex plane such as Fourier descriptors may provide satisfactory results in classification accuracy and efficiency. Among the 2D transformations, SPM and PSR have shown good characterization capabilities and performance for real-time applications.

In the classification of multiple PQDs and voltage sags, CNN have shown promising results in recent studies [121, 158, 179]. The potential of CNN lies in its ability to automatically extract the best set of features to obtain very accurate results in the classification of PQDs, even in noisy environments. However, CNN have some drawbacks such as the high computational requirement in training and large number of model parameters that hinder real-time application and implementation in embedded systems because of the required additional storage. Moreover, the physical interpretation of features is lost because the automatic extraction is performed within the CNN. A combination of CNN with other techniques may help to overcome the drawbacks. For instance, the few-shot learning technique [171], can be used to reduce the high computational and large dataset requirements for training. Alternatively, ELM is gaining popularity [108, 176], because of its simple operation, sufficiently accurate results, and requirement of fewer training data. The efficiency of ELM also facilitates real-time application.

In the context of the ongoing digitalization of the power system and smart grid paradigm, real-time detection and classification of PQDs play an important role. By online identification of PQDs, a rapid pinpoint of the root-causes can be achieved, and prompt automatic mitigation measures can be implemented to reduce negative technical and economic impacts, e.g., fault location and clearance, flicker source location and mitigation, harmonic resonance source location and mitigation, etc. For this purpose, large amounts of data provided by advanced metering infrastructures, i.e., smart meters, PQ monitors, phasor measurement units, etc., would require advanced algorithms to perform real-time detection and classification of PQDs.

Specific research on voltage sags also offers areas for contribution. For instance, methodologies for voltage sag classification and characterization have focused on single-or three-phase voltages but, to the extent of this review, there is no comprehensive method to automatically classify and characterize single-and three-phase voltage sags. Some approaches have addressed voltage sag root cause location (upstream or downstream), but a more precise location (pinpoint) of sag origin should be achieved. This could be useful for the operation of protection systems and mitigation measures. Also, a more accurate real-time characterization of POW features and phase angle jump can be achieved. The characterization of multistage voltage sags is also of emerging interest because of their common occurrence in real conditions and the limitations of single event voltage sag characterization. Progress in this direction has been made in [157] and [174] using SPM and the multidimensional characterization, respectively, but more studies are still required to improve real-time performance.

Regarding the lack of research in voltage notches, there are several opportunities for further work. For instance, more accurate tools for detecting and characterizing notches are required, including the calculation of single PQD features such as depth, width, area, number of notches, and other indices to assess severity and impact on end-user equipment. Furthermore, automatic classification of voltage notches has not been addressed in the literature. Some categories can be identified such as normal notching and notching ringing [5], while the categorization would allow a better understanding of the notch PQD and an improved severity and impact assessment.

Only supervised learning techniques have been analyzed in this review for the AI-based classification of PQDs. However, unsupervised techniques could be useful in PQD detection and classification because no labeled data is required. This can facilitate the process of training algorithms. Furthermore, unsupervised and supervised learning algorithms can be used together to exploit the potential of both approaches. For instance, simple unsupervised learning algorithms can be used to detect states different from the ideal voltage signal, and then more complex supervised learning algorithms can be used for classification. This approach can enhance efficiency for real-time application.

8 Conclusion

The comprehensive and systematic review conducted in this paper initially develops a methodology to identify the most relevant articles in the detection and classification of PQDs. This methodology results in a scalable and reproducible process that contributed to proposed indices to assess publications in terms of topic similarity

and quality of research. The narrow set of publications selected through the systematic process allows a comprehensive overview of the real-time detection and classification of PQDs.

The bibliometric analysis of the literature metadata demonstrates the increasing interest in PQD detection and classification. It also presents top publishing countries and researchers, and first quantitative insight into the relation of general topics, e.g., monitoring, detection, classification, and characterization with real-time applications. The need for further research in real-time approaches for PQD detection and classification is highlighted.

A comprehensive descriptive, qualitative, and quantitative review is performed throughout the stages for real-time detection and classification of PQDs, where techniques dealing with PQDs in general (multiple PQDs) or with specific (e.g., voltage sags and notches) are identified and described. The most relevant findings are summarized in taxonomy figures and tables. Also, more detailed quantitative analyses are provided for the most widely explored stages in the literature, i.e., transformation and classification.

The main remarks arising from the literature review are that transformation and classification techniques have been widely addressed and are very mature for offline applications. However, real-time applications still require more research to find efficient and accurate tools for real conditions in actual power systems. The computational burden is an essential aspect in this context, where embedded systems have limited resources. The proper integration of stages, e.g., preprocessing and feature engineering, and the development of new techniques can facilitate real-time applications.

Research gaps in voltage sags are addressed, including combined single-and three-phase analysis, sag root cause location, accurate and multistage sag characterization. Similarly, research gaps in voltage notches include accurate and unambiguous characterization and classification for severity and impact assessment.

Abbreviations

AF	Adaptive filter
AHP	Analytic hierarchy process
AI	Artificial intelligence
ANN	Artificial neural network
CNN	Convolutional neural network
CT	Chirplet transform
DBN	Deep belief network
DFT	Discrete Fourier transform
DVR	Dynamic voltage restorer
DWT	Discrete wavelet transform
ELM	Extreme learning machine
EMD	Empirical mode decomposition
FES	Fuzzy expert system
FFT	Fast Fourier transform

FT	Fourier transform
GA	Genetic algorithm
GT	Gabor transform
GWT	Gabor-Wigner transform
HHT	Hilbert-Huang transform
HT	Hilbert transform
HOS	Higher-order-statistics
ICA	Independent component analysis
IMF	Intrinsic mode function
KF	Kalman filter
k-NN	K-nearest neighbor
LSTM	Long short-term memory
MD	Mode decomposition
MM	Mathematical morphology
PCA	Principal component analysis
PLL	Phase-locked loop
PQ	Power quality
PQD	Power quality disturbance
PSR	Phase space reconstruction
SPM	Space phasor model
SSA	Singular spectrum analysis
SSD	Sparse signal decomposition
ST	Stockwell transform
STFT	Short-time Fourier transform
SVM	Support vector machine
TT	Time-time transform
VMD	Variational mode decomposition
WT	Wavelet transform

Acknowledgements

Not applicable.

Author contributions

JEC and DAM contributed significantly to methodology formulation, review development and analysis, and article drafting and writing. ERT and JM provided permanent supervision, contributed to quality improvement, and derive significant discussions and conclusions. All authors read and approved the final manuscript.

Funding

Not applicable.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Received: 27 July 2022 Accepted: 6 January 2023

Published online: 20 January 2023

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Article

Flexible Simulation Platform for Generating Realistic Waveforms with Voltage Notches

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Abstract: Voltage notches are steady-state sub-cycle waveform distortions caused by the normal operation of line-commutated power converters, significantly impacting power quality in industrial low-voltage (LV) networks. Despite their common occurrence, research on this phenomenon is still incipient, and realistic simulation platforms are lacking. This paper introduces a detailed MATLAB (R2024a)/Simulink-based simulation platform that models a benchmark low-voltage industrial installation, including a six-pulse controlled rectifier, linear loads, and a capacitor bank for power factor correction. Systematic simulations are performed with the platform to examine the sensitivity of notch characteristics to key parameters within plausible ranges, such as short-circuit power at the point of common coupling, commutation reactance, firing angle, snubber circuits, and rated power of the rectifier. In addition, parameters such as the rated power of linear loads and the compensation power of the capacitor bank are examined. Other influencing parameters including background voltage unbalance and distortion are also modeled and considered. A comparative analysis with field measurements from German industrial LV networks validates the plausibility and suitability of the simulations. Building upon this platform, a Monte Carlo simulation approach is adopted to generate extensive datasets of realistic voltage notch waveforms by randomly varying these key parameters. A case study conducted under conditions typical of German LV networks demonstrates the applicability of the simulations. To support further research, the simulation platform and exemplary synthetic waveforms are provided alongside the paper, serving as a valuable tool for testing and designing strategies for analysis, detection, and monitoring of voltage notches.

Keywords: low-voltage (LV) network; power quality disturbance (PQD); six-pulse controlled rectifier; voltage notch; waveform distortion



Citation: Caicedo, J.E.; Zyabkina, O.; Rivas, E.; Meyer, J. Flexible Simulation Platform for Generating Realistic Waveforms with Voltage Notches.

Appl. Sci. **2024**, *14*, 11031. <https://doi.org/10.3390/app142311031>

Academic Editor: Gerard Ghibaudo

Received: 31 October 2024

Revised: 23 November 2024

Accepted: 25 November 2024

Published: 27 November 2024



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1. Introduction

Voltage notches are steady-state voltage changes with a duration much shorter than the AC period, caused by the normal operation of line-commutated power converters [1–3]. This phenomenon distorts the voltage waveform and may excite the natural frequencies of the system, creating considerable high-frequency oscillations and significantly impacting power quality in distribution networks [4]. For instance, voltage notches lead to damaged capacitor banks, parallel resonances, radio disturbances, and electronic equipment malfunction [4]. Electronic equipment at low-voltage (LV) levels is particularly susceptible to voltage notches because of the resulting signal interference in logic or communication circuits [5]. Other power converters face challenges, including thermal stress in snubbers and filters caused by harmonic currents and control circuit malfunctions and timing errors in phase-locked-loop systems, due to distorted supply voltage [2,5].

Line-commutated power converters are extensively used in industrial applications such as electrochemical processes, motor drives, and controlled power supplies [6]. Consequently, voltage notch issues are particularly relevant to industrial LV networks [7–9].

Therefore, the analysis, monitoring, and mitigation of this disturbance are of paramount importance for the industrial sector.

Despite the relevance of voltage notches, the current state of research reveals a lack of comprehensive studies and practical tools to address this power quality disturbance (PQD) [1,7]. Previous efforts include formulating analytical expressions to model voltage and current waveforms during commutation notches [4,5,8], conducting measurement-based analyses at industrial installations [9,10], developing algorithms for detection and classification [7,11], and implementing mitigation measures [12]. While these works offer valuable insights, several aspects remain unexplored. For instance, though time-domain simulation of voltage notches has been developed for in-depth research, testing, and validation, many studies provide little detail on the simulation platform [4,8,12], limiting the wide application of these tools. Addressing this gap, [7] provides a simulation platform analyzing the impact of key parameters affecting the voltage notch waveform, such as firing angle, snubber circuit, and capacitor banks in the LV network. Moreover, the work in [7] adopts a systematic approach to generating synthetic datasets of voltage notches to test a classification algorithm. However, the dependence between the key parameters is not fully addressed in [7] for dataset generation, and the plausibility of simulations has not been validated. Moreover, influencing factors observed in real-world LV networks, such as background voltage unbalance and distortion, have not been considered.

The IEEE 519 standard emphasizes the necessity of realistic voltage notch simulations, stating that detailed simulations are required when direct measurements are unavailable to confirm an installation's compliance with recommended limits [5]. However, this concern has not been fully addressed, as previously described. To bridge this research gap, a MATLAB/Simulink-based platform is introduced to generate realistic synthetic waveforms with voltage notches, offering the following key contributions:

- To enable comprehensive analysis of voltage notches, a benchmark low-voltage (LV) installation model featuring a six-pulse controlled rectifier is introduced. This model allows for the examination of key parameters influencing voltage notch characteristics, such as short-circuit power at the point of common coupling (PCC), commutation reactance, firing angle, snubber circuit characteristics, rated power of the rectifier, and background voltage unbalance and distortion. The platform also facilitates simulations involving linear loads and capacitor banks in the LV network. While other factors like nonlinear loads can affect voltage waveforms, they are acknowledged but not included in this analysis. The plausibility and suitability of the simulations are validated against field measurements.
- To generate realistic synthetic waveform datasets for voltage notch studies, a Monte Carlo simulation approach is adopted. This method considers the dependence of key parameters and typical conditions of distribution networks, such as common short-circuit power levels, enhancing the realism and applicability of the generated data.

This paper is organized as follows. Section 2 introduces the fundamentals of voltage notches. Section 3 presents the design of the simulation platform in Simulink. Section 4 analyses the interrelations, limitations, and impact of key parameters on voltage notch waveforms. Section 5 validates the plausibility and suitability of simulations against field measurements from German industrial LV networks. Section 6 proposes a Monte Carlo simulation approach to generate realistic synthetic waveform datasets, and a case study is developed considering the conditions of German LV networks. Section 7 provides a discussion of the key findings, limitations, and implications of this study. Finally, Section 8 provides the closing remarks and offers potential further research.

2. Background on Voltage Notches

Voltage notches, also referred to as commutation notches [3,13] or notching [2,5], are caused by the normal operation of line-commutated power converters [1–3]. The notch occurs in the converter when the current is transferred from one conducting arm to another. In this finite time interval, both arms conduct simultaneously, leading also to a short-

duration overcurrent [13]. As depicted in Figure 1a, the main characteristics of a voltage notch include the following [5]:

- Depth: The average depth, ΔU_{com} , of the voltage notch from an ideal voltage sine wave. Common practice is the use of relative depth, d_{com} , with respect to the peak value of the fundamental component $\hat{U}_1$, i.e., $d_{com} = \Delta U_{com} / \hat{U}_1$ [14].
- Width: The duration of the notch, Δt_{com} , usually expressed in μs .
- Area: The product of the notch depth, ΔU_{com} , multiplied by the notch width, Δt_{com} .

Commutation oscillations, caused by capacitive elements, are another key characteristic of voltage notches. Oscillations in the kHz range, as depicted in Figure 1b, are linked to capacitor banks, filters, and capacitive loads in the LV network. Higher-frequency oscillations, as shown in Figure 1c, result from snubber capacitors in the power converter.

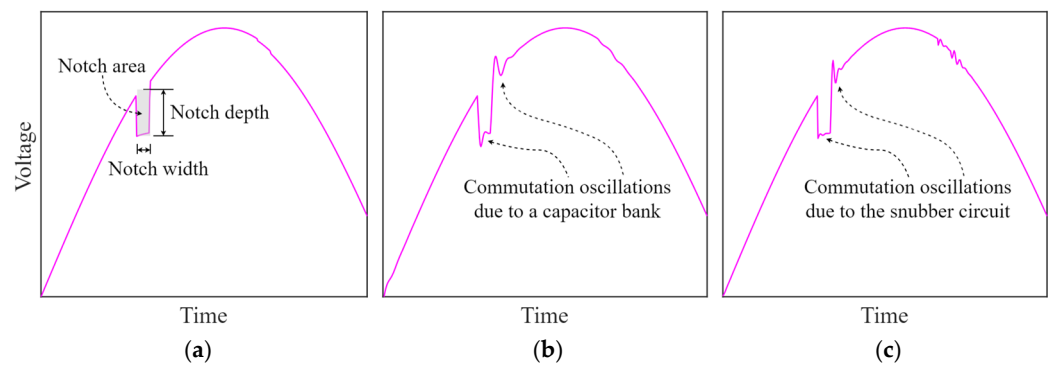


Figure 1. Characteristics of voltage notches. (a) Depth, width, and area of a classical notch (adapted from ref. [5]). (b) Notch with commutation oscillations due to a capacitor bank with no detuning. (c) Notch with commutation oscillations due to snubber capacitance.

Table 1 presents the notch limits according to the standard IEEE 519. Emission is restricted solely to notch depth and area. However, various studies, including [4,8], emphasize the need for a comprehensive analysis of commutation oscillations and the potential excitation of natural frequencies in the network impedance.

Table 1. Recommended limits on voltage notches according to IEEE standard 519 (data from ref. [5]).

Characteristic	Special Applications	General System	Dedicated System
d_{com}	0.1	0.2	0.5
A_{com} ($V \cdot \mu s$) *	16,400	22,800	36,500

* The values indicated for A_{com} are applicable to systems with a phase-to-phase voltage of 480 V. A conversion is then required for application at other voltage levels.

Commutation Phenomenon and Voltage Notch Generation

Voltage notches in a six-pulse controlled rectifier occur during the commutation process when current transfers between phases in the AC input.

- Ideal conditions: No commutation inductance leads to no voltage notches, with instantaneous commutation and an ideal six-pulse output DC voltage (U_D). This is shown in Figure 2a for a firing angle $\alpha = 30^\circ$.
- Non-ideal conditions: The presence of commutation inductance causes an overlap period where two SCRs conduct simultaneously, creating a transient AC short circuit and resulting in notches on phase-to-neutral and phase-to-phase voltages, also affecting U_D . This is shown in Figure 2b for $\alpha = 30^\circ$.

The firing angle α ranges from 0° to 180° , but the analysis in this paper focuses on the rectifier mode with $\alpha < 90^\circ$. Practical constraints, such as the need for sufficient gate triggering voltage, limit α to a range of 10° to 90° to ensure proper firing of SCRs [15].

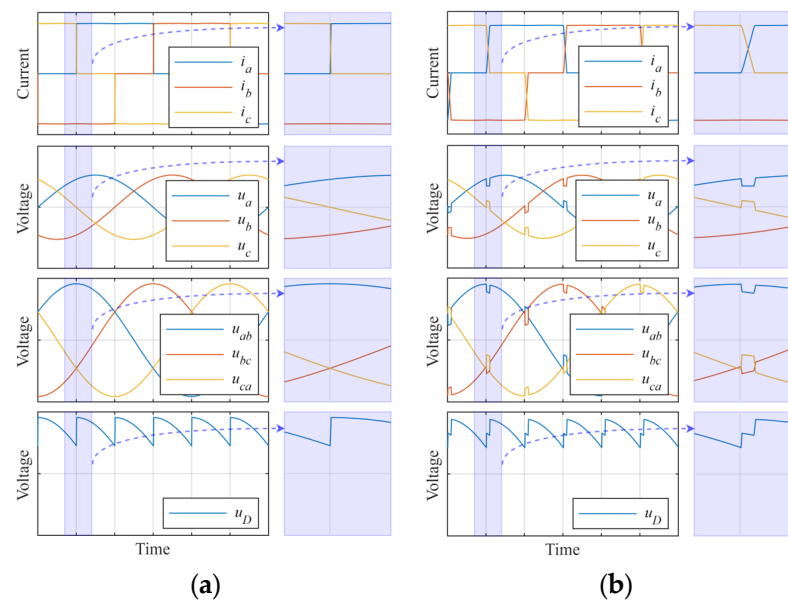


Figure 2. Waveforms of a six-pulse controlled rectifier under different commutation conditions. The graphs display (from top to bottom): three-phase line currents, phase-to-neutral voltages, phase-to-phase voltages, and DC output voltage. (a) Ideal conditions: Commutation occurs smoothly without voltage notches. (b) Non-ideal conditions: Commutation leads to voltage notches in the waveforms.

3. Simulation Platform Design

A benchmark LV installation for an industrial customer is modeled, featuring a six-pulse controlled rectifier as the primary component. The platform is designed for flexibility in configuring parameters that influence voltage notch waveforms, including short-circuit power at the PCC, commutation reactance, firing angle, snubber circuit, and the rectifier's rated power. The simulation platform also considers additional components typical in industrial LV networks, such as a capacitor bank (non-detuned or detuned) for power factor correction and aggregated linear loads. The upstream network is modeled to account for background voltage unbalance and distortion.

Figure 3 illustrates the benchmark LV installation implemented in Simulink. This installation provides a comprehensive framework for analyzing the key parameters influencing voltage notch waveforms. The detailed analysis and design of the main components and parameters are presented in the following sections.

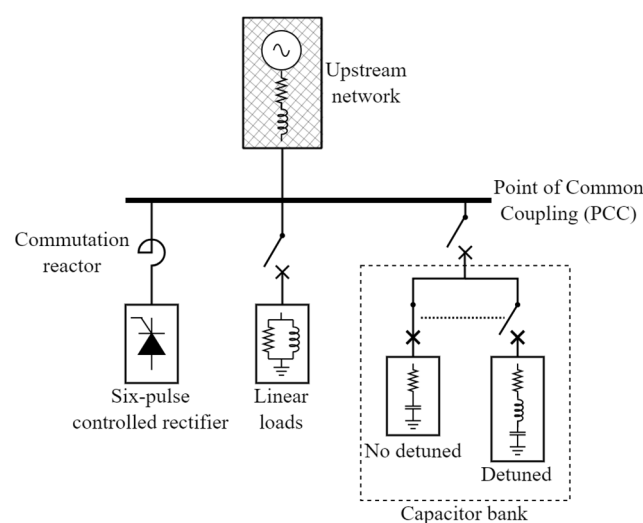


Figure 3. Benchmark LV installation implemented in Simulink for voltage notch simulation.

3.1. Upstream Network

The upstream network is modeled at the fundamental frequency by a three-phase source with nominal voltage U_s and a short-circuit impedance $Z_s = R_s + jX_s$, serving as a simplified (Thevenin) representation of the network upstream of the PCC. Z_s is derived from the short-circuit power S_{sc} using $Z_s = U_s^2 / S_{sc}$. S_{sc} is either provided by the distribution network operator or calculated as detailed in [14].

Reference values of Z_s for assessing PQDs at the LV level, as per IEC/TR 60725 [16], indicate typical short-circuit power levels derived from extensive surveys in European LV networks (230 V, 50 Hz). For example, devices with rated currents ≤ 16 A and ≤ 75 A have reference Z_s values of $0.24 + j0.15 \Omega$ and $0.15 + j0.15 \Omega$, corresponding to short-circuit powers of 0.57 MVA and 0.76 MVA, respectively.

3.1.1. Background Voltage Unbalance

In LV networks, three-phase voltages and currents often exhibit unbalance due to the extensive connection of single-phase devices and their unequal distribution among the phases. To analyze this unbalance, the method of symmetrical components is employed, which decomposes the unbalanced system into three sequence components: positive, negative, and zero.

Voltage unbalance (k_u) is quantified as the ratio of the negative sequence component ($\underline{U}_{neg}$) to the positive sequence component ($\underline{U}_{pos}$), expressed as $k_u = |\underline{U}_{neg}| / |\underline{U}_{pos}|$.

These symmetrical components can be calculated from the phase-to-neutral voltages using (1) [17].

$$\begin{bmatrix} \underline{U}_{pos} \\ \underline{U}_{neg} \\ \underline{U}_{zero} \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_a \\ \underline{U}_b \\ \underline{U}_c \end{bmatrix} \quad \text{with} \quad \underline{a} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}. \quad (1)$$

Voltage unbalance in most applications is less than 5%, with the IEC 61000-2-2 standard [18] recommending a 2% limit. Although the focus is on industrial LV networks, specific data on voltage unbalance in these settings is limited. Therefore, data from surveys conducted in public LV networks are used. For instance, a study involving 127 public LV networks in Germany [17] provides typical values of voltage unbalance, as shown in Table 2. This approach is justified because further surveys, including industrial and mixed LV networks, have demonstrated consistency and similar results in voltage unbalance levels [17]. Moreover, industrial LV networks are often connected to public LV networks, implying that the background voltage unbalance experienced in industrial networks can be similar to that in public networks. The referenced survey reports the 95th percentiles of 10 min aggregated measurements collected over two weeks, along with the respective percentiles from 127 different grids.

Table 2. Percentiles of $|\underline{U}_{neg}|$ for 10 min average values and 127 LV networks (Data from ref. [17]).

Percentile Over 127 Grids	1	5	50	95	99
$ \underline{U}_{neg} $ (V)	0.28	0.31	0.57	1.12	1.44
k_u (%)	0.12	0.13	0.25	0.49	0.63

The simulation platform in this paper considers background voltage unbalance by computing phase-to-neutral voltages for each phase of the three-phase source, based on predefined symmetrical component values. Only the negative sequence voltage is considered for voltage unbalance due to the delta winding of transformers, which prevents zero-sequence voltage transfer from the upstream network to LV networks [19].

3.1.2. Background Voltage Distortion

Voltage distortion, a common PQD, occurs due to nonlinear devices causing nonsinusoidal voltage drops while maintaining periodicity. These distortions are evaluated by using harmonic components through Fourier analysis. Nonlinear devices are present at all voltage levels of the power system, thus leading to an expected level of background voltage distortion transferred from the upstream network.

In LV networks, background harmonic distortion from the medium-voltage (MV) network is modeled using a constant harmonic voltage source (Thevenin voltage), as detailed in Table 3 for the LV side of distribution transformers in Germany [19]. Because the delta winding of MV/LV transformers blocks zero-sequence components, and due to limited measurement data, only the most significant harmonics with dominant positive or negative sequences (5th, 7th, 11th, and 13th harmonics) are considered. Higher-order harmonics are excluded from the analysis because they have smaller magnitudes and inconsistent measurement availability in distribution networks.

Table 3. Typical harmonic voltages at the LV side of distribution transformers in Germany (data from ref. [19]).

Distortion Level	Harmonic Voltage ($\underline{U}_h$)							
	$\underline{U}_5$		$\underline{U}_7$		$\underline{U}_{11}$		$\underline{U}_{13}$	
	RMS Mag. (V)	Phase (°)	RMS Mag. (V)	Phase (°)	RMS Mag. (V)	Phase (°)	RMS Mag. (V)	Phase (°)
Low	0.99	170 to 200	0.91	0 to 60	No data	No data	No data	No data
Medium	3.78	170 to 200	2.63	0 to 60	0.82	−10 to 20	0.56	80 to 110
High	7.62	170 to 200	6.11	0 to 60	2.72	−10 to 20	3.17	80 to 110

While specific data on background voltage distortion in industrial LV networks are scarce, data from public LV networks, as reported in Table 3, are used to model the background distortion. This approach is justified because industrial LV networks are often connected to public LV networks. Consequently, data from public networks can reasonably represent the background voltage distortion experienced in industrial settings.

3.2. Six-Pulse Controlled Rectifier

Figure 4 illustrates the general model of a six-pulse controlled rectifier connected to a PCC in an LV network. The main components of the model are as follows:

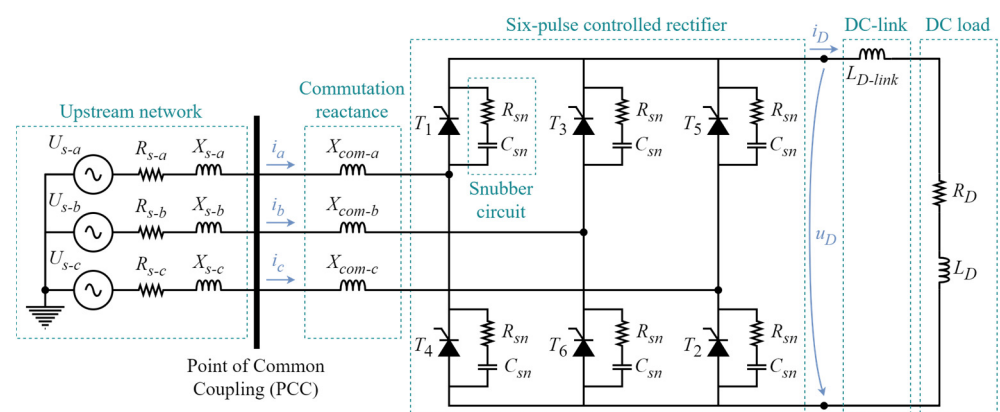


Figure 4. General model of a six-pulse controlled rectifier.

- Upstream network: modeled as a Thevenin equivalent as described in Section 3.1.
- Commutation reactance (X_{com}): the total reactance between the PCC and the power converter at power frequency.

- Six-pulse controlled rectifier: a three-phase full-wave rectifier that uses six sequentially firing Silicon Controlled Rectifiers (SCRs) (T_1 to T_6) arranged in bridge topology. The SCRs draw current in six pulses and produce a variable DC voltage (U_D) controlled by the firing angle. It also includes parallel RC-snubbers to protect SCRs from transients.
- DC smoothing reactor (L_{D-link}): an inductor at the DC-link to control harmonics and reduce ripple in U_D .
- DC load: represented as a resistor (R_D) and inductor (L_D) to model various DC loads like motors, drives, heaters, and welders.

Modeling Key Parameters of Controlled Rectifiers Affecting Voltage Notches

This section explores the key parameters affecting voltage notches in controlled rectifiers, highlighting the importance of modeling and sizing these parameters based on rectifier and DC load requirements and technical constraints of the LV network. For instance, [4,20] provides expressions for the d_{com} and commutation angle (μ) of voltage notches where these dependencies can be theoretically observed according to (2) and (3).

$$d_{com} = \frac{L_s}{2 \cdot (L_{com} + L_s)} \cdot \sqrt{2} \cdot U_s \cdot \sin(\alpha), \quad (2)$$

$$\mu = \alpha - \cos^{-1} \left(\frac{4 \cdot \pi \cdot f_s \cdot I_D \cdot (L_{com} + L_s)}{\sqrt{2} \cdot U_s} + \cos(\alpha) \right), \quad (3)$$

where L_s is the source inductance, L_{com} is the commutation inductance, U_s is the source RMS phase-to-phase voltage, f_s is the fundamental power frequency of the system, and I_D is the output DC current of the rectifier.

To generate realistic synthetic voltage notch waveforms using the scheme depicted in Figures 3 and 4, these key modeling parameters are analyzed below, and relevant expressions are considered and derived:

- Rectifier apparent power: The rated apparent power (S_r) and line current (I_s) are related by (4)–(6).

$$S_r = \sqrt{3} \cdot U_s \cdot I_s, \quad (4)$$

$$I_s = \sqrt{\frac{2}{3}} \cdot I_D. \quad (5)$$

$$S_r = \sqrt{2} \cdot U_s \cdot I_D. \quad (6)$$

- DC load: The DC load model includes an inductor (L_D) with typical values for DC motors provided in Table 4.

Table 4. Typical armature inductance values for DC motors in industrial LV applications.

Rated Power	Typical Armature Inductance
Up to 10 kW	From 15 mH to 100 mH
From 10 kW to 100 kW	From 5 mH to 15 mH
From 100 kW to 250 kW	From 1 mH to 5 mH
Above 250 kW	Up to 1 mH

- Commutation reactance: The commutation reactance (X_{com}) is inversely related to d_{com} , and directly related to Δt_{com} , as expressed in (7) to (10) [14].

$$X_{com} = u_{com} \cdot \frac{U_s^2}{S_r}, \quad (7)$$

where u_{com} is the relative short-circuit voltage given by (8).

$$u_{com} = \left[\frac{\sqrt{3}}{2 \cdot d_{com}} \cdot \sin(\alpha) - 1 \right] \cdot \frac{S_r}{S_{sc}}. \quad (8)$$

L_{com} is obtained as stated in (9).

$$L_{com} \approx L_s \cdot \left[\frac{\sqrt{3}}{2 \cdot d_{com}} \cdot \sin(\alpha) - 1 \right], \quad (9)$$

where L_s is calculated with (14).

$$L_s = \frac{U_s^2}{2 \cdot \pi \cdot f_s \cdot S_s}. \quad (10)$$

- Firing angle: U_D is controlled by adjusting α as described in (11). The displacement power factor of the rectifier (DPF_r) is related to α according to (12) [20].

$$U_D \approx \frac{3 \cdot \sqrt{2} \cdot U_s}{\pi} \cdot \cos(\alpha) - 6 \cdot f_s \cdot (L_s + L_{com}) \cdot I_D, \quad (11)$$

$$DPF_r \approx \cos\left(\alpha + \frac{\mu}{2}\right). \quad (12)$$

- Snubber circuit: Optimal RC-snubber component values, R_{sn} and C_{sn} , are determined based on the maximum reverse current (I_{rm}) and continuous reverse blocking voltage (U_r) of SCRs. Base values are calculated using Equations (13) and (14) [21].

$$R_{base} = \frac{U_r}{I_{rm}}, \quad (13)$$

$$C_{base} = 2 \cdot (L_s + L_{com}) \cdot \left(\frac{I_{rm}}{U_r} \right)^2. \quad (14)$$

The RC-snubber must maintain safe limits for reverse blocking voltage, with the recommended limit given by $U_{rm} = 0.8 \cdot U_{rrm}$ [22], where U_{rrm} is the maximum repetitive reverse blocking voltage of the thyristor.

A C_{sn} close to C_{base} is advised for optimal performance [21]. The overvoltage ratio U_{rm}/U_r varies with the normalized snubber resistor R_{sn}/R_{base} as shown in Figure 5.

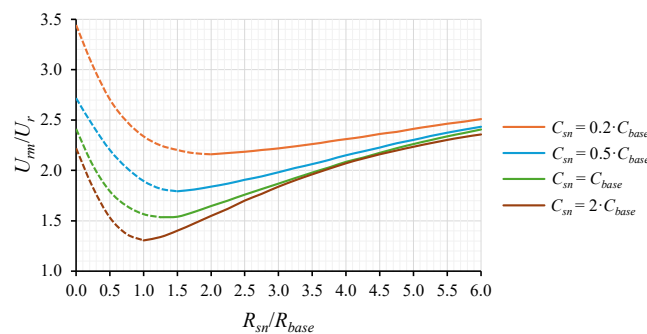


Figure 5. Overvoltage ratio U_{rm}/U_r as a function of the normalized snubber resistor R_{sn}/R_{base} for RC-snubbers. Curves for thyristors used in LV applications (adapted from ref. [21]).

Thyristor reverses recovery current decays exponentially, with a rate $-di/dt$ calculated by (15). The corresponding maximum reverse current I_{rm} is shown in Figure 6.

$$-di/dt = \frac{U_r}{2 \cdot (L_s + L_{com})}. \quad (15)$$

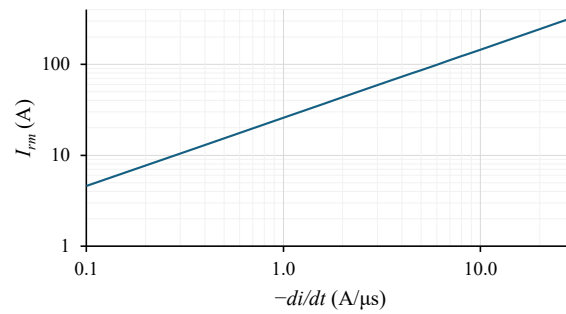


Figure 6. I_{rm} versus $-di/dt$ for thyristors in LV applications (adapted from ref. [23]).

For six-pulse rectifiers, the equivalent RC-snubber values observed by each thyristor are expressed in (16) and (17) [21].

$$R_{eq,sn} = \frac{3}{5} R_{sn}, \quad (16)$$

$$C_{eq,sn} = \frac{5}{3} C_{sn}. \quad (17)$$

- DC smoothing reactor: The smoothing reactor (L_{D-link}) is designed to smooth the output voltage and minimize discontinuous conduction according to (18) [24].

$$L_{D-link} = 0.296 \cdot 10^{-3} \cdot \sin(\alpha) \cdot I_D. \quad (18)$$

3.3. Linear Loads

The simulation scheme depicted in Figure 3 employs a three-phase symmetrical parallel RL load in a wye connection to represent aggregated linear loads in LV installations. This model reflects typical industrial loads, such as motors and lighting, which generally have a lagging power factor [25]. The model parameters include active power (P_{ll}) and reactive power (Q_{ll}), with power factors usually ranging from 0.8 to 0.95 [25]. The power factor is computed using (19).

$$PF_{ll} = \frac{P_{ll}}{|S_{ll}|} = \frac{P_{ll}}{\sqrt{P_{ll}^2 + Q_{ll}^2}}, \quad (19)$$

where S_{ll} is the apparent power of the aggregated linear loads.

3.4. Capacitor Bank

Capacitor banks are crucial in LV industrial facilities for voltage support and power factor (PF) correction. In the simulation scheme of Figure 3, a capacitor bank can be connected to the PCC. To determine the necessary capacitive reactive power (Q_{cb}) for achieving a new PF, (20) can be used.

$$Q_{cb} = P_{PCC} \left[\tan(\cos^{-1} PF_{existing}) - \tan(\cos^{-1} PF_{target}) \right], \quad (20)$$

where P_{PCC} is the active power at the PCC, $PF_{existing}$ is the current PF, and PF_{target} is the desired PF.

The capacitor bank's connection affects the system voltage, with voltage change (ΔU) calculated by (21). To ensure minimal impact on customer loads, the voltage change is typically constrained to around 2% to 3% for distribution networks [26].

$$\Delta U = \frac{Q_{cb}}{S_s}. \quad (21)$$

Capacitor banks may cause resonance, amplifying harmonics. The resonance frequency (f_{res}) is given by (22).

$$f_{res} = f_s \sqrt{\frac{S_s}{Q_{cb}}}, \quad (22)$$

Detuned reactors are added to prevent harmonic resonance. The tuning frequency (f_t) for detuned capacitor banks is expressed in (23).

$$f_t = h \cdot f_s = \frac{1}{2 \cdot \pi \cdot \sqrt{L_{cb} \cdot C_{cb}}}, \quad (23)$$

where f_s is the fundamental power frequency, h is the harmonic order, and L_{cb} and C_{cb} are the reactor inductance and capacitor bank capacitance, respectively. Typical h values range from 4.5 to 4.85.

Detuned capacitor banks include a resistor (R_{cb}) to dampen resonance and broaden the filter's bandwidth [27,28]. The quality factor (QF) measures damping and bandwidth are expressed in (24).

$$QF = \frac{X_{L,f_t}}{R_{cb}} = \frac{X_{C,f_t}}{R_{cb}}, \quad (24)$$

where $X_{L,f_t} = 2 \cdot \pi \cdot f_t \cdot L_{cb}$, and $X_{C,f_t} = 1 / (2 \cdot \pi \cdot f_t \cdot C_{cb})$.

Typical values of QF are categorized as follows [27–31]:

- Low: 5–20 (broader bandwidth and moderate attenuation).
- Medium: 20–50 (balanced attenuation and bandwidth).
- High: 50–100 (high attenuation and narrower bandwidth).

4. Analysis of Key Parameter Dependencies, Limitations, and Impacts

4.1. Dependencies and Limitations of Key Parameters

To generate realistic voltage notch waveforms in simulations, it is crucial to design, size, and model key parameters to closely reflect real-life LV network conditions. This requires analyzing the dependencies between parameters and considering their limitations, alongside external factors like standard limits, construction details, technological limitations, and technical constraints. Key points and dependencies observed in the conceptual diagram of Figure 7 are described as follows:

- S_{sc} : Determined by external factors such as upstream network characteristics provided by the distribution network operator.
- S_r : Limited by the ratio S_r/S_{sc} to ensure proper operation.
- X_{com} : Sized according to specified expressions to prevent excessive voltage notches.
- RC-snubber: Designed based on X_{com} , $-di/dt$, and U_{rrm} of the thyristors.
- α : Constrained by the minimum allowed or target power factor (PF) at the PCC.
- U_D and I_D : Depend on α , U_s , and other parameters.
- P_D and R_D : Computed from I_D and U_D values, with L_D derived from P_D .
- Linear Loads: The maximum apparent power ($P_{ll} + jQ_{ll}$) is constrained by the maximum allowable voltage drop at the PCC, which depends on S_{sc} and S_r . The PF of the LV installation depends on both α and the PF of linear loads.
- Q_{cb} : Improves PF and allows for a broader range of α . The maximum Q_{cb} is constrained by the voltage rise at the PCC, typically managed between 2% and 3%.
- Detuned capacitor bank: L_{cb} depends on Q_{cb} and the design f_t , while R_{cb} depends on Q_{cb} and QF .
- Non-detuned capacitor bank: P_{cb} is expressed as a percentage of Q_{cb} .

Other network parameters, such as background voltage unbalance and distortion, also impact notch waveforms and are therefore considered in the simulations. However, their dependency on the other key parameters is assumed negligible.

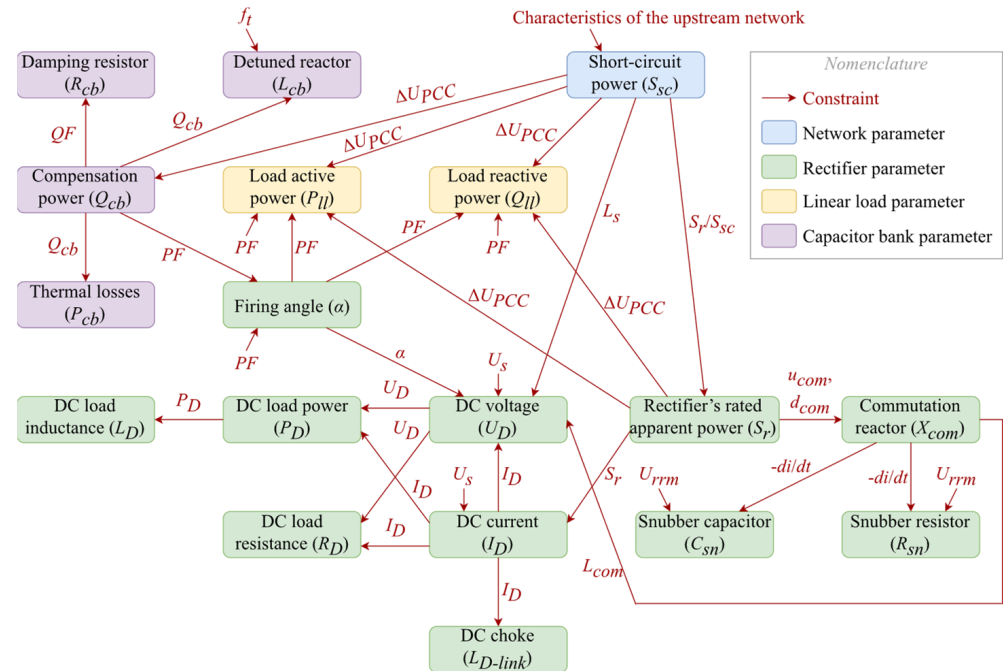


Figure 7. Conceptual diagram illustrating the interrelationships among key parameters.

4.2. Impact of Key Parameters on Voltage Notch Waveforms

To analyze the impact of key parameters on voltage notch waveforms, five scenarios are simulated using the scheme outlined in Figure 3, incorporating various components as described in Table 5. The six-pulse controlled rectifier is enabled in all scenarios, as it is the source of voltage notches. Simulations for each scenario are conducted under rated conditions of $U_s = 400$ V and $f_s = 50$ Hz. These values are chosen to align with available voltage notch measurements from German LV networks, though the simulation platform allows flexibility to modify these parameters for different rated conditions.

Table 5. Scenarios for analyzing the impact of key parameters on voltage notch waveforms through simulations.

Scenario	Component Incorporated in the Simulation				
	Six-Pulse Controlled Rectifier		Linear Loads	Capacitor Bank	Background Unbalance and Distortion
	Snubber Disabled	Snubber Enabled			
0	✓				
1		✓			
2		✓	✓		
3		✓		✓	
4		✓	✓	✓	
5		✓	✓	✓	✓

4.2.1. Scenario 0: S_{sc} , S_r , X_{com} , and α

This scenario considers only the six-pulse controlled rectifier in Figure 3, neglecting snubber circuits ($R_{sn} = 1$ M Ω , $C_{sn} = \infty$), yielding classical rectangular voltage notches for qualitative sensitivity analysis of the key parameters influencing voltage notch waveforms. The results of the simulations are depicted in Figure 8.

- Short-circuit power at the PCC (S_{sc}) (Figure 8a):
 - Simulations vary S_{sc} from 760 kVA to 5 MVA.
 - Fixed parameters are $S_r = 11.4$ kVA, $\alpha = 30^\circ$, $X_{com} = 245.3$ m Ω , and $I_D = 20.15$ A.

- Increasing S_{sc} decreases d_{com} , indicating a weaker PCC leads to more severe voltage notches.
- Rectifier's rated apparent power (S_r) (Figure 8b):
 - Simulations vary S_r from 10 kVA to 75 kVA with $S_{sc} = 5$ MVA.
 - Fixed parameters are $\alpha = 30^\circ$ and $X_{com} = 37.3$ m Ω .
 - Increasing S_r increases Δt_{com} (and line currents), resulting in more severe notches.
- Commutation reactance (X_{com}) (Figure 8c):
 - Simulations vary X_{com} from 37.3 m Ω to 111.8 m Ω
 - Fixed parameters are $S_{sc} = 5$ MVA, $S_r = 25$ kVA, and $\alpha = 30^\circ$.
 - Increasing X_{com} decreases d_{com} but increases Δt_{com} without affecting the voltage notch area significantly. It is therefore designed to prevent deep voltage notches while maintaining a trade-off with their duration.
- Firing angle (α) (Figure 8d):
 - Simulations vary α from 10° to 31°
 - Fixed parameters are $S_{sc} = 5$ MVA and $S_r = 25$ kVA.
 - Increasing α increases d_{com} and decreases Δt_{com} . It affects line current commutation time and phase angle of the fundamental component, directly impacting the DPF.

These simulations help understand how varying key parameters such as S_{sc} , S_r , X_{com} , and α influence voltage notch waveforms and line currents.

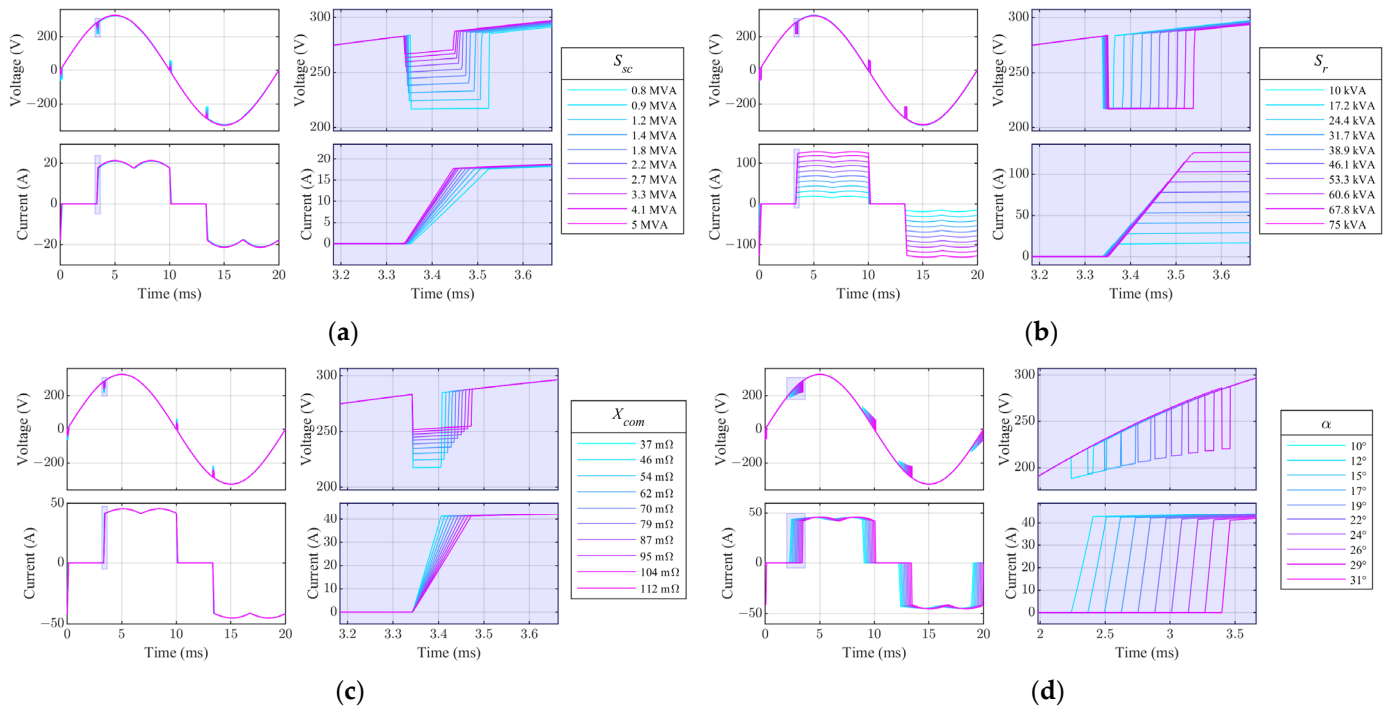


Figure 8. Impact of (a) S_{sc} , (b) S_r , (c) X_{com} , and (d) α on voltage notch waveforms (top) and input line currents (bottom).

4.2.2. Scenario 1: RC-Snubbers

This scenario examines the impact of RC-snubber circuits on commutation oscillations by varying both the snubber capacitor (C_{sn}) and the snubber resistor (R_{sn}).

- RC-Snubber (Figure 9a):
 - Simulations vary C_{sn} at $0.2 \cdot C_{base}$, $0.5 \cdot C_{base}$, C_{base} , and $2 \cdot C_{base}$.

- Fixed values include $S_{sc} = 5$ MVA, $S_r = 25$ kVA, $X_{com} = 37.3$ m Ω , $\alpha = 30^\circ$, $R_D = 10.45$ Ω , $U_{rrm} = 1400$ V, and $U_{rm} = 1120$ V.
- Increasing C_{sn} decreases the peak and frequency of commutation oscillations. The corresponding R_{sn} values are selected as optimal for each C_{sn} . Line currents increase in both peak magnitude and duration with increasing C_{sn} .
- Snubber resistor (R_{sn}) (Figure 9b):
 - Simulations vary R_{sn} within a range derived from the design curve (R_{sn}/R_{base} from 1.25 to 6).
 - Fixed values are $S_{sc} = 5$ MVA, $S_r = 25$ kVA, $X_{com} = 37.3$ m Ω , $\alpha = 30^\circ$, $R_D = 10.45$ Ω , and optimal $C_{sn} = 0.70$ μ F.
 - R_{sn} provides damping for commutation oscillations. Higher R_{sn} values reduce the snubber circuit's efficiency, limiting its significant impact within the suitable range. Current waveforms show slight damping.

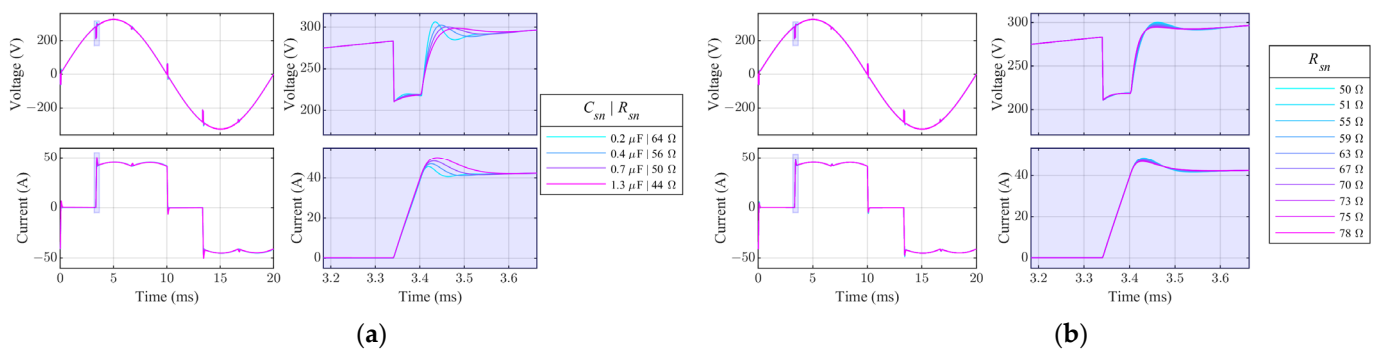


Figure 9. Impact of (a) RC-snubber and (b) R_{sn} on voltage notch waveforms (top) and input line currents (bottom).

4.2.3. Scenario 2: Rectifier and Linear Loads

This scenario examines the effect of aggregated linear loads on voltage notch waveforms while using a six-pulse rectifier with snubbers enabled.

- Apparent power of linear loads (S_{ll}) (Figure 10):
 - Simulations vary S_{ll} from 5 kVA to 170 kVA in ten steps, with $S_{ll,max}$ ensuring a PF above 0.85 and a voltage drop less than 5% at the PCC.
 - Fixed values include $S_{sc} = 5$ MVA, $S_r = 25$ kVA, $\alpha = 30^\circ$, $X_{com} = 37.3$ m Ω , $R_D = 10.45$ Ω , $R_{sn} = 49.58$ Ω , $C_{sn} = 0.70$ μ F (optimal RC-snubber), and $PF_{ll} = 0.9$.
 - Voltage waveforms (Figure 10, top) show that increasing S_{ll} smooths the voltage notch transitions and reduces oscillations. d_{com} slightly decreases, and the fundamental voltage drops linearly.
 - Line currents of the rectifier (Figure 10, middle) demonstrate that with increasing S_{ll} , commutation is slightly delayed. The line currents become more sinusoidal, reflecting improved commutation and increased linear load proportion.
 - Total line currents of LV installation (Figure 10, bottom) show improved sinusoidal quality due to the increased linear load and smoother commutations.

The simulations demonstrate that adding linear loads helps dampen voltage notches and smooth the transition segments, enhancing overall waveform quality while maintaining technical constraints.

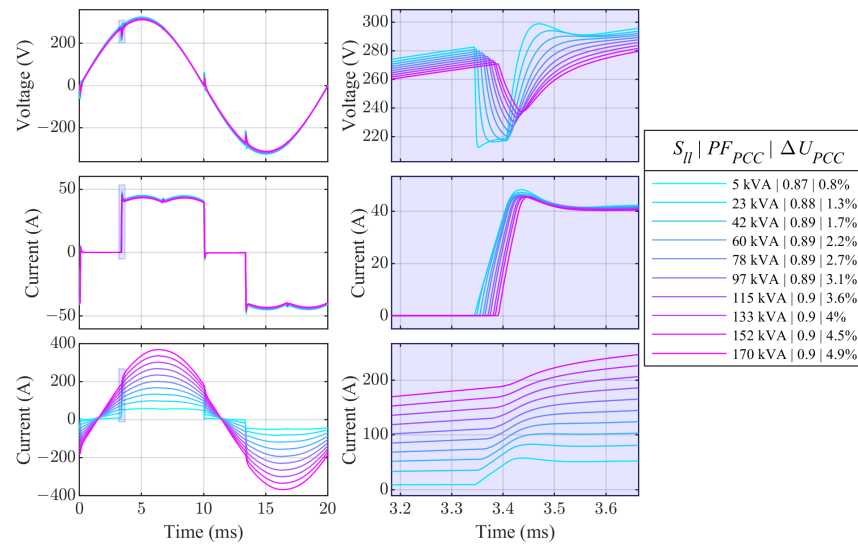


Figure 10. Impact of $S_{II} = P_{II} + jQ_{II}$, with a fixed $PF_{II} = 0.9$, on voltage notch waveforms (**top**), input line currents of the rectifier (**middle**), and total line currents of the LV installation (**bottom**).

4.2.4. Scenario 3: Rectifier and Capacitor Bank

In this scenario, a capacitor bank is added to the six-pulse rectifier, disabling linear loads. The goal is to maintain a PF above the desired limits ($PF > 0.95$) while allowing a broad range of firing angles (α).

- Compensation power of the capacitor bank (Q_{cb}) (Figure 11):
 - Q_{cb} varies from 50.1 kVar to 60.7 kVar. Two configurations are considered: non-detuned and detuned capacitor banks.
 - The fixed values are $S_{sc} = 5$ MVA, $S_r = 75$ kVA, $\alpha = 60^\circ$, $X_{com} = 106$ m Ω , $R_D = 1.88$ Ω , $R_{sn} = 83.18$ Ω , and $C_{sn} = 0.50$ μ F.

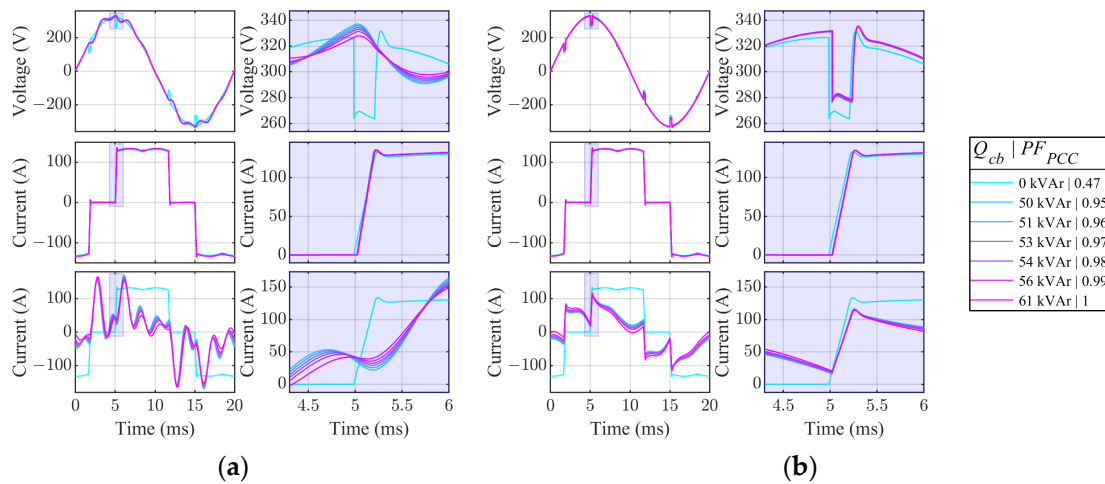


Figure 11. Impact of the capacitor bank Q_{cb} on voltage notch waveforms (**top**), input line currents of the rectifier (**middle**), and total line currents of the LV installation (**bottom**). (a) Non-detuned capacitor bank and (b) detuned capacitor bank.

- Non-detuned capacitor bank (Figure 11a):
 - Voltage waveforms are heavily distorted with indistinct notches, indicating excitation of the network's natural frequencies.
 - Total line currents are also significantly distorted.
 - Increasing Q_{cb} within the range slightly smooths the waveforms.

- Detuned capacitor bank (Figure 11b):
 - Notches are clear and distortion-free, as natural frequencies are not excited.
 - Q_{cb} variation within the range shows no significant impact.
 - Provides fundamental voltage support and slightly reduces notch depth.
 - Commutation oscillations are filtered by the detuning reactor.

The analysis shows that detuned capacitor banks effectively mitigate voltage distortions and maintain clear voltage notches, providing a more stable and efficient operation compared to non-detuned capacitor banks.

4.2.5. Scenario 4: All Components

In this scenario, all components are connected: the six-pulse rectifier, aggregated linear loads, and capacitor bank. The simulations examine the combined effects on voltage notch waveforms, considering both non-detuned and detuned capacitor banks.

- Firing angle (α) (Figure 12):
 - α varies from 30° to 85° . Two configurations are considered: non-detuned and detuned capacitor banks.
 - Fixed values from previous simulations are used. The corresponding R_D is calculated to maintain the full load operation of the rectifier. The target $PF = 0.95$ at the PCC is maintained by adjusting Q_{cb} .
- Non-detuned capacitor bank (Figure 12a):
 - Voltage and total current waveforms are heavily distorted due to the excitation of natural frequencies.
 - The presence of linear loads slightly smooths the distortion compared to previous results without linear loads.
- Detuned capacitor bank (Figure 12b):
 - Voltage and current waveforms, especially voltage notch waveforms, are smoother compared to non-detuned cases.
 - The required Q_{cb} is higher than in previous scenarios to maintain the target PF .
 - The rectifier operates over a wider range of α while maintaining the target PF .

The results indicate that integrating all components, including a detuned capacitor bank, results in smoother voltage and current waveforms and allows the rectifier to operate efficiently over a broader range of firing angles.

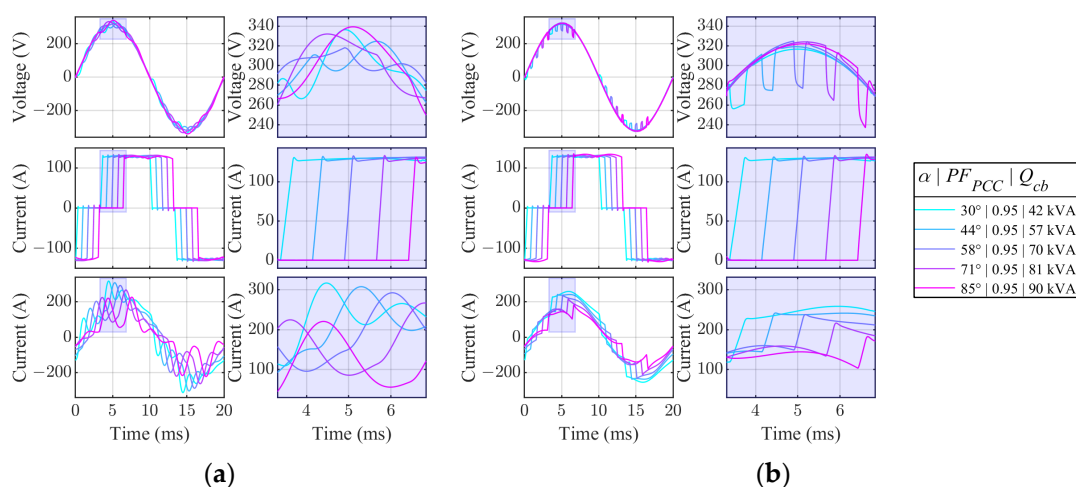


Figure 12. Impact of α , with linear loads enabled, while maintaining a target PF at the PCC by adjusting Q_{cb} on voltage notch waveforms (**top**), input line currents of the rectifier (**middle**), and total line currents (**bottom**). (a) Non-detuned capacitor bank and (b) detuned capacitor bank.

4.2.6. Scenario 5: Background Voltage Unbalance and Distortion

This scenario examines the impact of background voltage unbalance and distortion on voltage and current waveforms with all components connected, including the rectifier, linear loads, and capacitor bank.

- Background voltage unbalance (Figure 13):
 - Background unbalance is increased up to 2%, based on the IEC 61000-2-2 standard.
 - Both non-detuned (Figure 13a) and detuned (Figure 13b) capacitor banks are analyzed. Fixed parameters from previous simulations are used, with $\alpha = 60^\circ$.
 - Voltage waveforms show slight changes with background unbalance levels of 0%, 0.5%, 1%, and 2% for both capacitor banks.
 - Line currents exhibit no observable changes.
- Background voltage distortion (Figure 14):
 - The background distortion level is varied according to Table 4.
 - Simulations use fixed parameters from previous sets. Fixed phase angles for 5th, 7th, 11th, and 13th harmonic voltages are set to 185° , 30° , 5° , and 95° , respectively.
 - Voltage waveforms significantly change with increased background distortion, resulting in deeper notches and flat-top distortion at higher distortion levels.
 - When using a detuned capacitor bank, these effects are more pronounced.
 - Input currents in the rectifier are not significantly impacted.
 - Total line currents exhibit higher oscillations as distortion levels increase, likely due to more distorted currents drawn by the capacitor bank.

The results indicate that while background voltage unbalance has minimal impact, background voltage distortion significantly affects voltage waveforms and total line currents, particularly when a detuned capacitor bank is used.

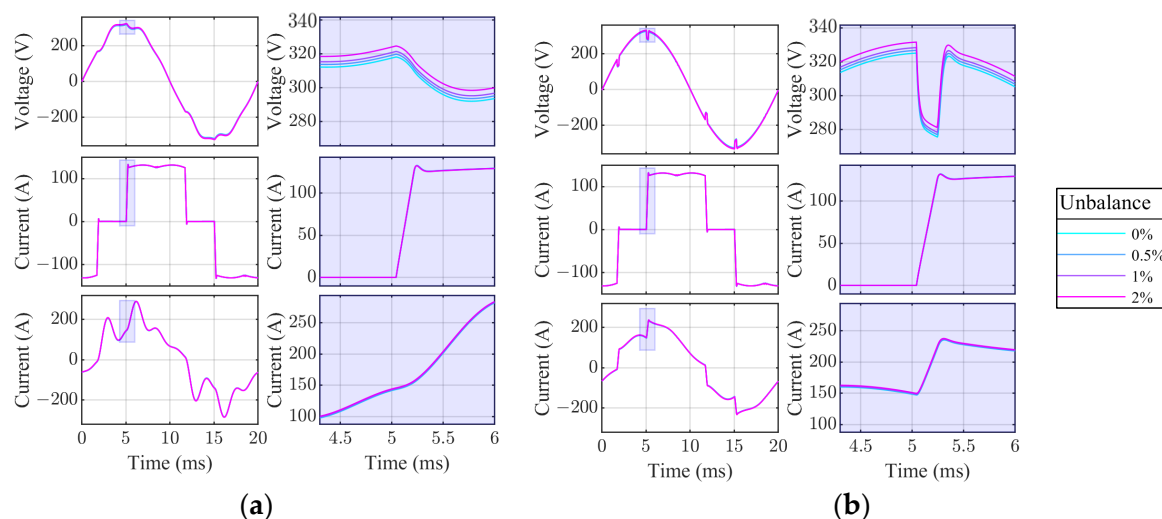


Figure 13. Impact of background unbalance on voltage notch waveforms (top), input line currents of the rectifier (middle), and total line currents of the LV installation (bottom). (a) Non-detuned capacitor bank and (b) detuned capacitor bank.

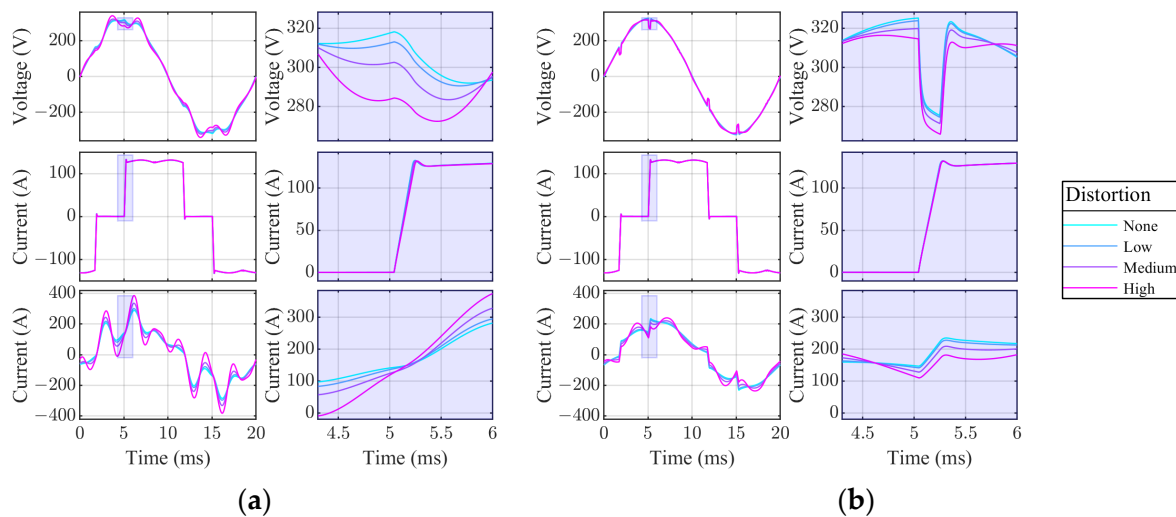


Figure 14. Impact of background distortion on voltage notch waveforms (**top**), input line currents of the rectifier (**middle**), and total line currents of the LV installation (**bottom**). (a) Non-detuned capacitor bank and (b) detuned capacitor bank.

5. Comparison of Simulations with Field Measurements

Synthetic voltage notch and current waveforms generated using the simulation platform are compared with field measurements to validate the platform's capability in emulating realistic conditions and waveforms. In addition, the analysis of the field measurement dataset provides key insights for proposing the Monte Carlo simulation platform. The approach to perform this validation is described below.

5.1. Selection of Field Data

Validation of the simulation platform is based on measurements of three-phase current and voltage waveforms with notches from a single industrial LV network installation in Germany. It is acknowledged that utilizing data from only one installation may limit the generalizability of the validation. However, due to data availability constraints, this installation provided the most comprehensive set of measurements over an extended period, capturing various operating conditions of a six-pulse controlled rectifier.

A set of 164 records from the industrial site are selected for analysis. The sampling frequency of voltage and current waveforms is 20.48 kHz. No additional data beyond the rated phase-to-phase voltage ($U_s = 400$ V), the rated frequency ($f_s = 50$ Hz), and the voltage and current waveforms are available; for instance, short-circuit power and network topology are unknown. Therefore, the first challenge consists of identifying a plausible network topology of the industrial site based on knowledge acquired from simulations.

Analysis of the field measurements indicates that the operating point of the rectifier does not vary significantly over time. This is likely due to the rectifier serving a specific, consistent load in the industrial process. Consequently, the variability in the measured voltage and current waveforms is limited, which presents a challenge for validating the simulation platform across a wide range of operating conditions.

5.2. Simulation Setup and Waveform Comparison

Through extensive analysis of voltage and current waveforms from simulations in Section 4, it was observed that Scenario 3 from Table 5, with a detuned capacitor bank as depicted in the network topology of Figure 15, matches the behavior of voltage and current waveforms from the field.

In setting up the simulation, a challenge was faced due to unknown specific parameters of the installation, such as the exact short-circuit power at the PCC and the rated power of the rectifier. While these parameters would typically remain constant for a given installation, they were varied within plausible ranges in the simulations to account for uncertainties

and to explore the sensitivity of the voltage notch characteristics to these parameters. This approach allowed a closer match to the observed waveforms to be achieved despite the lack of detailed installation data.

Unknown parameters in Figure 15 can be roughly estimated by closely examining field measurements. The resulting estimated parameters are then used to set up the simulation platform to obtain similar synthetic voltage and current waveforms. At first glance, α can be directly estimated by observing the Point-On-Wave (POW) of voltage notches. Likewise, the fundamental component of current in the field measurements indicates a possible range for the rated current drawn by the rectifier and linear load, and the corresponding values of S_r and S_{ll} . Significant values of d_{com} observed in the voltage measurements suggest that the recommended value of 0.015 in the ratio S_r/S_{sc} is exceeded or that the commutation reactor X_{com} is not sized according to technical recommendations. In this case, proper sizing of X_{com} is assumed for $d_{com} = 0.4$ (slightly below the recommended limit for dedicated systems, see Table 1), with $S_r/S_{sc} > 0.015$. Moreover, the phase shift between fundamental voltage and current indicates the necessity of a capacitor bank in the installation to prevent a very poor PF , probably with a detuning reactor, as natural frequencies seem not to be excited.

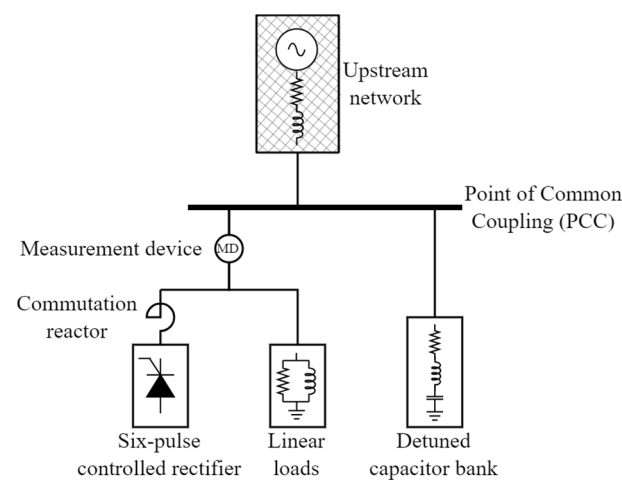


Figure 15. Case study from field measurements.

Although the short-circuit power at a particular installation would not change significantly over a short time frame, this parameter was adjusted in the simulations to account for possible variations in the network conditions that were not captured in the measurements, such as changes in upstream network impedance due to network reconfigurations or varying load conditions elsewhere in the network.

Ideally, the rated power of the rectifier should be fixed for a single installation. However, due to the absence of specific data regarding the rectifier's rated power, a range of values was simulated to find the best fit with the measured waveforms. This was necessary to compensate for the unknown actual loading conditions and potential deviations in the rectifier's performance over time.

Through this general analysis and considering the single field measurement record depicted in Figure 16 (left), a similar result is obtained from a simulation, as illustrated in Figure 16 (right), with parameters $S_{sc} = 4$ MVA, $S_r = 160$ kVA, $X_{com} = 44.3$ m Ω , $\alpha = 78^\circ$, $S_{ll} = 19$ kVA, $PF_{ll} = 0.85$, and $Q_{cb} = 120$ kVA. The simulation results show significant qualitative similarities in voltage and current waveforms. The depth and width of notches are very similar, as are the phase shift and shape of current waveforms. However, some apparent differences are observed in the commutation oscillations of both voltages and currents.

Figure 17 presents a more detailed comparison of the voltage and current waveforms from Figure 16. The top row in Figure 17 displays overlaid notches for each phase. These graphs focus on the period immediately surrounding the voltage notch, allowing for a precise comparison of notch characteristics between field measurements and simulations.

The overlaid waveforms demonstrate a high degree of similarity in terms of notch depth, width, and shape across all three phases. Some slight differences are observed in the POW, probably due to a certain level of unbalance in the field measurement record. Similar characteristics in commutation oscillation patterns are observed in phase *c*, whereas phases *a* and *b* show slight differences, suggesting a possible unbalance in network impedance.

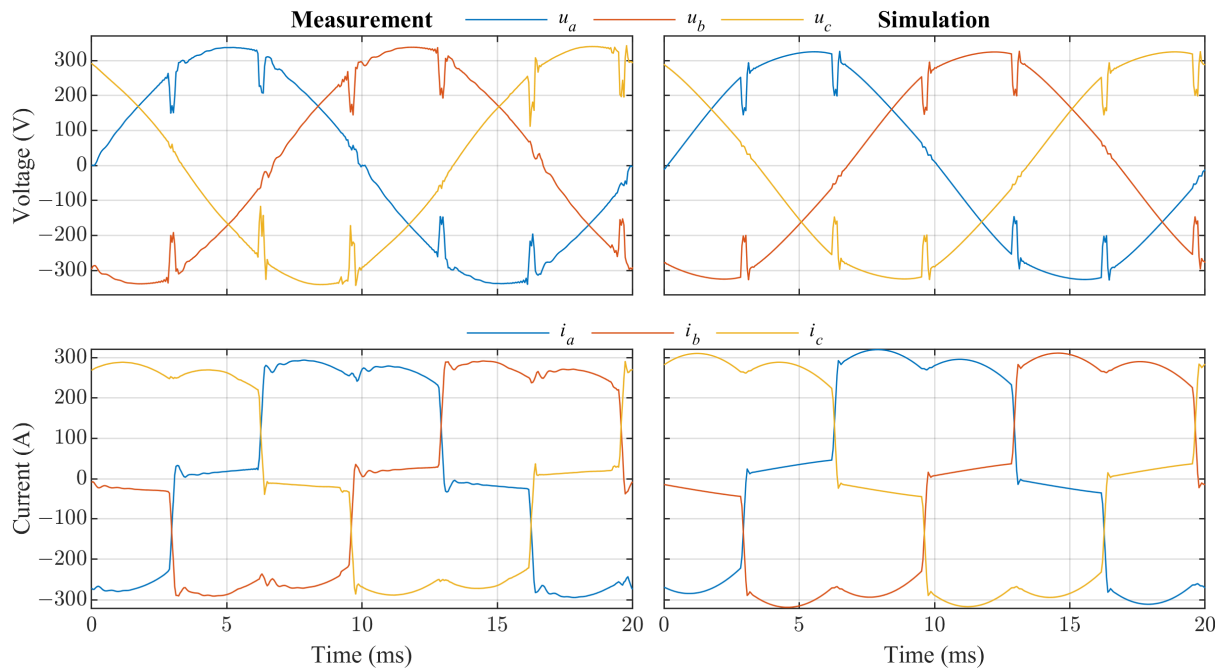


Figure 16. Example of three-phase voltage (top) and current (bottom) waveforms from a field measurement record (left) and a simulation (right).

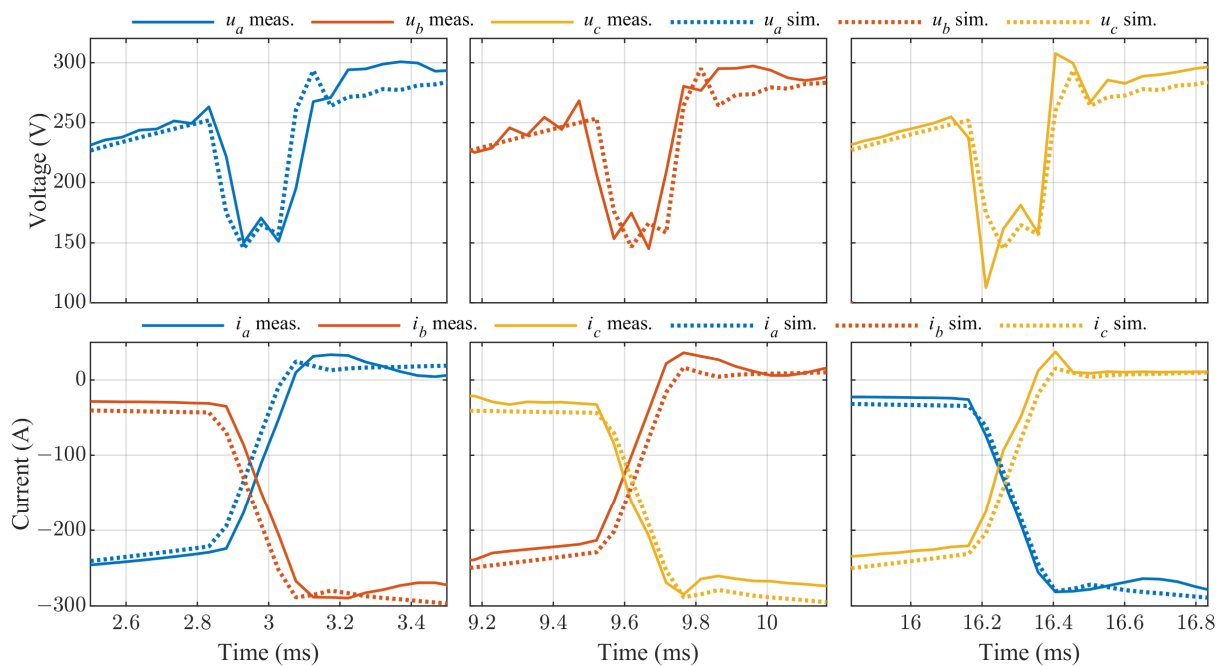


Figure 17. Overlaid voltage notch (top) and current (bottom) waveforms from the example of the field measurement record and simulation in Figure 16 for visual comparison. Phase *a* (left), phase *b* (middle), and phase *c* (right).

The bottom row in Figure 17 shows the corresponding current waveforms for each phase over the period surrounding the voltage notch. These graphs allow for a comprehensive comparison of current behavior, including the shape of the waveform, peak values, and timing of current pulses. The simulated current waveforms show good agreement with field measurements in terms of pulse timing and general shape. However, some minor differences in POW and commutation oscillations are noticeable.

Overall, Figure 17 provides visual evidence of the simulation platform's capability to generate synthetic waveforms that closely resemble real-world measurements. This comparison supports the validity of the simulation approach for studying voltage notches.

While the simulation approach involves varying parameters that are typically constant in a real-world single installation, this methodology was adopted due to the limited availability of detailed installation data. The primary goal was to demonstrate the simulation platform's capability to replicate the observed voltage notch phenomena under plausible operating conditions. It is acknowledged that this introduces limitations to the validation. Future work should aim to include measurements from multiple installations with known parameters to provide a more robust validation of the simulation platform.

5.3. Quantitative Validation

A more comprehensive comparison of field measurements with simulations is proposed through an approach for generating a similar set of synthetic waveforms. To facilitate quantitative characterization and subsequent comparison, a set of features is introduced to characterize voltage and current waveforms.

The selection of these features is based on their relevance to the behavior of voltage notches and their alignment with standard definitions and practices in power quality analysis [5]. The proposed features include the following:

- Voltage features:
 - Maximum relative depth of notches in a cycle of phase a : d_{com} (%): This feature represents the severity of voltage notches, as deeper notches can have more significant impacts on equipment performance.
 - Maximum width of notches in a cycle of phase a : Δt_{com} (μ s): The width of the notch indicates the duration of the disturbance, which affects the potential for interference with sensitive equipment.
 - POW of the first notch in a cycle of phase a : POW_{com} ($^\circ$): The POW provides insight into the timing of the notch occurrence within the voltage cycle, which is important for synchronization and timing-sensitive applications.
- Current features:
 - The magnitude of the fundamental component of current in a cycle of phase a : I_1 (A): The fundamental current magnitude is a key parameter in assessing the loading conditions and the power drawn by the rectifier.
 - Phase of the fundamental component of current in a cycle of phase a : angle (I_1) ($^\circ$): The phase angle of the current fundamental component relative to the voltage provides information on the displacement power factor, which is critical for power system operation and efficiency.
 - The magnitude of the fifth harmonic of current in a cycle of phase a : I_5 (A): The fifth harmonic is typically the most significant harmonic component in systems with six-pulse rectifiers [4], and it contributes to the overall distortion in the current waveform.

The selection of these features focuses on time-domain characteristics of voltage notches and the most significant frequency-domain component, aligning with standard practices for power quality assessment. Other harmonic orders, e.g., 7th, 11th, 13th, were considered for inclusion; however, preliminary analyses indicated that their additional contribution to the validation process was marginal, given that similar magnitudes of harmonic orders do not necessarily mean similarity in time-domain waveforms. Therefore,

the features selected provide a comprehensive yet efficient set of indices for comparing the essential characteristics of voltage notches in both the simulated and measured data.

This set of six features was extracted from 164 field measurement records. The resulting histograms are illustrated in Figure 18. The histograms derived from the field measurements show limited variability in certain features, reflecting the relatively constant operating point of the rectifier in the installation. This contrasts with the simulations, where a broader range of parameter variations was explored. Despite this, the simulations successfully captured the essential characteristics of the voltage notches and their dependencies on key parameters.

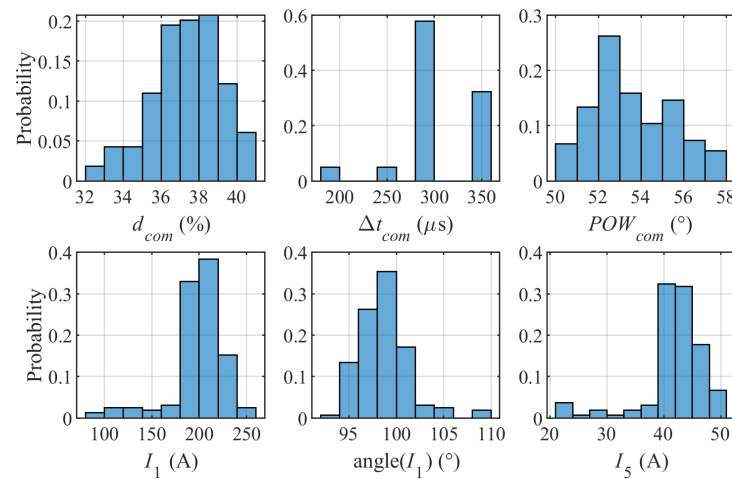


Figure 18. Histograms of features extracted from field measurements.

To obtain a comparable set of simulations, parameters are varied as follows:

- Short-circuit power S_{sc} : varied from 3.7 MVA to 4.3 MVA with uniform distribution.
- Firing angle α : varied from 78° to 84.5° .
- Rectifier's rated apparent power S_r : varied from 72 kVA to 197 kVA.
- Linear load parameters: S_{ll} is randomly varied with a normal distribution ($\mu = 25$ kVA, $\sigma = 2$ kVA), and PF_{ll} is randomly varied with a normal distribution ($\mu = 0.87$, $\sigma = 0.034$).
- RC-snubber: A fixed RC-snubber is designed with optimal values.
- Commutation reactance X_{com} : calculated with $\alpha_{max} = 84.5^{\circ}$ and $d_{com,max} = 0.4$.
- Commutation reactance Q_{cb} : varied in fixed steps of 10 kVA from 50 kVA to 140 kVA, according to the required PF correction.

A total of 164 simulations are performed using these conditions, with the same sampling frequency (20.48 kHz) for comparable results, and the set of features is extracted.

In Figure 19, a matrix of scatter plots and histograms comparing features from field measurements and simulations is presented. Strong similarities in distributions and ranges of features between measurements and simulations are observed. Notably, POW_{com} and I_1 distributions are very similar, as expected due to the use of comparable value combinations. The d_{com} distributions are consistent, with almost identical ranges. While the Δt_{com} range for simulations is found to fall within the slightly wider range of field measurements, its computation is noted to be affected by waveform resolution. Differences in Δt_{com} could be attributed to variations in network impedance or to unmodeled elements, such as parasitic inductances and capacitances, in the actual system. These factors can affect the commutation time and, consequently, the width of the notches observed in the field measurements.

Despite a slight phase shift, the fundamental current phase angle distributions appear similar. This slight phase shift could be due to differences in background harmonic distortion or unmodeled reactive power compensation devices in the field measurements. The simulation model assumes certain levels of background distortion and may not fully

capture the exact conditions present in the real-world system. Finally, strong similarities in I_5 distributions are observed, largely dependent on I_1 behavior.

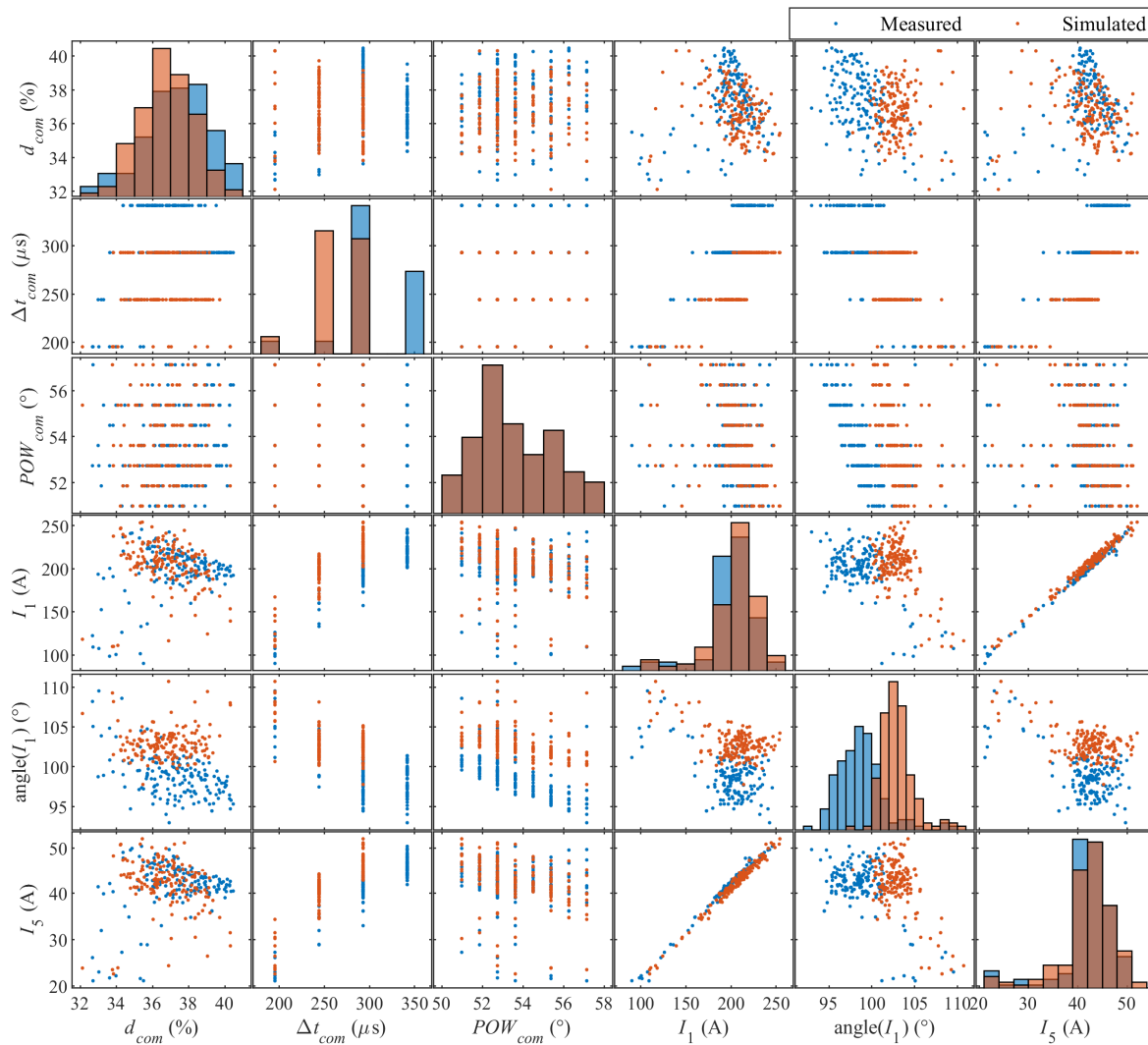


Figure 19. Matrix of scatter plots of features from field measurements and simulations.

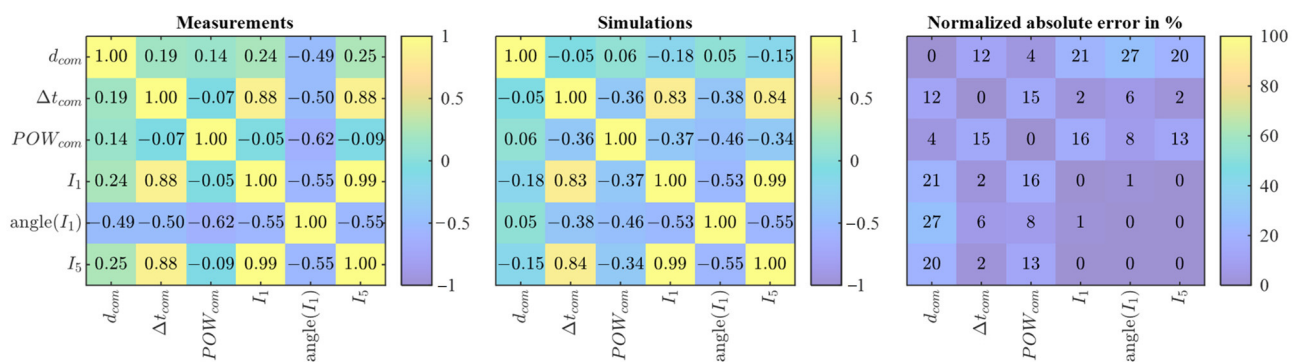
While the field measurements provided limited variability due to the consistent operating conditions of the rectifier, the simulations effectively replicate the observed voltage notch characteristics. The simulation model captures the essential dependencies between key parameters and the resulting voltage and current features, demonstrating its validity in representing real-world behavior under the given constraints.

Statistical measures, including mean and range, for the extracted features are presented in Table 6, along with relative errors. Very low relative errors are observed in the mean of features, with a slightly higher 12.83% in Δt_{com} , which can be attributed to waveform resolution and discretization errors. Range relative errors are also found to be reduced, with the highest errors noted in Δt_{com} and the fundamental current phase angle.

Figure 20 provides further insight into the relationship between features in both field measurements and simulations. The correlation matrices for measurements (left) and simulations (middle) show similar patterns, indicating that the simulation model captures the interdependence between features well. However, discrepancies are observed in the correlations involving d_{com} and Δt_{com} .

Table 6. Statistical metrics comparing features from measurements with those from simulations.

Statistical Metric	Group of Data	Features					
		d_{com} (%)	Δt_{com} (μ s)	POW_{com} ($^{\circ}$)	I_1 (A)	Angle (I_1) ($^{\circ}$)	I_5 (A)
Mean	Measurements	37.32	301.60	53.70	199.11	98.74	41.81
	Simulations	36.74	262.90	53.70	204.87	102.94	42.13
	Relative error (%)	1.55	12.83	0.00	2.89	4.25	0.76
Range	Measurements	7.79	146.48	6.15	155.45	16.59	29.27
	Simulations	8.19	97.66	6.15	144.27	12.97	29.42
	Relative error (%)	5.11	33.33	0.00	7.19	21.83	0.49

**Figure 20.** Correlation matrices of features extracted from field measurements (left) and simulations (middle) and normalized absolute error matrix (right).

From a technical standpoint, a negative correlation between d_{com} and Δt_{com} is expected, as the area of the voltage notch should remain approximately constant, implying that deeper notches should have shorter durations. However, field measurements show a weak positive correlation (0.19), whereas simulations indicate a slightly negative correlation (−0.05). This discrepancy may result from variations in network impedance and unmodeled factors, such as parasitic inductances and capacitances. In addition, uncertainties in accurately determining notch characteristics, potentially caused by disturbances like commutation swings or unbalanced network conditions, can simultaneously affect both the depth and duration of notches, leading to a positive correlation in the field data.

In contrast, the simulation model may not fully capture these unaccounted factors, as it assumes certain parameters to be constant or within specific ranges. In addition, the negative correlation in simulations could be due to the way the parameters were varied independently during the simulations, which might not reflect the exact interdependencies present in the real-world system.

Similarly, discrepancies in the correlations involving angle (I_1) are observed. The field measurements show a moderate negative correlation between angle (I_1) and both d_{com} and Δt_{com} , whereas, in the simulations, the correlations are weaker. This could be due to differences in background harmonic distortion or unmodeled reactive power compensation devices in the field measurements, which can affect the phase angle of the current.

These discrepancies highlight the limitations of the simulation model in capturing all the complexities of the real-world system. However, overall, the low relative errors in most of the statistical metrics indicate that the simulation model provides a reasonable approximation of the behavior observed in the field measurements.

6. Monte Carlo Simulation-Based Data Generation

6.1. Simulation Algorithm

A Monte Carlo simulation approach is proposed for generating realistic synthetic waveforms of voltage notches in industrial LV networks. The algorithm, illustrated in

Figure 21, is based on the theoretical analysis of voltage notches, exploration of parameter interrelations, and validation with field measurements presented in previous sections.

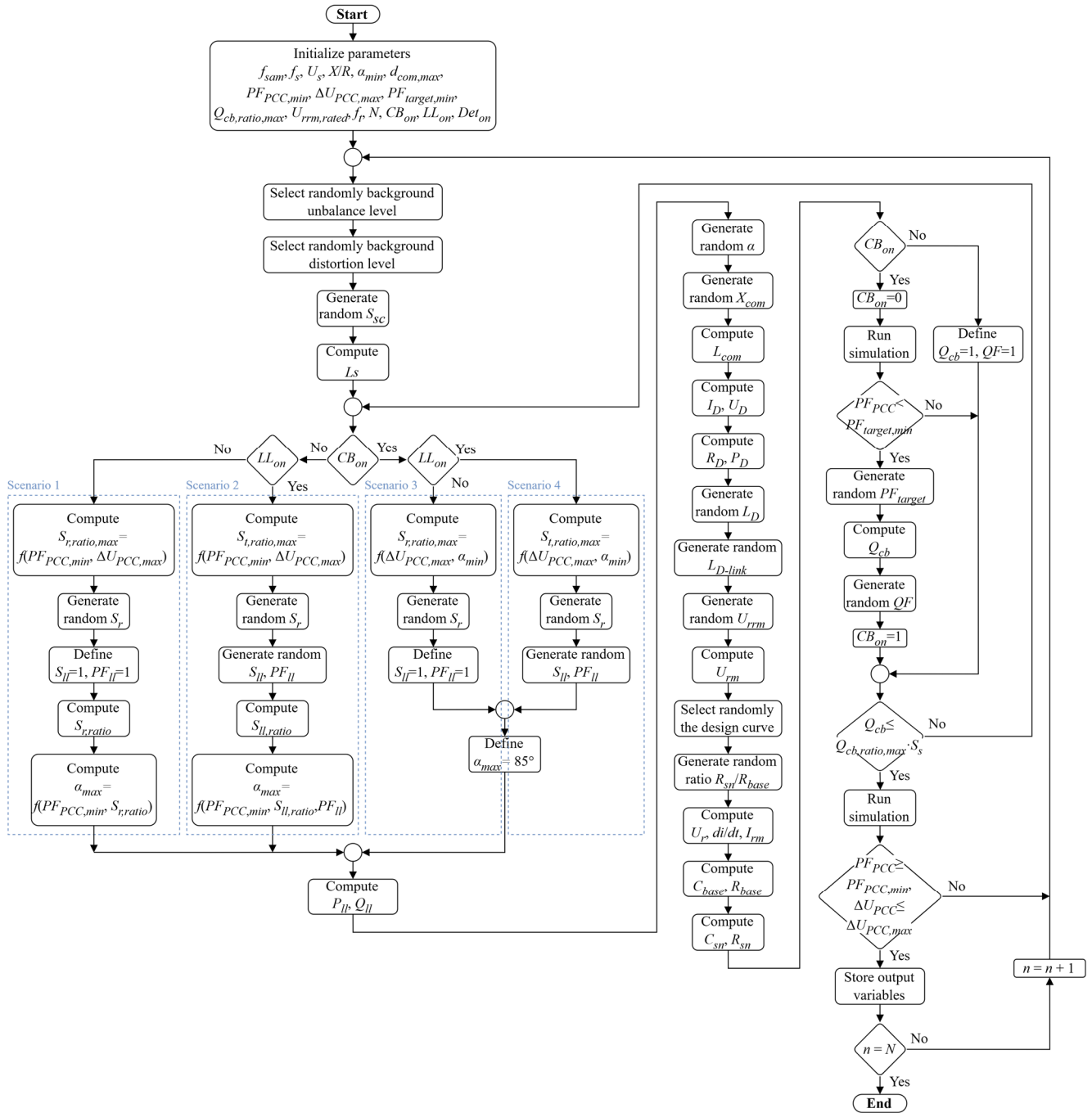


Figure 21. Flow diagram of Monte Carlo simulation-based data generation.

Key features of the algorithm include the following:

1. Initialization: Fixed parameters are set, including the sampling frequency f_{sam} , f_s , U_s , X/R ratio, α_{min} , $d_{com,max}$, $PF_{PCC,min}$, $PF_{target,min}$, $\Delta U_{PCC,max}$, $Q_{cb,ratio,max}$, $U_{rrm,rated}$, f_t , and the number of Monte Carlo simulations (N).
2. Scenario selection: The algorithm allows the simulation of different scenarios by enabling or disabling linear loads (LL_{on}) and the capacitor bank (CB_{on}). Moreover, the capacitor bank can be non-detuned or detuned according to Det_{on} .

3. Random parameter generation: Several system parameters are randomly generated within defined ranges and/or according to a probability distribution, including the background voltage unbalance and distortion, S_{sc} , S_r , α , X_{com} , L_D , snubber circuit parameters (U_{rrm} , R_{sn}/R_{base} ratio), linear load parameters (S_{ll} , PF_{ll}), and QF .
4. Computed parameters: Several parameters are computed according to expressions in Section 3 using other parameters as inputs, as summarized in Table 7.
5. Constraint handling: The algorithm incorporates various constraints and recommendations, such as $PF_{PCC,min}$ and $\Delta U_{PCC,max}$ limits, and $Q_{cb} < Q_{cb,ratio,max} \cdot S_{sc}$.
6. Adaptive parameter determination: Key limit parameters are derived using linearly regressed functions obtained from simulations of scenarios 1–4 (Table 5), including the maximum S_r/S_{sc} ratio ($S_{r,ratio,max}$), maximum $(S_r + S_{ll})/S_{sc}$ ratio ($S_{t,ratio,max}$), and maximum firing angle α_{max} . This approach ensures that most simulation cases avoid constraint violations while maintaining flexibility. Thus, the simulations are not strictly bound to recommended limits, such as the suggested 0.015 ratio for S_r/S_{sc} , as field measurements indicate that these limits are not universally adhered to in real-world scenarios.
7. Iteration and output: The simulation is repeated N times, storing output variables for each valid scenario.

Table 7. Computed parameters and corresponding equations.

Parameter	Equation	Parameter	Equation	Parameter	Equation
L_s	(10)	U_{rm}	$0.8 \cdot U_{rrm}$	R_{sn}	(16)
L_{com}	(9)	U_r	$U_{rm}/f(R_{sn}/R_{base})$; see Figure 5	C_{sn}	(17)
I_D	(5)–(6)	di/dt	(15)	P_{ll}	(19)
U_D	(11)	I_{rm}	$f(di/dt)$; see Figure 6	Q_{ll}	(19)
R_D	U_D/I_D	R_{base}	(13)	Q_{cb}	(20)
P_D	$U_D \cdot I_D$	C_{base}	(14)		

6.2. Case Study: Data Generation for German LV Networks

The Monte Carlo algorithm introduced in Section 6.1 is applied to generate realistic synthetic data of voltage notch waveforms within the context of German industrial LV networks. The fixed parameters are initialized as indicated in Table 8. Furthermore, synthetic data are generated for scenarios 1–4 from Table 6, with variants related to the inclusion of a detuned or non-detuned capacitor bank and different levels of background unbalance and distortion. LL_{on} , CB_{on} , and Det_{on} are set accordingly. A total of $N = 10$ exemplary simulations are run for each scenario and variant.

Table 8. Fixed parameters for the data generation in the case study (German LV networks).

Parameter	Value	Parameter	Value
f_{sam} (kHz)	50	$d_{com,max}$	0.2
f_s (Hz)	50	$PF_{PCC,min}$	0.85
U_s (V)	400	$\Delta U_{PCC,max}$ (%)	5
X/R	1	$U_{rrm,rated}$ (V)	1400
α_{min} (°)	10	f_t (Hz)	235

The flow diagram in Figure 21 shows that some parameters of the simulation are randomly generated, including S_{sc} , S_r , α , X_{com} , L_D , U_{rrm} , selection of the design curve, R_{sn}/R_{base} ratio, S_{ll} , PF_{ll} , and QF . Table 9 indicates the range for each random parameter, the type of probability distribution, and the probability corresponding to the respective range in case various ranges are adopted for the case study. Data on probabilities for S_{sc} obtained from extensive surveys at European LV networks is extracted from [16]. Ranges, types of probability distribution, and range probabilities of the rest of the parameters are computed and chosen according to the procedures and typical values described in Sections 3 and 4.

Table 9. Range, type of distribution, and range probability for random parameters in the case study.

Component	Parameter	Range	Probability Distribution	Range Probability
Network equivalent	S_{sc} (MVA)	[2, 3]	Uniform	0.25
		(3, 10]	Uniform	0.5
		(10, 15]	Uniform	0.25
Rectifier	S_r (kVA)	$[1, S_{r, ratio, max} \cdot S_{sc}]$	Uniform	1
	α (°)	$[10, \alpha_{max}]$	Uniform	1
	X_{com} (Ω)	$\mu = \text{optimal } X_{com}, \sigma = 0.15 \cdot \mu$	Normal	1
	L_D (mH)	[1, 5]	Uniform	1 for PD > 100 kW
		(5, 15]	Uniform	1 for 10 kW < PD ≤ 100 kW
		(15, 100]	Uniform	1 for PD ≤ 10 kW
	U_{rrm} (V)	$[U_{rrm, rated} = 1400, \infty]$	Exponential	1
	Design curve	0.2·Cbase	Uniform	0.1
		0.5·Cbase	Uniform	0.3
		1·Cbase	Uniform	0.5
		2·Cbase	Uniform	0.1
Linear loads	R_{sn}/R_{base}	$[\min R_{sn}/R_{base}, 6]$	Exponential	1
	S_{ll} (kVA)	$[5, S_{ll, max}]$	Uniform	1
	PF_{ll}	[0.8, 0.95]	Uniform	1
Capacitor bank	QF	[5, 20]	Uniform	0.25
		(20, 50]	Uniform	0.5
		(50, 100]	Uniform	0.25

6.3. Results

The Monte Carlo simulation approach introduced in this study generates a diverse range of realistic waveforms for voltage notches and associated currents in industrial LV networks. Figure 22 presents exemplary waveforms obtained from this simulation approach, illustrating various scenarios of increasing complexity.

Figure 22 comprises six subfigures (a–f), each depicting voltage (top) and total line current (bottom) waveforms for a specific network configuration:

- Scenario 1: represents the base case with only the controlled rectifier. The voltage waveform exhibits pronounced notches, while the current shows the characteristic quasi-square wave pattern of a six-pulse rectifier.
- Scenario 2: introduces linear loads alongside the rectifier. Voltage notches remain visible, but the current waveform appears smoother due to the linear load contribution.
- Scenario 3 with non-detuned capacitor bank: incorporates a non-detuned capacitor bank. Notches are not clearly observed given a heavy voltage distortion, and the current waveform displays increased high-frequency components due to capacitor effects.
- Scenario 3 with detuned capacitor bank: similar to (c) but employs a detuned capacitor bank. Voltage notches are clear and distortion-free, and the current waveform exhibits reduced high-frequency oscillations compared to the non-detuned case.
- Scenario 4 with non-detuned capacitor bank: combines the rectifier, linear loads, and a non-detuned capacitor bank. Notches are less prominent, and the current waveform reflects a complex interaction of rectifier, linear load, and capacitor bank characteristics.
- Scenario 4 with detuned capacitor bank: similar to (e) but with a detuned capacitor bank. Voltage notches are clearly observed, and the current waveform depicts some low-frequency oscillations.

These waveforms effectively illustrate the impact of various network configurations and power factor correction methods on voltage notches and current distortion in industrial LV networks. The progression from scenarios (a) to (f) showcases the increasing system

complexity and the corresponding evolution of voltage notch characteristics. This set of simulations provides valuable insights into the diversity of scenarios, conditions, and resulting realistic waveforms that can be obtained from the Monte Carlo simulation platform developed in this paper, facilitating testing/designing strategies for the detection, monitoring, classification, and mitigation of this PQD.

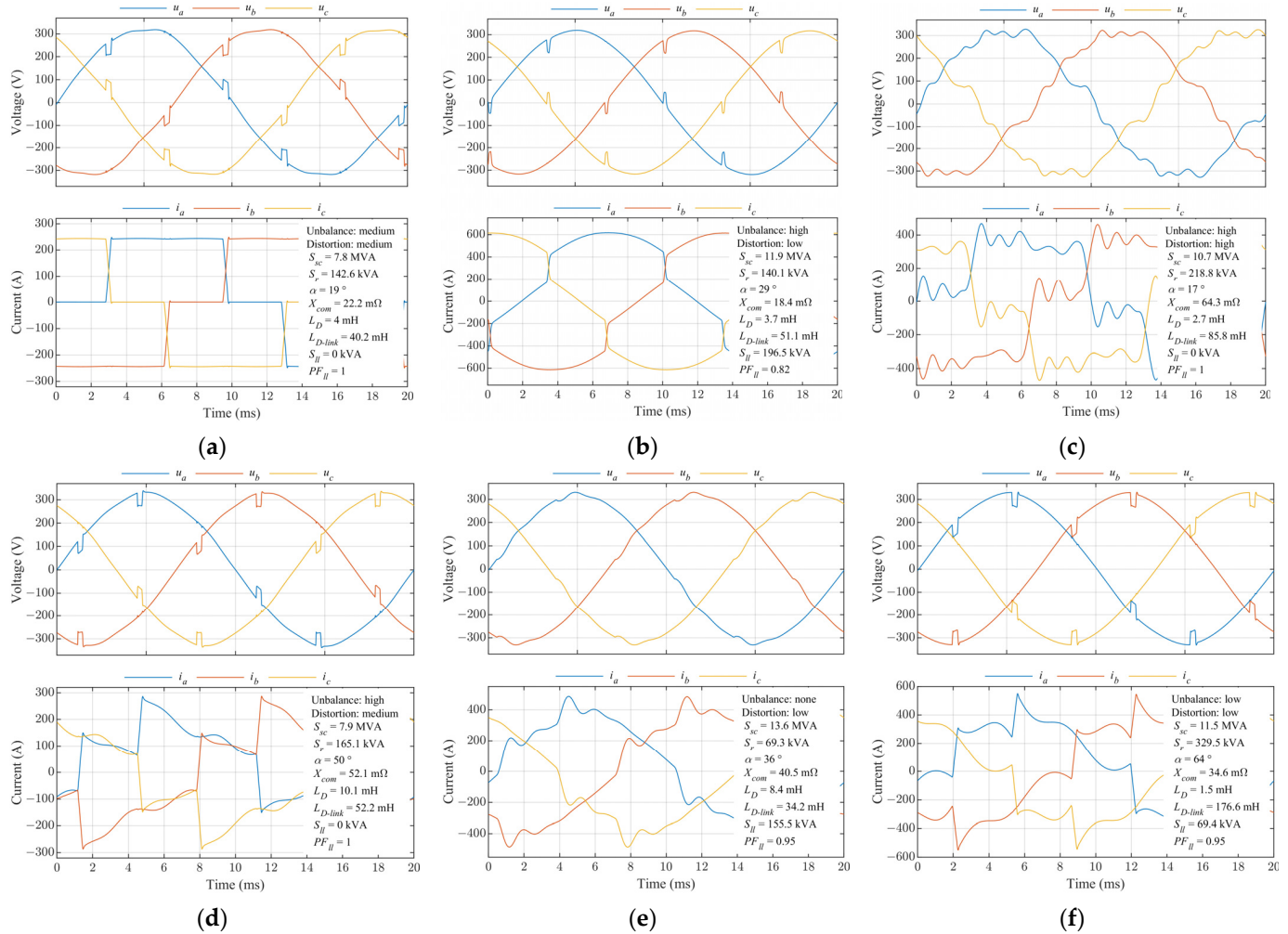


Figure 22. Examples of voltage notch (**top**) and current waveforms drawn by the LV installation (**bottom**) obtained from the Monte Carlo simulation for the case study. (a) Scenario 1, (b) Scenario 2, (c) Scenario 3 with non-detuned capacitor bank, (d) Scenario 3 with detuned capacitor bank, (e) Scenario 4 with non-detuned capacitor bank, and (f) Scenario 4 with detuned capacitor bank.

7. Discussion

The comparison of simulations with field measurements showed strong similarities in the distributions and ranges of key features, such as d_{com} , Δt_{com} , and current harmonic components. Statistical metrics, including mean, range, and relative error, indicated good agreement between the simulated and measured data. The correlation analysis revealed that the simulation model effectively captures the interdependencies among the features, although some discrepancies were observed, particularly involving d_{com} and Δt_{com} .

These discrepancies may be attributed to limitations in the simulation model, such as the assumptions made about network parameters or no consideration of additional impact factors in the field measurements. For instance, variations in network impedance, unaccounted reactive power compensation devices, or differences in background harmonic distortion could influence the observed correlations and feature distributions.

A limitation of the validation process presented in this paper is the limited variability of the field measurements, as only one rectifier in an industrial installation operating at almost steady conditions could be considered. This limitation impacts the generalizability of the results, as the validation is based on a narrow set of conditions. To enhance the robustness and applicability of the simulation platform, future work should involve the extension of the dataset with additional field measurements from different installations including varying operating conditions. This would allow for a more comprehensive validation and increased confidence in the simulation model's performance across different scenarios.

The slight phase shift observed in the fundamental current phase angle between simulations and measurements highlights the importance of accurately modeling all relevant factors, including background harmonic distortion and reactive power compensation devices. Addressing these discrepancies in future enhancements of the simulation platform could further improve its predictive accuracy.

8. Conclusions

This paper introduced a MATLAB/Simulink-based simulation platform for comprehensive analysis of voltage notch characteristics in industrial LV networks under various operating conditions. The platform models a benchmark LV installation, including a six-pulse controlled rectifier, linear loads, and a capacitor bank for power factor correction. Key parameters such as short-circuit power at the point of common coupling, commutation reactance, firing angle, snubber circuits, and the rated power of the rectifier and linear loads were systematically examined. Simulations incorporated background voltage unbalance and distortion to reflect real-world conditions.

Validation against field measurements from German industrial LV networks demonstrated that the simulated waveforms closely resemble actual voltage and current characteristics, confirming the platform's effectiveness in replicating complex system interactions. Employing a Monte Carlo simulation approach allowed for the generation of extensive datasets of synthetic voltage notch waveforms, accounting for dependencies and typical variations of key parameters in industrial settings.

To encourage further research and application, the simulation platform and exemplary synthetic waveforms have been included as Supplementary Material, providing valuable resources for the research community.

The developed simulation platform serves as a valuable tool for studying voltage notch phenomena, facilitating the testing and design of strategies for detection, monitoring, classification, and mitigation of voltage notches. This contributes to improved PQ in industrial LV networks. Insights from the parameter analyses can inform the design and operation of power converters and network components to minimize adverse effects.

Future work may involve incorporating a wider range of network configurations, including additional types of controlled rectifiers such as 12-pulse and 24-pulse converters, as well as exploring the parallel operation of controlled and uncontrolled rectifiers. Moreover, further validation of the simulations using additional field data, with precise knowledge of network configurations and from various regions, could enhance the platform's applicability and robustness.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/app142311031/s1>, Monte Carlo simulation platform in MATLAB/Simulink (R2024a), including the scheme, script, and necessary files: Platform.zip; 60 exemplary synthetic waveforms generated with the proposed platform (in spreadsheet format): Waveforms.zip.

Author Contributions: Conceptualization, J.E.C., O.Z., E.R. and J.M.; methodology, J.E.C., O.Z. and J.M.; software, J.E.C.; validation, J.E.C.; formal analysis, J.E.C.; investigation, J.E.C., O.Z. and J.M.; resources, O.Z. and J.M.; data curation, J.E.C. and O.Z.; writing—original draft preparation, J.E.C.; writing—review and editing, O.Z., E.R. and J.M.; visualization, J.E.C.; supervision, E.R. and J.M.; project administration, E.R.; funding acquisition, J.E.C., E.R. and J.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research was partially supported by the Oficina de Investigaciones (ODI) of the Universidad Distrital Francisco José de Caldas, Bogotá, Colombia, through the research project titled “A methodology for real-time detection and classification of power quality disturbances”.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The raw data derived from the simulations supporting the conclusions of this article will be made available by the authors upon request. The data set of field measurements used in this article is not readily available because of confidentiality agreements.

Acknowledgments: The authors gratefully acknowledge the support of KBR Energy Management GmbH for collecting and sharing extensive current and voltage field data, including voltage notch measurements. These data were essential for validating the simulation platform developed in this study. Their contribution significantly enhanced the quality and reliability of this research.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

A_{com}	Area of voltage notch
A	Firing angle of the controlled rectifier
C_{base}	Base capacitor for snubber design
C_{sn}	Snubber capacitor
d_{com}	Relative depth of voltage notch
di/dt	Current decay during turn-off of thyristor
DPF_r	Displacement power factor of the rectifier
ΔU	Voltage change (drop/rise)
Δt_{com}	Width of voltage notch
f_{res}	Resonance frequency
f_s	Rated frequency of the power system
f_{sam}	Sampling frequency of simulation
f_t	Tuning frequency of the detuned capacitor bank
h	Harmonic order
I_D	Output average DC current of the rectifier
I_{rm}	Maximum reverse current of thyristor
I_s	RMS line current
k_u	Voltage unbalance
L_{com}	Commutation reactor
L_s	Short-circuit inductance
L_D	DC load inductance
L_{D-link}	DC-link choke or smoothing reactor
M	Commutation angle
N	Number of Monte Carlo simulations
P_D	DC load power
P_{ll}	Active power of linear loads
PF_{ll}	Power factor of linear loads
PF_{PCC}	Power factor at the PCC
POW_{com}	Point-On-Wave of first notch
Q_{cb}	Compensation power of the capacitor bank
Q_{ll}	Reactive power of linear loads
QF	Quality factor of the detuned capacitor bank
R_{base}	Base resistor for snubber design
R_D	DC load resistance
R_{sn}	Snubber resistor
S_{ll}	Apparent power of linear loads
S_r	Rated apparent power of the rectifier
S_{sc}	Short-circuit power
u_{com}	Relative short-circuit voltage of the commutation reactor
U_D	Output average DC voltage of the rectifier

$\underline{U}_{neg}$	Complex value of the negative sequence voltage
$\underline{U}_{pos}$	Complex value of the positive sequence voltage
U_r	Reverse blocking voltage of thyristor
U_{rm}	Maximum reverse blocking voltage of thyristor
U_{rrm}	Maximum repetitive reverse blocking voltage of thyristor
U_s	Rated RMS phase-to-phase voltage of the power system
$\underline{U}_{zero}$	Complex value of the zero-sequence voltage
X_{com}	Commutation reactance

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Comparison of transformations and feature extraction techniques to characterize fault-induced voltage sags

Comparación de transformadas y técnicas de extracción de características para el análisis de huecos de tensión inducidos por fallas

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Recibido: 17/07/2023. Aceptado: 22/08/2023. Versión final: 12/10/2023

Abstract

This paper presents a comparative study of domain transformations and feature extraction techniques to characterize fault-induced voltage sags. For this purpose, synthetic signals of fault-induced voltage sags are generated through extensive simulations in MATLAB/Simulink. Next, some relevant transformations are applied to the synthetic signals, namely, the space phasor model, discrete Fourier transform, and short-time Fourier transform. A set of statistical, time series, and spectral features are extracted from transformation outputs to obtain signal characterization useful, for instance, for classification of voltage sags employing artificial intelligence techniques. The comparison of the applied domain transformations and feature extraction techniques covers quantitative and qualitative aspects including computation time, storage requirement, linear separability and physical interpretation of features, and suitability for characterizing voltage sags. Finally, the main findings of the work are discussed, and conclusions are remarked.

Keywords: Domain transformation; Feature extraction; Power quality; Voltage sag.

Resumen

Este artículo presenta una comparación de transformadas y técnicas de extracción de características para analizar huecos de tensión inducidos por fallas. Para este propósito, se generan señales sintéticas de huecos de tensión inducidos por fallas mediante simulaciones en MATLAB/Simulink. Luego, se aplican transformadas a las señales sintéticas como el espacio del modelo fasorial, la transformada discreta de Fourier y la transformada de Fourier de tiempo corto. Se extraen características estadísticas, de series temporales y espectrales de las salidas de las transformadas para obtener una caracterización de la señal, útil, por ejemplo, para clasificar huecos de tensión usando inteligencia artificial. La comparación de las transformadas y las técnicas de extracción de características cubre aspectos cuantitativos y

cualitativos como el tiempo de cómputo, requisitos de almacenamiento, separabilidad lineal e interpretación física de las características, e idoneidad para caracterizar huecos de tensión. Finalmente, se discuten los principales hallazgos y se presentan las conclusiones.

Palabras clave: Calidad de potencia; Extracción de características; Hueco de tensión; Transformada de dominio.

1. Introduction

Power Quality Disturbance (PQD) characterization, detection, and classification, is essential for applications in power systems including real-time monitoring and impact assessment, analysis of disturbance propagation, source location, and definition of mitigation measures [1], [2]. Among PQDs, voltage sags are of particular interest because of the huge economic losses caused during the operation of distribution networks [3], [4].

Voltage sags are mainly caused by the connection of heavy loads such as large induction motors, energization of transformers, and power system faults [5]. The latter is the most common root-cause of voltage sags and derive a classification of the disturbance according to the type of fault, i.e., three-phase, two-phase, two-phase to ground, and single phase to ground, which result in ten types considering the possible combinations [6]. The identification of the type of fault that originates the voltage sag, including the recognition of the faulted phase(s), and the characterization of the disturbance, facilitates the protection system to locate and isolate the affected area and thus, decreases voltage sag duration, provides more information on the severity of the event, and enables the proper operation of mitigation measures and devices such as dynamic voltage restorers [7]. Thus, the accurate and automatic detection, classification, and characterization of fault-induced voltage sags is important for the proper operation of protection systems, and the implementation of mitigation measures.

The main challenge in the process of detection, classification, and characterization of PQDs, including voltage sags, is to find efficient indices and metrics that provide reliable and suitable results according to the application. Those indices and metrics can be obtained using domain transformations and feature extraction techniques [1]. Transformations applied to characterize and identify types of voltage sags include techniques in the time domain such as the Phase Space Reconstruction (PSR) [8] and Space Phasor Model (SPM) [9], also referred to as the space vector [6], [7], frequency domain such as Discrete Fourier Transform (DFT) [10], and time-frequency domain such as Short-Time Fourier Transform (STFT) [5], Wavelet Transform (WT) [11]–[14], and Stockwell Transform (ST) [15]–[17]. Moreover, various types of features have been extracted for voltage sag characterization and classification such as

statistical [18], time series [19], [20], image-based [6], [7], [9], and spectral features [21]. It is worth to mention that the transformation step can be omitted, which means that features are not only extracted from transformation coefficients but also from the original waveforms. For instance, higher-order statistics [22], Euclidean distance [23], and gray images [24], have been directly extracted from original waveforms for PQD classification. Each type of features provides specific advantages according to the application and tools to be used. For instance, image-based features are effective when image processing techniques are used for classification, e.g., convolutional neural networks [8], [9].

The success in the stages following transformation and feature extraction depends on several aspects according to the required application. Therefore, some works in the literature have performed comparative studies considering diverse aspects of transformations and feature extraction techniques. Most of those studies focused on analyzing qualitative aspects of the approaches, such as mathematical representation, complexity of implementation, and general descriptions [2], [25]–[27], while few also considered quantitative aspects, including performance and computational burden [28]. Moreover, comparative studies in the literature are mainly dedicated to general-purpose methodologies for the classification of multiple PQD and not to specific disturbances, such as voltage sags.

In this context, this paper develops a comparative study of some relevant time-domain, frequency-domain, and time-frequency domain transformations applied specifically to fault-induced voltage sags, i.e., SPM, Discrete Fourier Transform (DFT), and STFT, respectively, and statistical, time series, and spectral feature extraction techniques, to demonstrate advantages and drawbacks and provide the following contributions:

- The analysis of quantitative and qualitative aspects of the techniques, such as computation time, storage requirement, linear separability of features, physical interpretation of features, and suitability for classification of fault-induced voltage sags.
- The proposal of feature engineering approaches for fault-induced voltage sags classification based on the combination of transformations and feature extraction techniques.

The remainder of the paper is organized as follows. Section 2 describes the background on domain transformations and feature extraction techniques. Section 3 presents the methodology for generation of extensive synthetic signals through simulations in MATLAB/Simulink, and the implementation of transformations and feature extraction techniques. Section 4 compares quantitative and qualitative aspects of transformations and features. Finally, Section 5 highlights the main conclusions of the paper.

2. Theoretical background

2.1. Domain transformations

There are different types of mathematical transformations for PQD detection, classification, and characterization, including techniques in the time domain, frequency domain, and time-frequency domain.

- **Time-domain** analysis involves observing the evolution of signal features over time, providing information on the signal's shape, duration, and amplitude. They are used to analyze events such as sags, and transients [1]. Examples of time-domain techniques are Kalman filters, PSR, and SPM.
- **Frequency-domain** analysis involves decomposing steady-state signals into their frequency components. This technique provides information on the spectral characteristics of the signal. Fourier Transform (FT) and its variants, including DFT, are the main frequency-domain techniques.
- **Time-frequency-domain** analysis involves tracking the evolution of the frequency content of a signal over time. This approach is useful for analyzing non-stationary signals. Examples of time-frequency-domain techniques include the WT, STFT, and ST.

In this paper, the application of SPM, DFT, and STFT to voltage sags is studied. SPM is analyzed because it has shown promising results in terms of accuracy and real-time implementation for classification of fault induced voltage sags [6], [7], [9]. Likewise, DFT is relevant because of its robustness to noise which is essential for accurate and reliable classification of sags in real-life environments [10]. Moreover, STFT has shown benefits of both time and frequency domains by representing spectral and time-series features, for instance, it allows the identification of the high-frequency components in the transition segments of voltage sags [5]. A brief theoretical description of the transformations mentioned above is presented in the following.

2.1.1. Space Phasor Model (SPM)

The SPM is directly derived from Clarke's transformation [9]. The resulting SPM from the three phase-to-neutral voltages, $u_a(t)$, $u_b(t)$, $u_c(t)$, is then given by Equation (1), [9].

$$\underline{s}(t) = \frac{2}{3} [u_a(t) + \alpha u_b(t) + \alpha^2 u_c(t)]; \quad t = 1, \dots, n, \quad (1)$$

where $\alpha = e^{j2\pi/3}$ and $\underline{s}(t)$ is a complex function.

2.1.2. Discrete Fourier Transform (DFT)

The DFT of a finite sequence of equally spaced samples of a time series can be calculated with the fast Fourier transform algorithm as expressed in Equation (2) [17].

$$X(k) = \sum_{n=0}^{N-1} x(n) \exp\left(-\frac{j2\pi kn}{N}\right); \quad k = 0, \dots, N-1, \quad (2)$$

where $X(k)$ is a set of complex numbers, and N is the length of the discrete signal $x(n)$.

2.1.3. Short-Time Fourier Transform (STFT)

The STFT is obtained by applying the DFT to the signal over short time intervals and a sliding window of length W_L , which can be overlapping [5]. The DFT of each segment is stored in a matrix as shown in Equation (3).

$$STFT(X) = [X_0(k), X_1(k), \dots, X_T(k)], \quad (3)$$

with,

$$X_f(k) = \sum_{n=-\infty}^{\infty} x(n) \cdot g(n - sR) e^{-j2\pi kn}, \quad (4)$$

where $X_f(k)$ is the DFT of the windowed signal centered at time sR , R is the distance between two adjacent windows and $g(n)$ is the window function.

2.2. Feature extraction techniques

The step of feature extraction refers to the computation of numerical indices (from transformation outputs or directly from the original waveforms) that are usable for tools in the decision space (detection, classification, and characterization) [1]. These indices comprise statistical, time series, spectral, and/or image-based features. It should be noted that there are limitations in the combination of transformations and features that can be extracted. For instance, the Total Harmonic Distortion

(THD), and the individual harmonics, which are spectral features, can only be extracted from frequency-domain, or time-frequency domain transformations. In this paper, statistical, time series, and spectral features are studied and briefly described in the following.

2.2.1. Statistical features

Statistical features include measures of dispersion of data such as maximum and minimum and statistical moments of different orders. For instance, lower-order statistical moments such as variance, and higher-order statistics such as skewness and kurtosis. These central statistical moments of order r , μ_r , are expressed in Equation (5).

$$\mu_r = \frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^r, \quad (5)$$

where N is the number of samples x_n , $\bar{x}$ is the arithmetic mean, and r is the moment order. Variance (σ^2), skewness, and kurtosis are given by Equations (6), (7), and (8), respectively.

$$\sigma^2 = \mu_2, \quad (6)$$

$$Skew = \frac{\mu_3}{\sigma^3}, \quad (7)$$

$$Kurt = \frac{\mu_4}{\sigma^4}. \quad (8)$$

2.2.2. Time series features

Sample-based or cycle-based features can be extracted from the time series data of a discrete signal, i.e., directly from the original waveform [1]. Euclidean distance, also referred to as momentary deviation [23], is a sample-based technique and is expressed in Equation (9).

$$u_{dev}(t) = u_s(t) - u_{ref}(t); t = 1, 2, \dots, N, \quad (9)$$

where $u_s(t)$ is an instant value of the original discrete voltage signal of length N , and $u_{ref}(t)$ is the corresponding instant value of a reference signal which is usually a rated power frequency (e.g., 60 Hz) undistorted sinusoidal voltage waveform.

Euclidean distance can also be calculated for time-domain transformations such as the SPM. In such a case, it can be expressed as the distance to the origin in the complex plane as represented by Equation (10).

$$s_{dev}(t) = |\underline{s}(t)|. \quad (10)$$

2.2.3. Spectral features

Spectral features contain information on the frequency content of the signal, which is obtained with frequency- or time-frequency-domain transformations. For instance, magnitude and phase angle of individual harmonics, and THD as expressed in Equation (11), are spectral features.

$$THD = \frac{\sqrt{\sum_{h=2}^{\infty} U_h^2}}{U_1}, \quad (11)$$

where U_1 is the amplitude of the fundamental component of voltage, and U_h is the h^{th} harmonic amplitude.

3. Methodology and numerical simulations

According to [1], the process for PQD monitoring and analysis consist of eight steps, namely, (i) input data preparation, (ii) data preprocessing, (iii) transformation, (iv) feature extraction, (v) feature selection, (vi) detection, (vii) classification, and (viii) characterization. This paper deals with the first four stages as explained in the following. Furthermore, the study is limited to the analysis of fault-induced voltage sags and focused on the comparison of domain transformations in step (iii), feature extraction techniques in step (iv), and the performance of several combinations of transformations and feature extraction techniques.

In the first step, synthetic time-domain discrete signals are obtained from simulations of fault-induced voltage sags in MATLAB/Simulink using the scheme of Figure 1, adapted from [29]. The scheme presents a solidly grounded radial distribution network at the Medium-Voltage (MV) level of 13.2 kV. The network consists of two parallel feeders of 1 km and 10 km supplying passive aggregated residential loads of 100 kVA, and 1 MVA, respectively, with lagging power factor of 0.95.

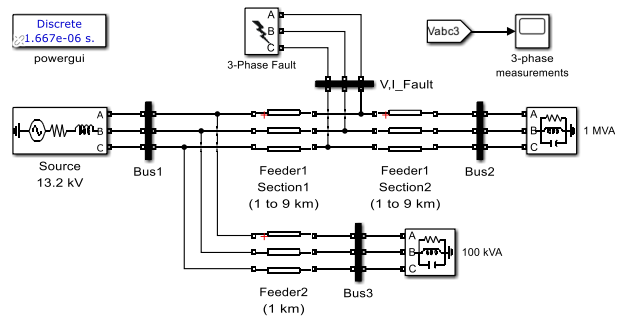


Figure 1. Schema in MATLAB/Simulink for the simulation of fault-induced voltage sags. Source: elaborated by the authors.

Ten types of sags are simulated, according to the type of fault as reported in Table 1. Similar identification performance is expected for the same groups of fault types, e.g., for all two-phase to ground faults. However, the distinction of the subtypes is made, i.e., abG, acG, and bcG, because the identification of the faulted phase(s) is useful for the proper operation of protection systems [7]. One hundred simulations are performed for each type varying randomly the location and duration of faults in Feeder1 for a total of 1,000 simulations. Thus, a database of 3,000 voltage signals (i.e., 1,000 for each phase) measured at Bus3 is generated. Only Bus3 is considered because end users at this bus would experience mainly voltage sags during faults in Feeder1, whereas end users at Bus2 would experience interruptions. A 1- μ s time step is used for the simulation of ten cycles at 60 Hz of the voltage signals. Signals are resampled with a 10- μ s time step because of storage issues. Therefore, each of the 3,000 voltage signals consists of 10,000 discrete values.

Table 1. Classification of sags according to fault type.

Fault type	Sag type
Three-phase	abc
Two-phase to ground (phases a, b)	abG
Two-phase to ground (phases a, c)	acG
Two-phase to ground (phases b, c)	bcG
Two-phase (phases a, b)	ab
Two-phase (phases a, c)	ac
Two-phase (phases b, c)	bc
Single-phase to ground (phase a)	aG
Single-phase to ground (phase b)	bG
Single-phase to ground (phase c)	cG

Source: elaborated by the authors.

Figure 2 shows examples of the simulated signals, illustrating the normalized three-phase voltages for a three-phase, and a two-phase to ground fault.

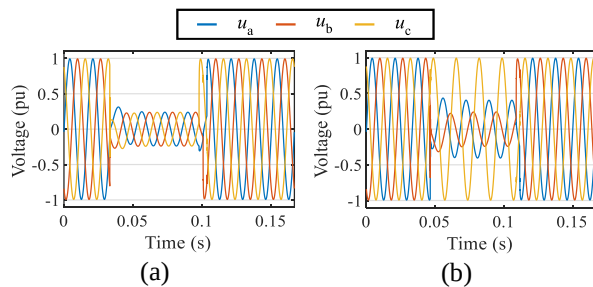


Figure 2. Synthetic sags caused by (a) a three-phase fault (abc), and (b) a two-phase to ground fault (abG). Source: elaborated by the authors.

The SPM, DFT, and STFT are applied to signals. Then, a set of statistical, time series, and spectral features are computed from transformation coefficients. A set of features is also extracted from the original time series signals to observe the results with no transformation. The combinations of applied transformations and extracted features are summarized in Figure 3.

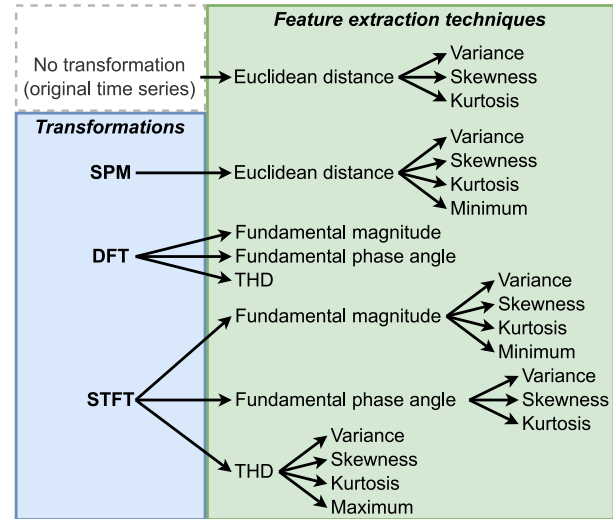


Figure 3. Applied transformations and features. Source: elaborated by the authors.

According to Figure 3, the Euclidean distance of the original time series is computed using Equation (9). Then, normalized variance, skewness and kurtosis of the computed Euclidean distance are calculated with Equations (6), (7), and (8), respectively. This process is applied to all synthetic signals of each of the phases, i.e., 3,000 signals (1,000 per phase), resulting in single values of variance, skewness, and kurtosis per signal, i.e., 3,000 values for each one. Figure 4 shows results of the process applied to the simulation case of Figure 2b.

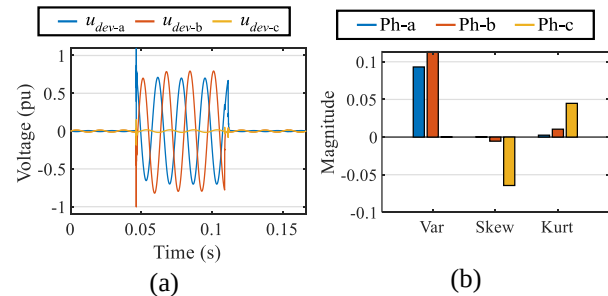


Figure 4. (a) Euclidean distance, (b) variance, skewness, and kurtosis for voltage signals of the fault in Figure 2b. Source: elaborated by the authors.

In the case of SPM, the signals of the three phases for each case simulated are converted to a single series of complex values using Equation (1), therefore, 1,000 series of complex values are obtained. The Euclidean distances for the 1,000 series are computed with Equation (10) resulting in 1,000 values for each dispersion measure (variance, skewness, kurtosis, and minimum). To illustrate the process, Figure 5 shows results of the two-phase to ground fault (abG) simulation case of Figure 2b.

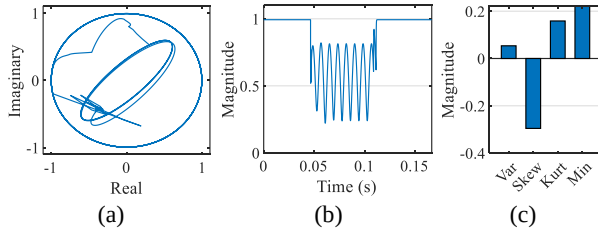


Figure 5. (a) SPM of the three phases, (b) corresponding Euclidean distance, and (c) variance, skewness, kurtosis, and minimum during the abG fault of Figure 2b. Source: elaborated by the authors.

Moreover, DFT is applied to signals of each of the phases with an interval of ten cycles and features are extracted as shown in Figure 3. A total of 3,000 values of fundamental magnitude, fundamental phase angle, and THD are calculated. Figure 6 shows an example with the computation of these features for the simulation case in Figure 2b. Fundamental magnitudes and phase angles of each phase are shown in Figure 6a.

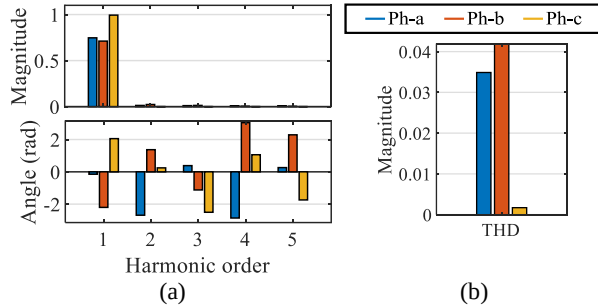


Figure 6. (a) DFT of each phase, magnitude (top), and phase angle (bottom), and (b) corresponding THD during the two-phase to ground fault (abG) of Figure 2b. Source: elaborated by the authors.

Finally, STFT is applied to signals of each of the phases, resulting in 3,000 matrices with information in the time and frequency domains using Equations (3), and (4). The window size is one cycle, i.e., 1,000 samples of the signal, and the overlap is half a cycle, i.e., the window hops over the original signal of ten cycles at intervals of

$R=500$ samples, resulting in 19 computations of the DFT. The spectrograms for each of the phases in the simulation case of Figure 2b are shown in Figure 7, where changes in the spectrum are apparent during voltage sag initiation and ending. Figure 8 presents the evolution over time of the corresponding fundamental magnitude and phase angle, with the respective measures of dispersion. The evolution over time of the calculated THD is shown in Figure 9.

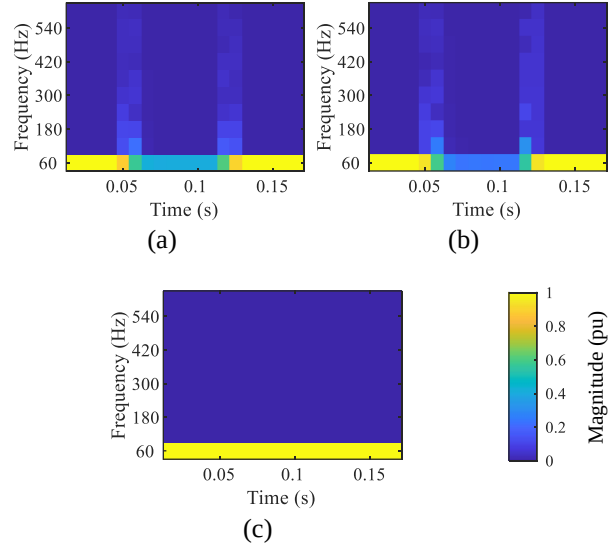


Figure 7. Spectrogram obtained with the STFT applied to synthetic signals of (a) phase a, (b) phase b, and (c) phase c during the abG fault of Figure 2b. Source: elaborated by the authors.

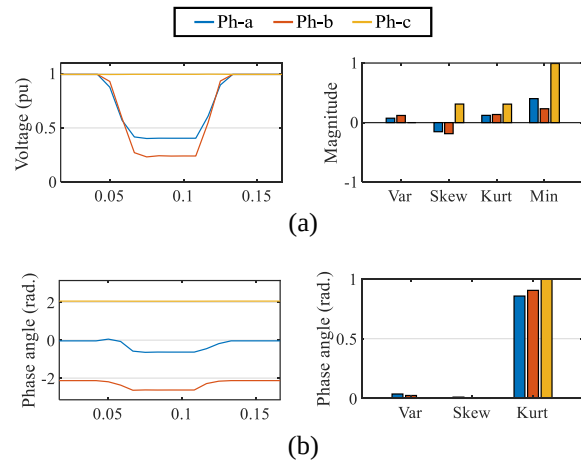


Figure 8. (a) Fundamental magnitude and (b) phase angle over time, obtained with the STFT applied to signals in Figure 2b (left), and dispersion measures (right). Source: elaborated by the authors.

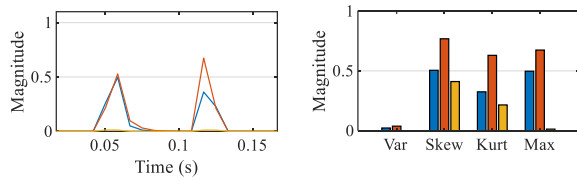


Figure 9. THD over time obtained with the STFT applied to signals in Figure 2b (left), and corresponding dispersion measures (right). Source: elaborated by the authors.

4. Comparison and results

The combinations of domain transformations and extracted features in Figure 3 are compared according to quantitative aspects including the computation time, storage requirement, and linear separability. Qualitative aspects such as physical interpretation, and suitability for fault-induced voltage sag characterization are also analyzed.

4.1. Computation time

The computation time reported in Table 2 is the median of 100 repeated calculations for each transformation and

feature extraction technique, executed on a CPU with a six-core processor, 2.20 GHz, and 12 GB RAM. The computation time covers the application of techniques on the 1,000 fault simulation cases, i.e., 3,000 voltage signals (1,000 per phase). The table shows that SPM is the fastest, and STFT is the slowest of the applied transformations, because SPM uses simple operations in a sample-based manner, whereas STFT calculates and stores the DFT on a sliding window which requires more calculation steps.

Furthermore, Table 2 shows that the calculations of the Euclidean distances of the original time series and SPM, and THD from DFT have reduced durations of similar order. Computations of fundamental magnitudes and phase angles from DFT and STFT are performed in 1 ms or less, whereas THD from STFT takes 257 ms. Computations of dispersion measures take longer for Euclidean distances (up to 915 ms) because more samples must be evaluated, whereas spectral features from STFT are calculated only 19 times per signal. The total computation times demonstrates that transformation and features extraction is faster with SPM, followed by DFT and no transformation, whereas the STFT is the slowest.

Table 2. Computation time for transformation and feature extraction.

Transformation		Feature extraction 1		Feature extraction 2		Total computation time (s)	
Technique	Computation time (s)	Feature	Computation time (s)	Feature	Computation time (s)		
None	--	Euclidean distance	0.058	Variance	0.074	1.897	
				Skewness	0.850		
				Kurtosis	0.915		
SPM	0.071	Euclidean distance	0.033	Variance	0.033	0.654	
				Skewness	0.294		
				Kurtosis	0.290		
				Minimum	0.004		
DFT	0.936	Fundamental magnitude	~ 0	--	--	0.915	
		Fundamental angle	~ 0	--	--		
		THD	0.015	--	--		
STFT	1.986	Fundamental magnitude	0.001	Variance	~ 0	1.99	2.27
				Skewness	0.002		
				Kurtosis	0.002		
				Minimum	~ 0		
		Fundamental angle	0.001	Variance	0.001	2.00	
				Skewness	0.005		
				Kurtosis	0.006		
		THD	0.257	Variance	~ 0	2.25	
				Skewness	0.002		
				Kurtosis	0.002		
				Maximum	~ 0		

Source: elaborated by the authors.

4.2. Storage requirement

Table 3 reports the size of matrices/vectors with data obtained after transformations and feature extraction techniques for one fault simulation case and a signal length of ten power cycles. In terms of application, this storage capability would be the requirement for a device in the field to perform the algorithms applied in this paper. For instance, SPM results in a $1 \times 10,000$ vector per fault simulated, whereas STFT results in a $3 \times 501 \times 19$ matrix per fault simulated, where dimensions correspond to phase, frequency data, and time data, respectively. After feature extraction with dispersion measures, SPM requires the smallest storage for four values (which would be the number of features used in a subsequent step, e.g., classification), whereas STFT requires the biggest for 11 vectors with size 3×1 .

4.3. Linear separability

Linear separability of features is tested with the implementation of a linear support vector machine [23] in the classification learner app in MATLAB/Simulink. To that end, the dataset of synthetic signals from the 1,000 simulated cases is used as input. Then, a holdout

validation dataset with 30% of the signals is defined randomly but considering an equal proportion to each type of sag. The linear SVM is automatically trained and tested in the classification learner app. If the features included are linearly separable for classification of fault-induced voltage sags according to Table 1, the results of accuracy tend to be 100%.

Results of the classification are reported in Table 4 where very good accuracy is observed in all cases except SPM. In that case, using only the Euclidean distance of SPM causes loss of information to distinguish between two-phase and single-phase faults. However, SPM provide more interesting image-based features [9] that are not analyzed in this paper. Spectral features provide useful information for sag classification, especially, THD. In the case of STFT, the individual performance of the dispersion measures applied only to the fundamental magnitudes is assessed resulting in a very satisfactory 99.9% accuracy. Similarly, performance of the dispersion measures of the fundamental phase angle and THD are also analyzed resulting in 96.3% and 100% accuracy. Finally, the combined performance of all dispersion measures obtained with STFT is assessed resulting also in 100% accuracy.

Table 3. Storage requirement of transformations and features.

Transformation		Feature extraction 1		Feature extraction 2	
Technique	Matrix size	Feature	Matrix size	Feature	Matrix size
None	$3 \times 10,000$	Euclidean distance	$3 \times 10,000$	Variance	3×1
				Skewness	3×1
				Kurtosis	3×1
SPM	$1 \times 10,000$	Euclidean distance	$1 \times 10,000$	Variance	1×1
				Skewness	1×1
				Kurtosis	1×1
				Minimum	1×1
DFT	$3 \times 5,001$	Fundamental magnitude	3×1	--	--
		Fundamental angle	3×1	--	--
		THD	3×1	--	--
STFT	$3 \times 501 \times 19$	Fundamental magnitude	3×19	Variance	3×1
				Skewness	3×1
				Kurtosis	3×1
				Minimum	3×1
		Fundamental angle	3×19	Variance	3×1
				Skewness	3×1
				Kurtosis	3×1
				Maximum	3×1
		THD	3×19	Variance	3×1
				Skewness	3×1
				Kurtosis	3×1
				Maximum	3×1

Source: elaborated by the authors.

Table 4. Linear separability of the extracted features.

Transformation	Feature extraction 1	Feature extraction 2	Accuracy (%)	
None	Euclidean distance	Variance, Skewness, Kurtosis	97.4	
SPM	Euclidean distance	Variance, Skewness, Kurtosis, Minimum	27.6	
DFT	Fundamental magnitude, Fundamental angle, THD	--	100	
STFT	Fundamental magnitude	Variance, Skewness, Kurtosis, Minimum	99.9	100
	Fundamental angle	Variance, Skewness, Kurtosis	96.3	
	THD	Variance, Skewness, Kurtosis, Maximum	100	

Source: elaborated by the authors.

4.4. Physical interpretation

Interpretability of features is very useful to obtain a better understanding of the voltage sag phenomenon to provide better applications of the analysis. For instance, duration and magnitude are relevant features for sag impact assessment. In that sense, STFT provide very detailed information as can be seen in spectrograms of Figure 7 where sag initiation and ending can be identified with changes in the spectral content and the fundamental magnitude. Moreover, SPM shows potentialities in image-based features (see Figure 5a).

It should be noted that physical features such as duration and magnitude in voltage sags can be obtained through well-established and widely accepted methods indicated in standards, e.g., IEC 61000-4-30. Therefore, the implementation of this capability would be very valuable in proposed detection, classification, and characterization methodologies to comply with standards. Moreover, to analyze these capabilities in different transformation and feature extraction techniques is relevant because they can provide enhanced performance compared to standard algorithms. For instance, DFT and STFT are robust against noise, and SPM has demonstrated improved capabilities for the identification of voltage sag characteristics such as phase angle jump [30], and shape in multistage voltage sags [31].

4.5. Suitability for voltage sag characterization

Results of dispersion measures applied in the Euclidean distance of the original time series show very good performance for sag characterization and classification. However, the interpretation of features is not straightforward, and the storage requirements can hinder the real-time application in embedded systems. Despite the Euclidean distance of the SPM shows inaccurate classification results, potentialities are observed in the use of the image-based features that are interpretable and seemed to provide better classification performance [9].

Furthermore, DFT shows that frequency-domain information provides a very good performance in terms of classification but limited interpretable features. That drawback is overcome with the implementation of time-frequency domain transformations such as STFT, where very useful information in both time and frequency domains is obtained at the cost of higher computational time and storage requirements.

4.6. Summary and comparative tables

To summarize the results of the comparison discussed above, Table 5 presents advantages and drawbacks of each of the analyzed transformations in the time domain, i.e., none, SPM; the frequency domain, i.e., DFT; and the time-frequency domain, i.e., STFT. In contrast to the comparison presented in the previous tables, Table 5 intends to compare qualitative aspects of the individual domain transformations and not the combined performance. It should also be noted that the comparison is made in the context of the work developed in this paper, therefore, the observations are obtained from the simulations and algorithms implemented, as well as for the specific application of detection, classification, and characterization of fault-induced voltage sags.

Similarly, Table 6 reports advantages and drawbacks of the time-series, spectral, and statistical feature extraction techniques studied in this work.

Some examples in the literature illustrate the specific advantages of each domain transformation and feature extraction technique. For instance, SPM is useful for real-time applications including the operation of protection systems and the implementation of mitigation measures such as dynamic voltage restorer operation [6], [7]. Moreover, robustness of DFT and STFT, and spectral and statistical features is relevant in the application of detection, classification, and characterization of fault-induced voltage sags in noisy environments [5], [10].

Table 5. Comparison of domain transformations.

Domain	Transformation	Advantages	Drawbacks
Time	None	- No computational burden - Simplicity - Domain transformation knowledge is not required - No loss of information	- Involve more complex subsequent decision space - Lack of spectral information - High storage requirement
	SPM	- Low computational burden - Simplicity - Low loss of information - Capability of representing three-phase voltages in one signal - It provides useful image-based features	- High/medium storage requirement - Lack of spectral information - Lack of single-phase information
Frequency	DFT	- Reduced storage requirement - Higher resolution of spectral information (compared to the application of one-cycle-based STFT)	- Loss of time information - Inaccurate for non-periodic signals - Medium/high computational burden - It is prone to spectral leakage and windowing
Time-frequency	STFT	- Flexibility in window selection - It provides information in both time and frequency domains - Robust to noise	- Complexity - Computational burden - Trade-off between time and frequency resolution - It is prone to spectral leakage and windowing

Source: elaborated by the authors.

5. Conclusion

This paper provides a comparative study of domain transformations and feature extraction techniques applied to synthetic signals of fault-induced voltage sags aimed at providing input data for subsequent steps in the process of detection, classification, and characterization of the disturbance. Domain transformations including the Space Phasor Model (SPM), Discrete Fourier Transform (DFT), and Short-Time Fourier Transform (STFT) are implemented, and feature extraction techniques such as Euclidean distance, variance, skewness, kurtosis, minimum, and maximum are applied. The results show that Euclidean distance of the original signal provides very good and simple performance at the cost of relatively high storage requirement, and computation time. Similarly, STFT provides high linear separability and physical interpretability of features using more storage and computation time. DFT allows accurate classification results with reduced storage requirement and low/medium computational burden, however, physical interpretability of features is very limited. Moreover, Euclidean distance of the SPM shows loss of information for classification, but image-based features, that are not analyzed in detail in this paper, seem promising for characterization and classification of fault-induced voltage sags.

Based on the conclusions mentioned above, accurate and simple classifiers, i.e., based on Euclidean distance of the original waveform + dispersion measures, and DFT + spectral features are suitable for real-time applications where only classification is required (limited interpretability of features). In addition, a more complex approach, e.g., STFT + spectral features + dispersion measures, is useful for robust and reliable classification (required, e.g., in noisy environments caused by different disturbances), and characterization when physical interpretability is needed.

Finally, the application of SPM in combination with complementary transformation steps and additional feature extraction techniques applied to the resulting image-based features in the complex plane (see Figure 5a) seem to be a promising future work. Research in this direction has been found in the literature related to sags [6], [7], [9], but a more comprehensive methodology is still required to provide results in all steps of the decision space, i.e., detection, classification, and characterization.

Funding

This work was supported by Centro de Investigaciones y Desarrollo Científico (CIDC) of the Universidad Distrital Francisco José de Caldas, Bogotá, Colombia.

Table 6. Comparison of feature extraction techniques.

Type	Feature	Advantages	Drawbacks
Time-series	Euclidean distance	- Low computational burden - Simple and intuitive - Robust to noise - Suitable for low-dimensional spaces	- Sensitive to scale and normalization. - Limited in high-dimensional spaces. - It may not capture complex relationships
Spectral	Fundamental magnitude	- No computational burden - Robust to noise - High capability for analysing sags	- No time information - Lack of location information
	Fundamental phase angle	- No computational burden - Robust to noise - Phase information during events	- Lack of time information - No information on event severity
	THD	- Low computational burden - Spectral information on a wide range of frequencies - Robust to noise	- No time information - Sensitive to signal bandwidth - It may not capture transient behaviour (sag start/end) - Limited dynamic range
Statistical	Variance	- It captures signal dispersion - Easy to interpret - Sensitive to signal amplitude changes - Robust to outliers	- Limited information - May not capture shape information - Limited dynamic range
	Skewness	- Captures signal asymmetry - Sensitive to changes in signal shape - Easy to interpret	- Limited information - Sensitive to outliers
	Kurtosis	- Captures signal shape information - Can differentiate between signal types - Sensitive to changes in signal shape - Provides complementary information to other measures	- Limited information - Sensitive to outliers
	Minimum / Maximum	- Provide information about the lowest and highest points - Easy to compute - Can be used to detect outliers - Complement other measures	- Limited information - Sensitive to noise - May not capture information about signal dynamics

Source: elaborated by the authors.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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